ELITE IGCSE ACADEMY

Edexcel IGCSE Mathematics A - Unit 1 (4WM1)

Higher Tier

Elite IGCSE Classified Unit 1 Expertise Answers (Q20+)

131 Questions | Question page then worked-solution page

PREPARED & CLASSIFIED BY

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Modular palette | **Unit 1 / 4WM1**

Student worked-solution book - Q20+ expertise.

Each question is followed by its worked solution on the next page.

Contents

Topic 1

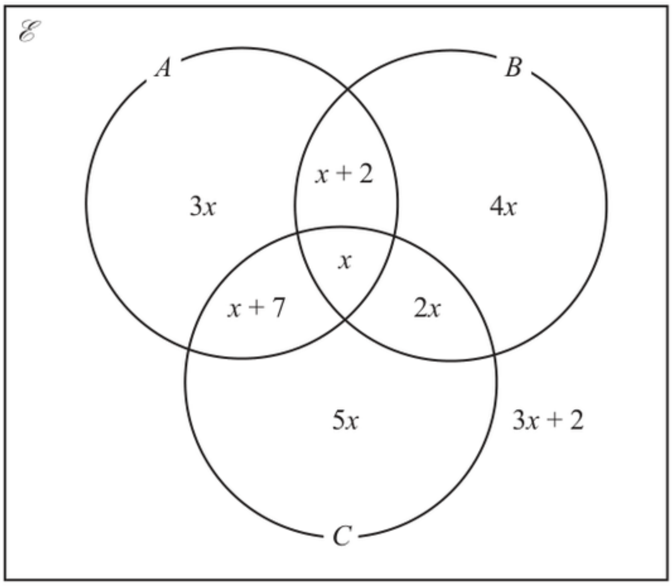
Set Notation & Venn Diagrams

CLASSIFIED PROBLEM Paper: P2H | Session: Nov 2024 | Q#: 21 | Marks: 4

TOPIC: Set Notation & Venn Diagrams

Source: Nov 2024 P2H

21 The Venn diagram shows information about the numbers of items in set A, set B and set C, where x is an integer.



Given that $n(A \cup B)' = 26$

find $n(A' \cap C)$

$n(A' \cap C) =$ ....

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P2H | Question 21

Set Notation & Venn Diagrams

1. The complement of contains the

The complement of $A \cup B$ contains the regions outside both A and B. These are $5x$ and $3x + 2$.

2. Calculate value

So $5x + (3x + 2) = 26$, hence $8x + 2 = 26$ and $x = 3$.

3. Calculate statistic

Now $A' \cap C$ means inside C, but not inside A. These are the regions $5x$ and $2x$.

4. Calculate value

So $n(A' \cap C) = 5x + 2x = 7x = 7(3) = 21$.

Final answer

21

Topic 2

Rounding, Estimation & Bounds

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jan 2022 | Q#: 22 | Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Source: Jan 2022 P1HR

22 The diagram shows triangle *ABC*

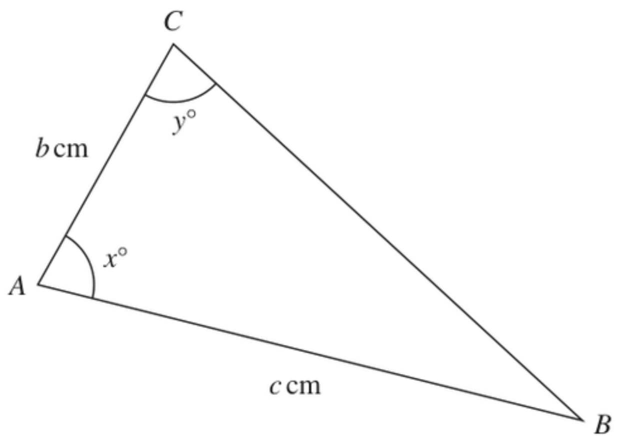


Diagram **NOT** accurately drawn

$c = 11.5$ correct to one decimal place
$x = 80$ correct to the nearest whole number
$y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of b
Show your working clearly.
Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 22
Rounding, Estimation & Bounds

1. Find upper bound

For an upper bound for b, use the upper bound of c:
$c = 11.55$

2. Use sine rule

Use the lower bounds of x and y, because this makes angle B as large as possible and makes $\sin y$ as small as possible.
$x = 79.5°, \quad y = 74.5°$

3. Use trigonometry

So
$\angle B = 180° - 79.5° - 74.5° = 26°$

4. Use trigonometry

Using the sine rule,

$$\frac{b}{\sin 26°} = \frac{11.55}{\sin 74.5°}$$

$$b = \frac{11.55 \sin 26°}{\sin 74.5°} = 5.254\ldots$$

Final answer
5.25 cm to 3 significant figures.

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2023 | Q#: 25 | Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Source: Jun 2023 P2H

25 A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in g/cm³ correct to 2 decimal places.
Show your working clearly.

.... g/cm^3

(Total for Question 25 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 25

Rounding, Estimation & Bounds

1. Find upper bound

For the upper bound of density, use the upper bound of the mass and the lower bound of the volume.

2. The radius is cm correct

The radius is 2.8 cm correct to 1 decimal place, so the lower bound is
$r = 2.75$

3. Find upper bound

$M = 260$ correct to 2 significant figures, so the upper bound is
$M = 265$

4. The mass is given as

The mass is given as $M\pi$ grams, where $M = 265$ for the upper bound.

5. The volume of the sphere

The volume of the sphere is

$$\frac{4}{3}\pi r^3$$

6. Rearrange formula

So

$$\text{density} = \frac{M\pi}{\frac{4}{3}\pi r^3}$$

7. The cancels

The π cancels:

$$\text{density} = \frac{3M}{4r^3}$$

8. Upper bound

Upper bound:
$\dfrac{3(265)}{4(2.75)^3} = 9.5567\ldots$

Final answer

9.56 g/cm^3 to 2 decimal places.

CLASSIFIED PROBLEM

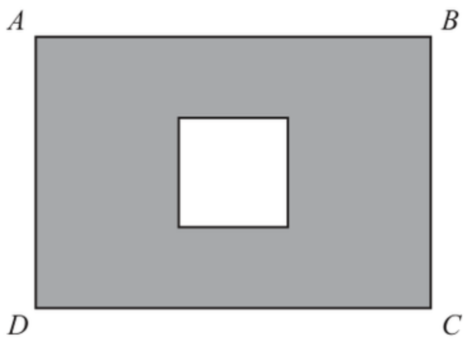
Paper: P2H | Session: Jun 2025 | Q#: 22 | Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Source: Jun 2025 P2H

22 The diagram shows a square inside rectangle *ABCD*

Diagram **NOT** accurately drawn



The total area of the region shown shaded in the diagram is X cm²

$AB = 11.5$ cm correct to the nearest 0.5 cm
$BC = 9.2$ cm correct to 2 significant figures
side of square $= 4.1$ cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

X = ...

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 22

Rounding, Estimation & Bounds

1. The shaded area is

The shaded area is
$X =$ area of rectangle $-$ area of square

2. Bounds

Bounds:
$11.25 \le AB < 11.75$
$9.15 \le BC < 9.25$
$4.05 \le$ side of square < 4.15

3. Find lower bound

Lower bound for X:
$11.25(9.15) - 4.15^2 = 85.715$

4. Find upper bound

Upper bound for X:
$11.75(9.25) - 4.05^2 = 92.285$

5. Estimate the value

So $85.715 \le X < 92.285$. Both bounds round to 90 to the nearest 10.

Final answer
$X = 90$ cm^2 to a suitable degree of accuracy.

CLASSIFIED PROBLEM

Paper: P1H | Session: May 2021 | Q#: 20 | Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Source: May 2021 P1H

20 $P = \dfrac{t - w}{y}$

$t = 9.7$ correct to 1 decimal place

$w = 5.9$ correct to 1 decimal place

$y = 3$ correct to 1 significant figure

Calculate the upper bound for the value of P.
Show your working clearly.

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 20
Rounding, Estimation & Bounds

1. Estimate the value

$$P = \frac{t - w}{y}$$

2. Find upper bound

For the upper bound of P, make the numerator as large as possible and the denominator as small as possible.
$t = 9.75, \quad w = 5.85, \quad y = 2.5$

$$P = \frac{9.75 - 5.85}{2.5}$$

$$P = \frac{3.90}{2.5} = 1.56$$

Final answer
1.56

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2024 | Q#: 21 | Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Source: Nov 2024 P1H

21 $T = \dfrac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.
Show your working clearly.

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | Question 21

Rounding, Estimation & Bounds

1. Estimate the value

$$T = \frac{x^2 + y^2}{w}$$

2. Find upper bound

For the upper bound of T, use the upper bounds of x and y, and the lower bound of w.

$x = 28.45, \quad y = 17.5, \quad w = 87.5$

$$T = \frac{28.45^2 + 17.5^2}{87.5}$$

$$T = 12.7503...$$

Final answer

12.8 to 3 significant figures.

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2025 | Q#: 24 | Marks: 5

TOPIC: **Rounding, Estimation & Bounds**

Source: Nov 2025 P2H

24 $D = \dfrac{n}{p - q}$

$n = 10.3$ correct to 1 decimal place
$p = 7.24$ correct to 2 decimal places
$q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 24
Rounding, Estimation & Bounds

1. Estimate the value

$$D = \frac{n}{p-q}$$

2. Bounds

Bounds:
$10.25 \le n < 10.35$
$7.235 \le p < 7.245$
$4.385 \le q < 4.395$

3. Lower bound

Lower bound:

$$\frac{10.25}{7.245 - 4.385} = 3.5839...$$

4. Upper bound

Upper bound:

$$\frac{10.35}{7.235 - 4.395} = 3.6443...$$

5. Estimate the value

So $3.5839... \le D < 3.6443....$ Both bounds round to 3.6 to 1 decimal place.

Final answer
$D = 3.6$ to a suitable degree of accuracy.

Topic 3

Surds

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2020 | Q#: 20 | Marks: 3

TOPIC: **Surds**

Source: Jan 2020 P2H

20 The area of a rectangle is 18 cm^2

The length of the rectangle is $\left(\sqrt{7}+1\right)$ cm.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.
Give your answer in the form $a\sqrt{b}+c$ where a, b and c are integers.

.... cm

(Total for Question 20 is 3 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jan 2020 P2H | Question 20
Surds

1. Area of rectangle

Area of rectangle:

$$18 \text{ cm}^2$$

2. Length

Length:

$$\sqrt{7}+1$$

3. Width

Width:

$$\frac{18}{\sqrt{7}+1}$$

4. Rationalise the denominator

Rationalise the denominator:

$$\frac{18}{\sqrt{7}+1} \times \frac{\sqrt{7}-1}{\sqrt{7}-1}$$

$$= \frac{18(\sqrt{7}-1)}{7-1}$$

$$= \frac{18(\sqrt{7}-1)}{6}$$

$$= 3(\sqrt{7}-1)$$

$$= 3\sqrt{7}-3$$

Final answer
$\boxed{3\sqrt{7}-3 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jun 2022 | Q#: 21 | Marks: 4

TOPIC: **Surds**

Source: Jun 2022 P1HR

21 Express $\dfrac{3+\sqrt{8}}{\left(\sqrt{2}-1\right)^2}$ in the form $p+\sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 21
Surds

1. Simplify surd

$$\frac{3+\sqrt{8}}{(\sqrt{2}-1)^2}$$

2. Simplify surd

First simplify:

$$\sqrt{8} = 2\sqrt{2}$$

3. Simplify surd

and

$$(\sqrt{2}-1)^2 = 2 - 2\sqrt{2} + 1 = 3 - 2\sqrt{2}$$

4. Simplify surd

So

$$\frac{3+2\sqrt{2}}{3-2\sqrt{2}}$$

5. Rationalise the denominator

Rationalise the denominator:

$$\frac{3+2\sqrt{2}}{3-2\sqrt{2}} \times \frac{3+2\sqrt{2}}{3+2\sqrt{2}} = \frac{(3+2\sqrt{2})^2}{9-8}$$

$$(3+2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$$

Final answer
$17 + 12\sqrt{2}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2023 | Q#: 23 | Marks: 4

TOPIC: **Surds**

Source: Jun 2023 P2H

23 The diagram shows a cuboid with a square cross section.

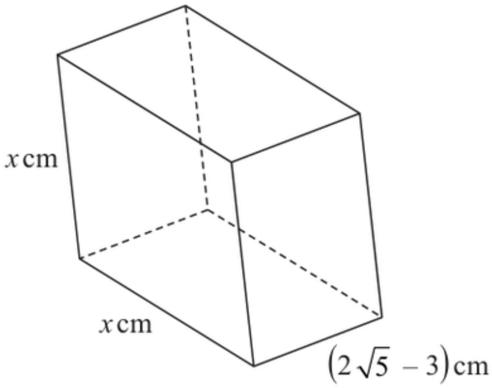


Diagram **NOT** accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\,\text{cm}^3$

Without using a calculator, find the value of x
Give your answer in the form $a + \sqrt{b}$ where a and b are integers.
Show your working clearly.

$x =$ ....

(Total for Question 23 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | **Question 23**

Surds

1. The cuboid has square cross

The cuboid has square cross section, so its volume is

$$x^2(2\sqrt{5}-3)$$

2. Given

Given

$$x^2(2\sqrt{5}-3) = 13+6\sqrt{5}$$

3. Use trigonometry

So

$$x^2 = \frac{13+6\sqrt{5}}{2\sqrt{5}-3}$$

4. Rationalise the denominator

Rationalise the denominator:

$$x^2 = \frac{13+6\sqrt{5}}{2\sqrt{5}-3} \cdot \frac{2\sqrt{5}+3}{2\sqrt{5}+3}$$

$$x^2 = \frac{(13+6\sqrt{5})(2\sqrt{5}+3)}{20-9}$$

$$x^2 = \frac{99+44\sqrt{5}}{11}$$

$$x^2 = 9+4\sqrt{5}$$

5. Use trigonometry

Now

$$(2+\sqrt{5})^2 = 4+4\sqrt{5}+5 = 9+4\sqrt{5}$$

6. Use trigonometry

Therefore

$$x = 2+\sqrt{5}$$

Final answer

$\boxed{2+\sqrt{5}}$

Topic 4

Algebraic Roots & Indices

CLASSIFIED PROBLEM

Paper: 4WM1H **Session: Specimen** **Q#: 22** **Marks: 3**

TOPIC: **Algebraic Roots & Indices**

Source: Specimen 4WM1H

22 $\dfrac{18 \times \left(\sqrt{27}\right)}{6 \times 9^{2n+8}} = 3^x$

Express x in terms of n

Show your working clearly and simplify your expression.

$x =$ ....

WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM1H | Question 22
Algebraic Roots & Indices

1. Write everything as powers of 3

$$\sqrt{27} = 3^{3/2}, \qquad 9 = 3^2, \qquad \frac{18}{6} = 3$$

2. Simplify the powers

$$\frac{18(\sqrt{27})^{4n+6}}{6(9)^{2n+8}} = 3 \times \frac{\left(3^{3/2}\right)^{4n+6}}{(3^2)^{2n+8}}$$

$$= 3^1 \times \frac{3^{6n+9}}{3^{4n+16}}$$

3. Subtract the indices

$$3^{1+6n+9-(4n+16)} = 3^{2n-6}$$

So

$$x = 2n - 6$$

Final answer
$x = 2n - 6$.

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2020 | Q#: 21 | Marks: 7

TOPIC: **Algebraic Roots & Indices**

Source: Jan 2020 P2HR

21 (a) Simplify fully $\dfrac{10x^2 + 23x + 12}{4x^2 - 9}$

....

(3)

$$2^{2y} \times 2^{3y+2} = \frac{8^{5y}}{4^n}$$

(b) Find an expression for n in terms of y.
Show clear algebraic working and simplify your expression.

....

(4)

(Total for Question 21 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 21
Algebraic Roots & Indices

1. (a)

(a)

$$\frac{10x^2+23x+12}{4x^2-9} = \frac{(5x+4)(2x+3)}{(2x-3)(2x+3)}$$
$$= \frac{5x+4}{2x-3}$$

2. (b)

(b)

$$2^{2y} \times 2^{3y+2} = 2^{5y+2}$$

3. Evaluate fraction

Also,

$$\frac{8^{5y}}{4^n} = \frac{(2^3)^{5y}}{(2^2)^n} = 2^{15y-2n}$$

4. Equate powers

Equate powers:

$$5y + 2 = 15y - 2n$$

$$2n = 10y - 2$$

$$n = 5y - 1$$

Final answer
(a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2022 | Q#: 24 | Marks: 3

TOPIC: **Algebraic Roots & Indices**

Source: Jun 2022 P2H

24

$$\frac{18 \times \left(\sqrt{27}\right)}{6 \times 9^{2n+8}} = 3^{x}$$

Express x in terms of n
Show your working clearly and simplify your expression.

$x =$ ....

(Total for Question 24 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2H | Question 24

Algebraic Roots & Indices

1. Simplify surd

$$\frac{18(\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

2. Use powers of 3

Use powers of 3:

$$18 \div 6 = 3$$

$$\sqrt{27} = (3^3)^{1/2} = 3^{3/2}$$

$$(\sqrt{27})^{4n+6} = 3^{6n+9}$$

$$9^{2n+8} = (3^2)^{2n+8} = 3^{4n+16}$$

3. Evaluate fraction

So

$$\frac{3 \cdot 3^{6n+9}}{3^{4n+16}} = 3^{1+6n+9-4n-16} = 3^{2n-6}$$

4. Calculate value

Therefore,

$$x = 2n - 6$$

Final answer

$x = 2n - 6$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2024 | Q#: 23 | Marks: 3

TOPIC: **Algebraic Roots & Indices**

Source: Jun 2024 P2H

23 Simplify $\dfrac{30 \times 25^{2x+7}}{\sqrt{180} \times \left(\sqrt{5}\right)^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 23
Algebraic Roots & Indices

1. Simplify surd

$$\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$$

$$\frac{30}{\sqrt{180}} = \frac{30}{6\sqrt{5}} = \sqrt{5} = 5^{1/2}$$

$$25^{2x+7} = (5^2)^{2x+7} = 5^{4x+14}$$

$$(\sqrt{5})^{4x+9} = 5^{2x+\frac{9}{2}}$$

2. Evaluate fraction

So

$$5^{1/2} \times 5^{4x+14} \div 5^{2x+\frac{9}{2}} = 5^{\frac{1}{2}+4x+14-2x-\frac{9}{2}}$$

$$= 5^{2x+10}$$

Final answer
5^{2x+10}, so $w = 2x + 10$

CLASSIFIED PROBLEM

Paper: P1HR | Session: May 2023 | Q#: 24 | Marks: 5

TOPIC: **Algebraic Roots & Indices**

Source: May 2023 P1HR

24 Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n
State any restrictions on n

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 24
Algebraic Roots & Indices

1. Calculate value

$$2^n = 2^{x^2} \times 16^x \times 8$$

$$16^x = (2^4)^x = 2^{4x}$$

$$8 = 2^3$$

2. Calculate value

So

$$2^n = 2^{x^2+4x+3}$$

$$n = x^2 + 4x + 3$$

3. Complete the square

Complete the square:

$$n = (x+2)^2 - 1$$

$$(x+2)^2 = n + 1$$

4. Simplify surd

Since $x > 0$, take the positive root:

$$x + 2 = \sqrt{n+1}$$

$$x = \sqrt{n+1} - 2$$

5. Simplify surd

For $x > 0$,

$$\sqrt{n+1} > 2$$

$$n > 3$$

Final answer
$x = \sqrt{n+1} - 2$, with $n > 3$

CLASSIFIED PROBLEM

Paper: P1HR | Session: MayNov 2020 | Q#: 21 | Marks: 5

TOPIC: **Algebraic Roots & Indices**

Source: May-Nov 2020 P1HR

21 Given that $M = \dfrac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | Question 21

Algebraic Roots & Indices

1. Evaluate fraction

$$M = \frac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$$

2. Use index laws

Write everything in powers of 2 and 3:

$$18^{4n} = (2 \cdot 3^2)^{4n} = 2^{4n}3^{8n}$$

$$2^{3(n^2-6n)} = 2^{3n^2-18n}$$

$$3^{2(1-4n)} = 3^{2-8n}$$

$$12^2 = (2^2 \cdot 3)^2 = 2^4 3^2$$

3. Calculate value

So

$$M = 2^{4n+3n^2-18n-4} \times 3^{8n+2-8n-2}$$

$$M = 2^{3n^2-14n-4}$$

4. Calculate value

Given $M = 2$,

$$2^{3n^2-14n-4} = 2^1$$

$$3n^2 - 14n - 4 = 1$$

$$3n^2 - 14n - 5 = 0$$

$$(3n+1)(n-5) = 0$$

Final answer

$n = 5$ or $n = -\frac{1}{3}$

Topic 5

Completing the Square

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: May2025 | Q#: 25 | Marks: 5

TOPIC: **Completing the Square**

Source: May 2025 4WM1H

25 (a) Express $8 - 12x - 4x^2$ in the form $a - b(x + c)^2$
where a, b and c are constants.

....

(3)

(b) Hence solve $8 - 12x - 4x^2 = 0$
Show your working clearly.
Give your solutions in surd form.

....

(2)

(Total for Question 25 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | **Question 25**

Completing the Square

1. Simplify surd

$$8x^2 - 12x - 4 = 8\left(x^2 - \frac{3}{2}x\right) - 4$$

2. Complete the square

Complete the square.

$$x^2 - \frac{3}{2}x = \left(x - \frac{3}{4}\right)^2 - \frac{9}{16}$$

3. Simplify surd

So

$$8x^2 - 12x - 4 = 8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$$

4. Simplify surd

Now solve

$$8x^2 - 12x - 4 = 0$$

$$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2} = 0$$

$$\left(x - \frac{3}{4}\right)^2 = \frac{17}{16}$$

$$x - \frac{3}{4} = \pm\frac{\sqrt{17}}{4}$$

$$x = \frac{3 \pm \sqrt{17}}{4}$$

Final answer

$8\left(x - \frac{3}{4}\right)^2 - \frac{17}{2}$, and $x = \frac{3\pm\sqrt{17}}{4}$.

CLASSIFIED PROBLEM

Paper: 4WM1H Session: Nov2025 Q#: 22 Marks: 2

TOPIC: **Completing the Square**

Source: Nov 2025 4WM1H

22 $9x^2 - 12x + q$ can be written in the form $(px - r)^2$ where p, q and r are integers.

Find the value of q

$q =$ ....

(Total for Question 22 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 22

Completing the Square

1. Calculate value

$$9x^2 - 12x + q = (3x - 2)^2$$

$$(3x - 2)^2 = 9x^2 - 12x + 4$$

Final answer

$q = 4.$

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 24 | Marks: 3

TOPIC: **Completing the Square**

Source: Nov 2025 4WM1H

24 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a, b and c are integers.

(Total for Question 24 is 3 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Nov 2025 4WM1H | Question 24
Completing the Square

1. Calculate value

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5((x-2)^2 - 4) + 23$$

$$= 5(x-2)^2 + 3$$

Final answer
$5(x-2)^2 + 3.$

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Specimen | Q#: 20 | Marks: 4

TOPIC: **Completing the Square**

Source: Specimen 4WM1H

20 Find the values of a, b and c so that

$$7 + 12x - 2x^2$$

is written as $a - b(x - c)^2$

$a =$ ....

$b =$ ....

$c =$ ....

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM1H | **Question 20**
Completing the Square

1. Factor out -2 from the quadratic terms

$$7 + 12x - 2x^2 = -2(x^2 - 6x) + 7$$

2. Complete the square

$$x^2 - 6x = (x-3)^2 - 9$$

3. Simplify the expression

$$-2((x-3)^2 - 9) + 7 = -2(x-3)^2 + 18 + 7$$

$$= 25 - 2(x-3)^2$$

4. Read the values

$$a = 25, \qquad b = 2, \qquad c = 3$$

Final answer
$a = 25,\ b = 2,\ c = 3.$

Topic 6

Algebraic Fractions

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: May2025 | Q#: 20 | Marks: 3

TOPIC: **Algebraic Fractions**

Source: May 2025 4WM1H

20 Given that $k = m + n$ and $m = \dfrac{3}{4n}$

express $\dfrac{7k}{5 - m}$ in the form $\dfrac{a + bn^2}{cn - d}$ where a, b and c are integers and d is prime.

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 20

Algebraic Fractions

1. Simplify fraction

$$k = m + n, \qquad m = \frac{3}{4n}$$

2. Simplify fraction

So

$$k = \frac{3}{4n} + n = \frac{3 + 4n^2}{4n}$$

3. Simplify fraction

Then

$$7k = \frac{21 + 28n^2}{4n}$$

4. Simplify fraction

and

$$5 - m = 5 - \frac{3}{4n} = \frac{20n - 3}{4n}$$

5. Simplify fraction

Therefore

$$\frac{7k}{5 - m} = \frac{\frac{21+28n^2}{4n}}{\frac{20n-3}{4n}}$$
$$= \frac{21 + 28n^2}{20n - 3}$$

Final answer

$\frac{21+28n^2}{20n-3}$

CLASSIFIED PROBLEM

Paper: 4WM1HR | Session: May2025 | Q#: 24 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: May 2025 4WM1HR

24 Write $(2x-1)+\dfrac{2x^2-5x-12}{x^2-16}\times\dfrac{x^2+4x}{6x^2+11x+3}$ as a single fraction in its simplest form.

Show clear algebraic working.

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 24
Algebraic Fractions

1. Factor the fractional part first

Factor the fractional part first:

$$\frac{2x^2-5x-12}{x^2-16} \times \frac{x^2+4x}{6x^2+11x+3}$$
$$= \frac{(2x+3)(x-4)}{(x-4)(x+4)} \times \frac{x(x+4)}{(3x+1)(2x+3)}$$
$$= \frac{x}{3x+1}$$

2. Evaluate fraction

So

$$(2x-1) + \frac{x}{3x+1} = \frac{(2x-1)(3x+1)+x}{3x+1}$$
$$= \frac{6x^2-1}{3x+1}$$

Final answer
$\frac{6x^2-1}{3x+1}$.

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 25 | Marks: 3

TOPIC: **Algebraic Fractions**

Source: Nov 2025 4WM1H

25 $a = \dfrac{2x+5}{1-x} \qquad x = \dfrac{5-2y}{3y}$

Write a in the form $\dfrac{m+ny}{p(y-1)}$ where m, n and p are integers.

Show your working clearly.

$a =$ ....

(Total for Question 25 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 25
Algebraic Fractions

1. Split the ratio

$$a = \frac{2x+5}{1-x}, \quad x = \frac{5-2y}{3y}$$

2. Split the ratio

Substitute for x.

3. Numerator

Numerator:

$$2x + 5 = 2\left(\frac{5-2y}{3y}\right) + 5$$

$$= \frac{10 - 4y + 15y}{3y} = \frac{10+11y}{3y}$$

4. Denominator

Denominator:

$$1 - x = 1 - \frac{5-2y}{3y}$$

$$= \frac{3y - 5 + 2y}{3y} = \frac{5y-5}{3y} = \frac{5(y-1)}{3y}$$

5. Split the ratio

Therefore

$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10+11y}{5(y-1)}$$

Final answer

$$\frac{10+11y}{5(y-1)}$$

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 28 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Nov 2025 4WM1H

28 Solve $\dfrac{6x^3 + 5x^2 + x}{4x^2 - 1} \div \dfrac{15x^2 - x - 2}{8x - 4} = 2x$

Show clear algebraic working.

$x =$ ....

(Total for Question 28 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 28

Algebraic Fractions

1. Factor each expression

Factor each expression:

$$\frac{x(3x+1)(2x+1)}{(2x-1)(2x+1)} \div \frac{(5x-2)(3x+1)}{4(2x-1)} = 2x$$

$$\frac{x(3x+1)}{2x-1} \times \frac{4(2x-1)}{(5x-2)(3x+1)} = 2x$$

$$\frac{4x}{5x-2} = 2x$$

$$4x = 2x(5x-2)$$

$$10x^2 - 8x = 0$$

$$2x(5x-4) = 0$$

2. The original fractions do not

The original fractions do not allow $x = 0$, so

$$x = \frac{4}{5}$$

Final answer

$x = \frac{4}{5}$.

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Specimen | Q#: 21 | Marks: 3

TOPIC: **Algebraic Fractions**

Source: Specimen 4WM1H

21 Express $\left(\dfrac{20}{x^2-36}-\dfrac{2}{x-6}\right)\times\dfrac{1}{4-x}$ as a single fraction in its simplest form.

Show clear algebraic working.

....

WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM1H | Question 21
Algebraic Fractions

1. Factorise the denominator

$$x^2 - 36 = (x-6)(x+6)$$

2. Simplify the bracket

$$\frac{20}{(x-6)(x+6)} - \frac{2}{x-6} = \frac{20-2(x+6)}{(x-6)(x+6)}$$

$$= \frac{8-2x}{(x-6)(x+6)} = \frac{-2(x-4)}{(x-6)(x+6)}$$

3. Cancel the common factor

Since $4 - x = -(x-4)$,

$$\frac{-2(x-4)}{(x-6)(x+6)} \times \frac{1}{4-x} = \frac{2}{(x-6)(x+6)}$$

Final answer
$\frac{2}{x^2-36}$.

Topic 7

Completing the Square

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2020 | Q#: 23 | Marks: 3

TOPIC: **Completing the Square**

Source: Jan 2020 P2HR

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 23

Completing the Square

1. Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

$$= -2(x^2 + 6x) + 7$$

$$= -2\left((x+3)^2 - 9\right) + 7$$

$$= -2(x+3)^2 + 18 + 7$$

$$= 25 - 2(x+3)^2$$

Final answer

$25 - 2(x+3)^2$

CLASSIFIED PROBLEM

Paper: P1H | Session: Jan 2022 | Q#: 20 | Marks: 4

TOPIC: **Completing the Square**

Source: Jan 2022 P1H

20 (a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

....

(3)

C is the curve with equation $y = 7 + 12x - 3x^2$

The point A is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of A

(..................... , ...)

(1)

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1H | Question 20

Completing the Square

1. (a)

(a)

$$7 + 12x - 3x^2 = -3x^2 + 12x + 7$$

$$= -3(x^2 - 4x) + 7$$

$$= -3\left((x-2)^2 - 4\right) + 7$$

$$= 19 - 3(x-2)^2$$

2. This is the same as

This is the same as

$$19 - 3(x + (-2))^2$$

3. (b) The maximum point is

(b) The maximum point is read from the completed square form.

$$y = 19 - 3(x-2)^2$$

4. Calculate value

The maximum occurs when $x = 2$, giving $y = 19$.

Final answer

(a) $19 - 3(x-2)^2$, (b) $A = (2, 19)$

CLASSIFIED PROBLEM

Paper: P1H | Session: Jun 2022 | Q#: 24 | Marks: 4

TOPIC: **Completing the Square**

Source: Jun 2022 P1H

24 Express each of a, b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$a =$ ....

$b =$ ....

$c =$ ....

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1H | Question 24

Completing the Square

1. Evaluate fraction

$$q + 12x - qx^2 = -q\left(x^2 - \frac{12}{q}x\right) + q$$

2. Complete the square

Complete the square:

$$x^2 - \frac{12}{q}x = \left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}$$

3. Evaluate fraction

So

$$q + 12x - qx^2 = -q\left[\left(x - \frac{6}{q}\right)^2 - \frac{36}{q^2}\right] + q$$
$$= q + \frac{36}{q} - q\left(x - \frac{6}{q}\right)^2$$

4. Compare coefficients

Compare with

$$a - b(x - c)^2$$

Final answer

$a = q + \dfrac{36}{q}$, $b = q$, $c = \dfrac{6}{q}$

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jun 2022 | Q#: 23 | Marks: 5

TOPIC: **Completing the Square**

Source: Jun 2022 P1HR

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a, b and c are integers.

....

(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(.......................... , ...)

(2)

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 23

Completing the Square

1. (a)

(a)

$$2x^2 - 12x + 3 = 2(x^2 - 6x) + 3$$

$$= 2\left((x-3)^2 - 9\right) + 3$$

$$= 2(x-3)^2 - 18 + 3$$

$$= 2(x-3)^2 - 15$$

2. (b)

(b)

$$y = 2(x+4)^2 - 12(x+4) + 3$$

3. Let

Let $u = x + 4$. From part (a),

$$2u^2 - 12u + 3 = 2(u-3)^2 - 15$$

4. Calculate value

So

$$y = 2((x+4) - 3)^2 - 15$$

$$y = 2(x+1)^2 - 15$$

5. The minimum point is when

The minimum point is when $x = -1$, giving $y = -15$.

Final answer

(a) $2(x-3)^2 - 15$, (b) $M = (-1, -15)$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2025 | Q#: 25 | Marks: 6

TOPIC: Completing the Square

Source: Jun 2025 P2H

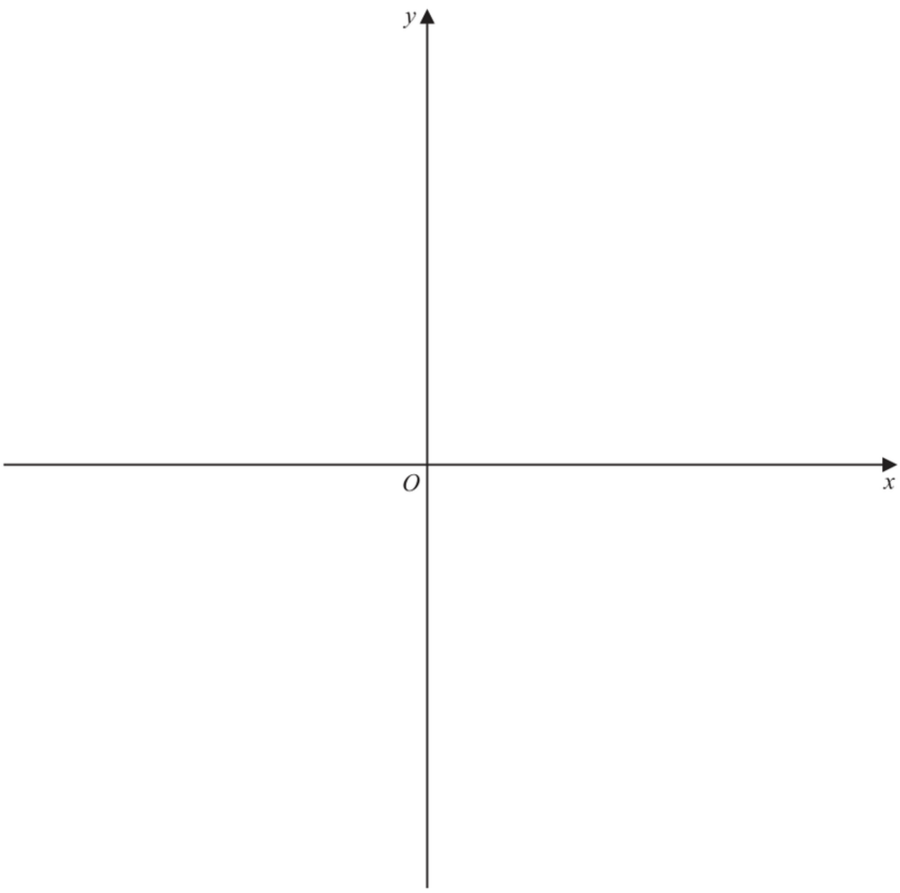
25 (a) Write $28+24x-6x^2$ in the form $a-b(x-c)^2$ where a, b and c are integers.

....

(3)

(b) On the axes below, sketch the graph of $y=28+24x-6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y-axis.



(3)

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 25

Completing the Square

1. (a)

(a)

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

$$= -6(x^2 - 4x) + 28$$

$$= -6\left((x-2)^2 - 4\right) + 28$$

$$= 52 - 6(x-2)^2$$

2. (b)

(b)

$$y = 52 - 6(x-2)^2$$

3. The graph is a downward

The graph is a downward-opening parabola.

4. The turning point is

The turning point is

$$(2, 52)$$

5. Calculate value

The point of intersection with the y-axis is found by putting $x = 0$:

$$y = 28$$

6. Find intercepts

So the y-intercept is

$$(0, 28)$$

Final answer

(a) $52 - 6(x-2)^2$. For the sketch, show a downward parabola with turning point $(2, 52)$ and y-intercept $(0, 28)$.

CLASSIFIED PROBLEM

Paper: P2H | Session: May 2021 | Q#: 22 | Marks: 4

TOPIC: **Completing the Square**

Source: May 2021 P2H

22 The curve **S** has equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = x^2$
The curve **T** has equation $y = \mathrm{g}(x)$ where $\mathrm{g}(x) = 2x^2 - 12x + 13$

By writing $\mathrm{g}(x)$ in the form $a(x - b)^2 - c$, where a, b and c are constants, describe fully a series of transformations that map the curve **S** onto the curve **T**.

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P2H | Question 22
Completing the Square

1. Solve inequality

$$g(x) = 2x^2 - 12x + 13$$

$$= 2(x^2 - 6x) + 13$$

$$= 2\left((x-3)^2 - 9\right) + 13$$

$$= 2(x-3)^2 - 18 + 13$$

$$= 2(x-3)^2 - 5$$

2. Calculate value

Starting with $S : y = x^2$:

3. Translate 3 units to the

- Translate 3 units to the right to get $y = (x-3)^2$.
- Stretch parallel to the y-axis by scale factor 2 to get $y = 2(x-3)^2$.
- Translate 5 units down to get $y = 2(x-3)^2 - 5$.

Final answer
$g(x) = 2(x-3)^2 - 5$. Transformations: right 3, stretch parallel to the y-axis by scale factor 2, then down 5.

CLASSIFIED PROBLEM

Paper: P1H | Session: May 2024 | Q#: 25 | Marks: 4

TOPIC: **Completing the Square**

Source: May 2024 P1H

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a, b and c are constants.

$f(x) =$ ....

(Total for Question 25 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 25

Completing the Square

1. Solve inequality

$$f(x) = 17 - 3x^2 + 12x$$

$$= -3x^2 + 12x + 17$$

$$= -3(x^2 - 4x) + 17$$

$$= -3\left((x-2)^2 - 4\right) + 17$$

$$= -3(x-2)^2 + 12 + 17$$

$$= 29 - 3(x-2)^2$$

Final answer

$f(x) = 29 - 3(x-2)^2$

CLASSIFIED PROBLEM

Paper: P1HR **Session: May 2024** **Q#: 20** **Marks: 5**

TOPIC: **Completing the Square**

Source: May 2024 P1HR

20 (a) Express $2x^2 - 11x + 9$ in the form $a(x - b)^2 - c$ where a, b and c are numbers to be found.

....

(3)

The curve **C** has equation $y = 2(x - 3)^2 - 11(x - 3) + 9$

The point P is the minimum point on **C**

(b) Find the coordinates of P

(.................... , ...)

(2)

(Total for Question 20 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1HR | Question 20

Completing the Square

1. Use Moved Coordinate Geometry Completing

For part (a),

$$2x^2 - 11x + 9$$

2. Factor out 2 from the

Factor out 2 from the quadratic terms:

$$2\left(x^2 - \frac{11}{2}x\right) + 9$$

3. Complete the square

Complete the square:

$$2\left[\left(x - \frac{11}{4}\right)^2 - \frac{121}{16}\right] + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + 9$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{121}{8} + \frac{72}{8}$$

$$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

4. Evaluate fraction

So

$$2x^2 - 11x + 9 = 2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$$

5. Calculate value

For part (b),

$$y = 2(x-3)^2 - 11(x-3) + 9$$

6. Let

Let

$$u = x - 3$$

7. Evaluate fraction

From part (a), the minimum occurs when

$$u = \frac{11}{4}$$

8. Evaluate fraction

So

$$x - 3 = \frac{11}{4}$$

$$x = \frac{23}{4}$$

9. The minimum value of is

The minimum value of y is

$$-\frac{49}{8}$$

Final answer

$2\left(x - \frac{11}{4}\right)^2 - \frac{49}{8}$, $P = \left(\frac{23}{4}, -\frac{49}{8}\right)$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2025 | Q#: 21 | Marks: 3

TOPIC: **Completing the Square**

Source: Nov 2025 P1H

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a, b and c are integers.

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 21

Completing the Square

1. Solve inequality

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5\left((x-2)^2 - 4\right) + 23$$

$$= 5(x-2)^2 - 20 + 23$$

$$= 5(x-2)^2 + 3$$

Final answer

$5(x-2)^2 + 3$

Topic 8

Algebraic Fractions

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2020 | Q#: 24 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jan 2020 P2H

24 Express

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

as a single fraction in its simplest form.

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 24
Algebraic Fractions

1. Use trigonometry

$$\left(\frac{4}{2x-5}-\frac{3}{2x-3}\right)\div\frac{9x-4x^3}{6x^2-17x+5}$$

2. Use trigonometry

First simplify the bracket:

$$\frac{4}{2x-5}-\frac{3}{2x-3}=\frac{4(2x-3)-3(2x-5)}{(2x-5)(2x-3)}$$

$$=\frac{8x-12-6x+15}{(2x-5)(2x-3)}$$

$$=\frac{2x+3}{(2x-5)(2x-3)}$$

3. Use trigonometry

Now factor the second fraction:

$$9x-4x^3=-x(2x-3)(2x+3)$$

$$6x^2-17x+5=(2x-5)(3x-1)$$

4. Dividing by a fraction means

Dividing by a fraction means multiplying by its reciprocal:

$$\frac{2x+3}{(2x-5)(2x-3)}\times\frac{(2x-5)(3x-1)}{-x(2x-3)(2x+3)}$$

5. Cancel common factors

Cancel common factors:

$$=-\frac{3x-1}{x(2x-3)^2}$$

$$=\frac{1-3x}{x(2x-3)^2}$$

Final answer

$$\boxed{\frac{1-3x}{x(2x-3)^2}}$$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2021 | Q#: 21 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jan 2021 P2H

21 Write $\dfrac{25x^2 - 64}{5x^2 - 13x - 6} \times \dfrac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

(Total for Question 21 is 4 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jan 2021 P2H | Question 21
Algebraic Fractions

1. Use trigonometry

$$\frac{25x^2-64}{5x^2-13x-6} \times \frac{x^2-8x+15}{5x+8} - (x-7)$$

2. Factor

Factor:

$$25x^2-64=(5x-8)(5x+8)$$

$$5x^2-13x-6=(5x+2)(x-3)$$

$$x^2-8x+15=(x-3)(x-5)$$

3. Use trigonometry

So

$$\frac{(5x-8)(5x+8)}{(5x+2)(x-3)} \times \frac{(x-3)(x-5)}{5x+8} - (x-7)$$

4. Cancel common factors

Cancel common factors:

$$\frac{(5x-8)(x-5)}{5x+2} - (x-7)$$

5. Use the common denominator

Use the common denominator $5x+2$:

$$\frac{(5x-8)(x-5)-(x-7)(5x+2)}{5x+2}$$

6. Expand

Expand:

$$(5x-8)(x-5)=5x^2-33x+40$$

$$(x-7)(5x+2)=5x^2-33x-14$$

7. Use trigonometry

So the numerator is

$$5x^2-33x+40-(5x^2-33x-14)=54$$

Final answer

$$\boxed{\frac{54}{5x+2}}$$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2022 | Q#: 23 | Marks: 3

TOPIC: **Algebraic Fractions**

Source: Jun 2022 P2HR

23 Express $\left(\frac{20}{x^2 - 36} - \frac{2}{x - 6}\right) \times \frac{1}{4 - x}$ as a single fraction in its simplest form.

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 23

Algebraic Fractions

1. Use trigonometry

$$\left(\frac{20}{x^2-36}-\frac{2}{x-6}\right)\times\frac{1}{4-x}$$

2. Factor

Factor:

$$x^2-36=(x-6)(x+6)$$

3. Use trigonometry

So

$$\frac{20}{(x-6)(x+6)}-\frac{2}{x-6}$$

4. Use the common denominator

Use the common denominator $(x-6)(x+6)$:

$$\frac{20-2(x+6)}{(x-6)(x+6)}$$

$$=\frac{20-2x-12}{(x-6)(x+6)}$$

$$=\frac{8-2x}{(x-6)(x+6)}$$

$$=\frac{-2(x-4)}{(x-6)(x+6)}$$

5. Use trigonometry

Now multiply by

$$\frac{1}{4-x}=-\frac{1}{x-4}$$

$$\frac{-2(x-4)}{(x-6)(x+6)}\times\left(-\frac{1}{x-4}\right)$$

6. Cancel

Cancel $x-4$:

$$\frac{2}{(x-6)(x+6)}$$

$$=\frac{2}{x^2-36}$$

Final answer

$$\boxed{\frac{2}{x^2 - 36}}$$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2024 | Q#: 20 | Marks: 3

TOPIC: **Algebraic Fractions**

Source: Jun 2024 P2H

20 Given that $k = x - y$ and $x = \dfrac{1}{4y}$

express $\dfrac{5k}{x+2}$ in the form $\dfrac{a - by^2}{c + dy}$ where a, b, c and d are integers.

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 20
Algebraic Fractions

1. Simplify fraction

$$k = x - y, \quad x = \frac{1}{4y}$$

2. Simplify fraction

So

$$k = \frac{1}{4y} - y = \frac{1 - 4y^2}{4y}$$

3. Simplify fraction

Also

$$x + 2 = \frac{1}{4y} + 2 = \frac{1 + 8y}{4y}$$

4. Simplify fraction

Therefore

$$\frac{5k}{x+2} = \frac{5\left(\frac{1-4y^2}{4y}\right)}{\frac{1+8y}{4y}}$$

5. Cancel

Cancel $4y$:

$$\frac{5(1-4y^2)}{1+8y}$$

$$= \frac{5 - 20y^2}{1 + 8y}$$

Final answer

$$\frac{5 - 20y^2}{1 + 8y}$$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2024 | Q#: 26 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jun 2024 P2HR

26 Write $4 - \left[(3x - 5) \div \dfrac{3x^2 + x - 10}{4x - 1}\right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2HR | **Question 26**
Algebraic Fractions

1. Use trigonometry

$$4 - \left[(3x-5) \div \frac{3x^2+x-10}{4x-1}\right]$$

2. Factor the quadratic

Factor the quadratic:

$$3x^2 + x - 10 = (3x-5)(x+2)$$

3. Use trigonometry

So the expression inside the square brackets is

$$(3x-5) \div \frac{(3x-5)(x+2)}{4x-1}$$

4. Dividing by a fraction means

Dividing by a fraction means multiplying by its reciprocal:

$$(3x-5) \times \frac{4x-1}{(3x-5)(x+2)}$$

5. Cancel

Cancel $3x-5$:

$$\frac{4x-1}{x+2}$$

6. Use trigonometry

Therefore the full expression is

$$4 - \frac{4x-1}{x+2}$$

7. Use the common denominator

Use the common denominator $x+2$:

$$\frac{4(x+2)-(4x-1)}{x+2}$$

$$= \frac{4x+8-4x+1}{x+2}$$

$$= \frac{9}{x+2}$$

Final answer

$$\boxed{\frac{9}{x+2}}$$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2025 | Q#: 23 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: Jun 2025 P2H

23 $\dfrac{4x^2 - 4x - 120}{5x^2 - 180} \div \dfrac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

$p =$ ....

(Total for Question 23 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 23
Algebraic Fractions

1. Split the ratio

$$\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x}$$

2. Factorise

Factorise:

$$4x^2 - 4x - 120 = 4(x-6)(x+5)$$

$$5x^2 - 180 = 5(x-6)(x+6)$$

$$x^2 + 5x = x(x+5)$$

$$10x^2 + 60x = 10x(x+6)$$

3. Split the ratio

So

$$\frac{4(x-6)(x+5)}{5(x-6)(x+6)} \div \frac{x(x+5)}{10x(x+6)}$$

$$= \frac{4(x+5)}{5(x+6)} \times \frac{10(x+6)}{x+5} = 8$$

Final answer

$p = 8$

CLASSIFIED PROBLEM

Paper: P1H | Session: May 2021 | Q#: 21 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: May 2021 P1H

21 Given that $x = \dfrac{5}{9y+5}$ and that $y = \dfrac{5}{5a-2}$

find an expression for x in terms of a.
Give your expression as a single fraction in its simplest form.

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 21
Algebraic Fractions

1. Given

Given

$$x = \frac{5}{9y+5}$$

2. Use trigonometry

and

$$y = \frac{5}{5a-2}$$

3. Use cosine rule

Substitute $y = \frac{5}{5a-2}$ into the expression for x:

$$x = \frac{5}{9\left(\frac{5}{5a-2}\right)+5}$$

$$x = \frac{5}{\frac{45}{5a-2}+5}$$

4. Write the denominator as a

Write the denominator as a single fraction:

$$\frac{45}{5a-2}+5 = \frac{45+5(5a-2)}{5a-2}$$

$$= \frac{45+25a-10}{5a-2}$$

$$= \frac{25a+35}{5a-2}$$

5. Use trigonometry

So

$$x = 5 \div \frac{25a+35}{5a-2}$$

$$x = 5 \times \frac{5a-2}{25a+35}$$

$$x = \frac{5(5a-2)}{5(5a+7)}$$

$$x = \frac{5a-2}{5a+7}$$

Final answer

$$\boxed{\frac{5a-2}{5a+7}}$$

CLASSIFIED PROBLEM

Paper: P1HR | Session: May 2023 | Q#: 23 | Marks: 4

TOPIC: **Algebraic Fractions**

Source: May 2023 P1HR

23 Simplify $(x^2 - 4) \div \left(\dfrac{4x^2 - 7x - 2}{x}\right) - 2x$

Give your answer in the form $\dfrac{ax^2}{bx + c}$ where a, b and c are integers.

(Total for Question 23 is 4 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

May 2023 P1HR | Question 23
Algebraic Fractions

1. Simplify surd

$$(x^2-4) \div \left(\frac{4x^2-7x-2}{x}\right) - 2x$$

2. Change division into multiplication by

Change division into multiplication by the reciprocal:

$$(x^2-4) \times \frac{x}{4x^2-7x-2} - 2x$$

3. Factorise

Factorise:

$$x^2-4=(x-2)(x+2)$$

$$4x^2-7x-2=(4x+1)(x-2)$$

4. Simplify surd

So

$$\frac{x(x-2)(x+2)}{(4x+1)(x-2)} - 2x = \frac{x(x+2)}{4x+1} - 2x$$

$$\frac{x(x+2)-2x(4x+1)}{4x+1} = \frac{x^2+2x-8x^2-2x}{4x+1} = \frac{-7x^2}{4x+1}$$

Final answer

$$\frac{-7x^2}{4x+1}$$

CLASSIFIED PROBLEM

Paper: P1H | Session: MayNov 2020 | Q#: 21 | Marks: 5

TOPIC: **Algebraic Fractions**

Source: May-Nov 2020 P1H

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 21
Algebraic Fractions

1. Use trigonometry

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

2. Factor

Factor:

$$9x^2 - 4 = (3x-2)(3x+2)$$

$$3x^2 - 13x - 10 = (3x+2)(x-5)$$

3. Use trigonometry

So the first product becomes

$$\frac{1}{3x-2} \times \frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$$

4. Cancel common factors

Cancel common factors:

$$\frac{1}{x-5}$$

5. Use trigonometry

Now subtract:

$$\frac{1}{x-5} - \frac{7}{x-1}$$

6. Use the common denominator

Use the common denominator $(x-5)(x-1)$:

$$\frac{x-1-7(x-5)}{(x-5)(x-1)}$$

$$= \frac{x-1-7x+35}{(x-5)(x-1)}$$

$$= \frac{34-6x}{(x-5)(x-1)}$$

Final answer

$$\boxed{\frac{34-6x}{(x-5)(x-1)}}$$

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2023 | Q#: 24 | Marks: 5

TOPIC: **Algebraic Fractions**

Source: Nov 2023 P2H

24 Solve $$\frac{45x^3 - 80x}{3x^2 + x - 4} \times \left(\frac{1}{3x-4} + \frac{1}{x}\right) = \frac{4(x+2)}{5x-8}$$

Show clear algebraic working.

$x =$ ....

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | **Question 24**

Algebraic Fractions

1. Solve quadratic equation

$$\frac{45x^3 - 80x}{3x^2 + x - 4}\left(\frac{1}{3x-4} + \frac{1}{x}\right) = \frac{4(x+2)}{5x-8}$$

2. Factorise

Factorise:

$$45x^3 - 80x = 5x(3x-4)(3x+4)$$

$$3x^2 + x - 4 = (3x+4)(x-1)$$

3. Solve quadratic equation

Also,

$$\frac{1}{3x-4} + \frac{1}{x} = \frac{x + 3x - 4}{x(3x-4)} = \frac{4(x-1)}{x(3x-4)}$$

4. Solve quadratic equation

So the left side becomes

$$\frac{5x(3x-4)(3x+4)}{(3x+4)(x-1)} \times \frac{4(x-1)}{x(3x-4)} = 20$$

5. Solve quadratic equation

Therefore,

$$20 = \frac{4(x+2)}{5x-8}$$

$$20(5x-8) = 4(x+2)$$

$$100x - 160 = 4x + 8$$

$$96x = 168$$

$$x = \frac{7}{4}$$

Final answer

$x = \frac{7}{4}$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2025 | Q#: 23 | Marks: 3

TOPIC: **Algebraic Fractions**

Source: Nov 2025 P1H

23 $a = \dfrac{2x+5}{1-x}$ $\quad x = \dfrac{5-2y}{3y}$

Write a in the form $\dfrac{m+ny}{p(y-1)}$ where m, n and p are integers.

Show your working clearly.

$a =$ ....

(Total for Question 23 is 3 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Nov 2025 P1H | Question 23
Algebraic Fractions

1. Split the ratio

$$a = \frac{2x+5}{1-x}, \quad x = \frac{5-2y}{3y}$$

2. Split the ratio
Substitute for x.

3. Numerator
Numerator:

$$2x + 5 = 2\left(\frac{5-2y}{3y}\right) + 5$$

$$= \frac{10 - 4y + 15y}{3y} = \frac{10 + 11y}{3y}$$

4. Denominator
Denominator:

$$1 - x = 1 - \frac{5-2y}{3y}$$

$$= \frac{3y - 5 + 2y}{3y} = \frac{5y - 5}{3y} = \frac{5(y-1)}{3y}$$

5. Split the ratio
Therefore

$$a = \frac{\frac{10+11y}{3y}}{\frac{5(y-1)}{3y}} = \frac{10 + 11y}{5(y-1)}$$

Final answer
$$\frac{10 + 11y}{5(y-1)}$$

Topic 9

Coordinate Geometry

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: May2025 | Q#: 24 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: May 2025 4WM1H

24 *PQRS* is a square.

PR is a diagonal of the square.

P is the point with coordinates (4, 7)
R is the point with coordinates (8, –5)

Find an equation of the straight line that passes through the points Q and S
Give your answer in the form $ay = bx + c$ where a, b and c are integers.

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 24

Coordinate Geometry

1. The diagonals of a square

The diagonals of a square bisect each other and are perpendicular.

2. Find the midpoint

Midpoint of $P(4, 7)$ and $R(8, -5)$:

$$\left(\frac{4+8}{2}, \frac{7-5}{2}\right) = (6, 1)$$

3. Find the gradient

Gradient of PR:

$$\frac{-5-7}{8-4} = -3$$

4. Find the gradient

So the perpendicular gradient is $\frac{1}{3}$.

5. The line through and passes

The line through Q and S passes through $(6, 1)$, so

$$y - 1 = \frac{1}{3}(x - 6)$$

$$3y - 3 = x - 6$$

$$3y = x - 3$$

Final answer

$3y = x - 3$

CLASSIFIED PROBLEM

Paper: 4WM1HR | Session: May2025 | Q#: 23 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: May 2025 4WM1HR

23 A circle C has centre $(3q, 4q)$ where q is an integer.

The straight line with equation $y = -\frac{5}{4}x + \frac{21}{4}$ is the tangent to C at the point $(p, p+3)$ where p is an integer.

Work out the radius of C
Give your answer in the form $\sqrt{a}$ where a is an integer.
Show your working clearly.

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | **Question 23**

Coordinate Geometry

1. The tangent point lies on

The tangent point lies on the tangent line, so

$$p + 3 = -\frac{5}{4}p + \frac{21}{4}$$

$$p = 1$$

2. Find the gradient

The tangent gradient is $-\frac{5}{4}$, so the radius gradient is $\frac{4}{5}$.

3. Use trigonometry

Using centre $(3q, 4q)$ and tangent point $(1, 4)$,

$$\frac{4 - 4q}{1 - 3q} = \frac{4}{5}$$

$$q = 2$$

4. Simplify surd

So the centre is $(6, 8)$. Therefore

$$r = \sqrt{(6-1)^2 + (8-4)^2} = \sqrt{41}$$

Final answer

$\sqrt{41}$.

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 27 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Nov 2025 4WM1H

27 $ABCD$ is a rhombus with $AB = BC = CD = DA$

A has coordinates $(16, -9)$

C has coordinates $(24, 15)$

Find an equation of the line BD

Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

Show your working clearly.

(Total for Question 27 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 27
Coordinate Geometry

1. Diagonals of a rhombus bisect

Diagonals of a rhombus bisect each other at right angles.

2. Find the midpoint

Midpoint of AC:

$$\left(\frac{16+24}{2}, \frac{-9+15}{2}\right) = (20, 3)$$

3. Find the gradient

Gradient of AC:

$$\frac{15-(-9)}{24-16} = 3$$

4. Find the gradient

So gradient of BD is $-\frac{1}{3}$.

$$y - 3 = -\frac{1}{3}(x - 20)$$

$$x + 3y - 29 = 0$$

Final answer
$x + 3y - 29 = 0$.

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Specimen | Q#: 23 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Specimen 4WM1H

23 *ABCD* is a kite.

$AB = AD$ and $CB = CD$

The point B has coordinates $(k, 1)$ where k is a negative constant.
The point D has coordinates $(8, 7)$

The straight line **L** passes through the points B and D and has a gradient of $\frac{3}{5}$

Find an equation of AC
Give your answer in the form $px + qy = r$ where p, q and r are integers.
Show your working clearly.

....

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Specimen 4WM1H | **Question 23**
Coordinate Geometry

1. Find k from the gradient

The gradient of BD is

$$\frac{7-1}{8-k} = \frac{3}{5}$$
$$\frac{6}{8-k} = \frac{3}{5}$$
$$30 = 24 - 3k$$
$$k = -2$$

2. Find the midpoint of BD

So $B = (-2, 1)$ and $D = (8, 7)$.

$$\text{midpoint of } BD = \left(\frac{-2+8}{2}, \frac{1+7}{2}\right) = (3, 4)$$

3. Use the perpendicular gradient

In this kite, AC is the perpendicular bisector of BD.
The gradient of BD is $\frac{3}{5}$, so the gradient of AC is

$$-\frac{5}{3}$$

4. Form the equation

Using point $(3, 4)$:

$$y - 4 = -\frac{5}{3}(x - 3)$$
$$3y - 12 = -5x + 15$$
$$5x + 3y = 27$$

Final answer
$5x + 3y = 27$.

CLASSIFIED PROBLEM

Paper: P1H | Session: Jan 2020 | Q#: 22 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jan 2020 P1H

22 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$
J is the point with coordinates $(j, 15)$ where $j < 0$
K is the point with coordinates $(6, k)$

M is the midpoint of JK.
The gradient of HM is 2

Find the value of j and the value of k.

$j =$ ....

$k =$ ....

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1H | Question 22
Coordinate Geometry

1. Find the midpoint

Since M is the midpoint of JK,

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2}\right)$$

2. Find the gradient

The gradient of HM is 2, and $H = (-4, 1)$.

$$\frac{\frac{15+k}{2} - 1}{\frac{j+6}{2} + 4} = 2$$

$$\frac{k+13}{j+14} = 2$$

$$k = 2j + 15$$

3. Simplify surd

Also,

$$JK = \sqrt{80}$$

4. Calculate value

So

$$(6-j)^2 + (k-15)^2 = 80$$

5. Substitute

Substitute $k = 2j + 15$:

$$(6-j)^2 + (2j)^2 = 80$$

$$j^2 - 12j + 36 + 4j^2 = 80$$

$$5j^2 - 12j - 44 = 0$$

$$(5j+10)(j-4.4) = 0$$

6. Solve inequality

Since $j < 0$,

$$j = -2$$

7. Calculate value

Then

$$k = 2(-2) + 15 = 11$$

Final answer
$j = -2,\ k = 11$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2020 | Q#: 22 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2020 P2H

22 The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B.

Find the coordinates of A and B.
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2H | Question 22

Coordinate Geometry

1. The line is

The line is

$$y = x + 2$$

2. Substitute this into the circle

Substitute this into the circle equation:

$$x^2 + y^2 - 2y = 24$$

$$x^2 + (x + 2)^2 - 2(x + 2) = 24$$

$$x^2 + x^2 + 4x + 4 - 2x - 4 = 24$$

$$2x^2 + 2x - 24 = 0$$

$$x^2 + x - 12 = 0$$

$$(x + 4)(x - 3) = 0$$

3. Calculate value

So

$$x = -4 \quad \text{or} \quad x = 3$$

4. Calculate value

Using $y = x + 2$:

$$x = -4 \Rightarrow y = -2$$

$$x = 3 \Rightarrow y = 5$$

Final answer

$(-4, -2)$ and $(3, 5)$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2020 | Q#: 24 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2020 P2HR

24 $\mathbf{L}_1$ and $\mathbf{L}_2$ are two straight lines.
The origin of the coordinate axes is O.

$\mathbf{L}_1$ has equation $5x + 10y = 8$
$\mathbf{L}_2$ is perpendicular to $\mathbf{L}_1$ and passes through the point with coordinates (8, 6)

$\mathbf{L}_2$ crosses the x-axis at the point A.
$\mathbf{L}_2$ intersects the straight line with equation $x = -3$ at the point B.

Find the area of triangle AOB.
Show your working clearly.

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P2HR | Question 24

Coordinate Geometry

1. Evaluate fraction

For L_1,

$$5x + 10y = 8$$

$$10y = -5x + 8$$

$$y = -\frac{1}{2}x + \frac{4}{5}$$

2. Find the gradient

So the gradient of L_1 is $-\frac{1}{2}$.

3. Find the gradient

Since L_2 is perpendicular to L_1, the gradient of L_2 is 2.

4. Calculate value

Using the point $(8, 6)$:

$$y - 6 = 2(x - 8)$$

$$y = 2x - 10$$

5. Calculate value

At A, the line crosses the x-axis, so $y = 0$:

$$0 = 2x - 10$$

$$x = 5$$

6. Calculate value

So $A = (5, 0)$.

7. Calculate value

At B, $x = -3$:

$$y = 2(-3) - 10 = -16$$

8. Calculate value

So $B = (-3, -16)$.

9. Triangle has base and perpendicular

Triangle AOB has base $OA = 5$ and perpendicular height 16.

$$\text{Area} = \frac{1}{2} \times 5 \times 16 = 40$$

Final answer

40

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jan 2021 | Q#: 22 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2021 P1HR

22 ABC is an isosceles triangle with $AB = AC$.

B is the point with coordinates $(-1, 5)$
C is the point with coordinates $(2, 10)$
M is the midpoint of BC.

Find an equation of the line through the points A and M.
Give your answer in the form $py + qx = r$ where p, q and r are integers.

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P1HR | Question 22
Coordinate Geometry

1. Find the gradient

Since $AB = AC$, point A lies on the perpendicular bisector of BC.

2. The midpoint of is

The midpoint of BC is

$$M = \left(\frac{-1+2}{2}, \frac{5+10}{2}\right)$$

$$M = \left(\frac{1}{2}, \frac{15}{2}\right)$$

3. The gradient of is

The gradient of BC is

$$\frac{10-5}{2-(-1)} = \frac{5}{3}$$

4. Find the gradient

So the gradient of the perpendicular bisector is

$$-\frac{3}{5}$$

5. Find line equation

Use the line through M:

$$y - \frac{15}{2} = -\frac{3}{5}\left(x - \frac{1}{2}\right)$$

6. Multiply by 10

Multiply by 10:

$$10y - 75 = -6x + 3$$

$$10y + 6x = 78$$

7. Divide by 2

Divide by 2:

$$5y + 3x = 39$$

Final answer
$5y + 3x = 39$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2022 | Q#: 22 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jan 2022 P2HR

22 ABC is a triangle in which angle $ABC = 90°$

p and q are integers such that

the coordinates of A are $(p, 10)$
the coordinates of B are $(-1, -5)$
the coordinates of C are $(8, q)$

Given that the gradient of AC is $-\dfrac{6}{7}$

work out the value of p and the value of q

$p =$ ....

$q =$ ....

(Total for Question 22 is 5 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jan 2022 P2HR | Question 22
Coordinate Geometry

1. The gradient of is

The gradient of AC is $-\frac{6}{7}$.

$$\frac{q-10}{8-p} = -\frac{6}{7}$$

$$7(q-10) = -6(8-p)$$

$$7q - 70 = -48 + 6p$$

$$6p - 7q = -22$$

2. Find the gradient

Since $\angle ABC = 90°$, lines AB and BC are perpendicular.

$$\text{gradient of } AB = \frac{10-(-5)}{p-(-1)} = \frac{15}{p+1}$$

$$\text{gradient of } BC = \frac{q-(-5)}{8-(-1)} = \frac{q+5}{9}$$

3. The product of perpendicular gradients

The product of perpendicular gradients is -1:

$$\frac{15}{p+1} \times \frac{q+5}{9} = -1$$

$$15(q+5) = -9(p+1)$$

$$5(q+5) = -3(p+1)$$

$$3p + 5q = -28$$

4. Solve

Now solve

$$6p - 7q = -22$$

$$3p + 5q = -28$$

5. Double the second equation

Double the second equation:

$$6p + 10q = -56$$

6. Subtract

Subtract:

$$17q = -34$$

$$q = -2$$

7. Calculate value

Then

$$3p + 5(-2) = -28$$

$$3p = -18$$

$$p = -6$$

Final answer

$p = -6,\ q = -2$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2022 | Q#: 22 | Marks: 7

TOPIC: **Coordinate Geometry**

Source: Jun 2022 P2H

22 $ABCD$ is a kite, with diagonals AC and BD, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D.

(.......................... , ...)

(.......................... , ...)

(Total for Question 22 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2H | Question 22

Coordinate Geometry

1. The midpoint of is

The midpoint of BD is $M = (0, 2)$.

2. Find the gradient

Since $A = (-3, 4)$, the gradient of AM is

$$\frac{2-4}{0-(-3)} = -\frac{2}{3}$$

3. Find the gradient

For the kite, diagonal AC is perpendicular to BD, so the gradient of BD is

$$\frac{3}{2}$$

4. Use direction vector along

Use direction vector $(2, 3)$ along BD.

5. Let

Let

$$B = (2t, 2 + 3t)$$

6. Calculate value

and

$$D = (-2t, 2 - 3t)$$

7. Calculate value

Since $AB = 6.5$,

$$(2t - (-3))^2 + (2 + 3t - 4)^2 = 6.5^2$$

$$(2t + 3)^2 + (3t - 2)^2 = 42.25$$

$$4t^2 + 12t + 9 + 9t^2 - 12t + 4 = 42.25$$

$$13t^2 + 13 = 42.25$$

$$13t^2 = 29.25$$

$$t^2 = 2.25$$

$$t = 1.5$$

8. Use

The x-coordinate of B is positive, so use $t = 1.5$.

$$B = (3, 6.5)$$

$$D = (-3, -2.5)$$

Final answer

$B = (3, 6.5),\ D = (-3, -2.5)$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2022 | Q#: 20 | Marks: 4

TOPIC: **Coordinate Geometry**

Source: Jun 2022 P2HR

20 The centre O of a circle has coordinates (4, 7)

The point A, on the circle, has coordinates (6, 11) and AOP is a diameter of the circle.

Find an equation of the tangent to the circle at the point P

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2HR | Question 20

Coordinate Geometry

1. The centre of the circle

The centre of the circle is

$$O = (4, 7)$$

2. The point is

The point A is

$$A = (6, 11)$$

3. Find the midpoint

Since AOP is a diameter, O is the midpoint of AP.

4. Let

Let $P = (x, y)$.

$$\left(\frac{6+x}{2}, \frac{11+y}{2}\right) = (4, 7)$$

$$6 + x = 8, \qquad 11 + y = 14$$

$$P = (2, 3)$$

5. The gradient of is

The gradient of OP is

$$\frac{3-7}{2-4} = 2$$

6. Use circle theorem

The tangent is perpendicular to the radius, so its gradient is

$$-\frac{1}{2}$$

7. Evaluate fraction

Using point $P = (2, 3)$,

$$y - 3 = -\frac{1}{2}(x - 2)$$

$$y = -\frac{1}{2}x + 4$$

Final answer

$y = -\frac{1}{2}x + 4$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2023 | Q#: 24 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jun 2023 P2H

24 *ABCD* is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\dfrac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(.......................... , ...)

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 24

Coordinate Geometry

1. Calculate value

In the kite, $AB = AD$ and $CB = CD$, so AC is the line of symmetry.

2. Apply transformation

Therefore D is the reflection of B in the line AC.

$$A = (-2, 10), \qquad B = \left(-\frac{27}{5}, 4\right), \qquad C = (4, -5)$$

3. A direction vector for is

A direction vector for AC is

$$\begin{pmatrix} 4-(-2) \\ -5-10 \end{pmatrix} = \begin{pmatrix} 6 \\ -15 \end{pmatrix}$$

4. Use the simpler direction vector

Use the simpler direction vector

$$\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

5. Find the gradient

Let H be the foot of the perpendicular from B to AC.

$$\vec{AB} = \begin{pmatrix} -\frac{17}{5} \\ -6 \end{pmatrix}$$

6. The fraction along to is

The fraction along AC to H is

$$\frac{\vec{AB} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix}}{2^2 + (-5)^2} = \frac{-\frac{34}{5} + 30}{29}$$

$$= \frac{\frac{116}{5}}{29} = \frac{4}{5}$$

7. Use matrices

So

$$H = A + \frac{4}{5}\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

$$H = \left(-2 + \frac{8}{5}, 10 - 4\right)$$

$$H = \left(-\frac{2}{5}, 6\right)$$

8. Find the midpoint

Since H is the midpoint of BD,

$$D = 2H - B$$

$$D = \left(-\frac{4}{5}, 12\right) - \left(-\frac{27}{5}, 4\right)$$

$$D = \left(\frac{23}{5}, 8\right)$$

Final answer

$D = \left(\frac{23}{5}, 8\right)$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2023 | Q#: 21 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jun 2023 P2HR

21 Work out the coordinates of the points of intersection of

$$y - 2x = 1 \quad \text{and} \quad y^2 + xy = 7$$

Show clear algebraic working.

(.......................... , ...)

(.......................... , ...)

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 21

Coordinate Geometry

1. The line is

The line is

$$y - 2x = 1$$

2. Calculate value

So

$$y = 2x + 1$$

3. Calculate value

Substitute into

$$y^2 + xy = 7$$

$$(2x + 1)^2 + x(2x + 1) = 7$$

$$4x^2 + 4x + 1 + 2x^2 + x = 7$$

$$6x^2 + 5x - 6 = 0$$

4. Factorise

Factorise:

$$(3x - 2)(2x + 3) = 0$$

5. Evaluate fraction

So

$$x = \frac{2}{3} \quad \text{or} \quad x = -\frac{3}{2}$$

6. Evaluate fraction

Using $y = 2x + 1$:

$$x = \frac{2}{3} \Rightarrow y = \frac{7}{3}$$

$$x = -\frac{3}{2} \Rightarrow y = -2$$

Final answer

$\left(\frac{2}{3}, \frac{7}{3}\right)$ and $\left(-\frac{3}{2}, -2\right)$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2023 | Q#: 25 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jun 2023 P2HR

25 The straight line with equation $y - 2x = 7$ is the perpendicular bisector of the line AB where A is the point with coordinates $(j, 7)$ and B is the point with coordinates $(6, k)$

Find the coordinates of the midpoint of the line AB
Show clear algebraic working.

(.......................... , ...)

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | **Question 25**

Coordinate Geometry

1. The perpendicular bisector is

The perpendicular bisector is

$$y - 2x = 7$$

2. Calculate value

so

$$y = 2x + 7$$

3. The gradient of the perpendicular

The gradient of the perpendicular bisector is 2, so the gradient of AB is

$$-\frac{1}{2}$$

4. Evaluate fraction

Since $A = (j, 7)$ and $B = (6, k)$,

$$\frac{k-7}{6-j} = -\frac{1}{2}$$

$$2(k-7) = -(6-j)$$

$$j = 2k - 8$$

5. The midpoint of is

The midpoint of AB is

$$\left(\frac{j+6}{2}, \frac{7+k}{2}\right)$$

6. Substitute

Substitute $j = 2k - 8$:

$$\left(k-1, \frac{k+7}{2}\right)$$

7. Find the midpoint

The midpoint lies on $y = 2x + 7$, so

$$\frac{k+7}{2} = 2(k-1) + 7$$

$$k + 7 = 4k + 10$$

$$k = -1$$

8. Calculate value

Then

$$j = 2(-1) - 8 = -10$$

9. Find the midpoint

Therefore the midpoint is

$$\left(\frac{-10+6}{2}, \frac{7+(-1)}{2}\right) = (-2, 3)$$

Final answer

$\boxed{(-2, 3)}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2024 | Q#: 22 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jun 2024 P2H

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.................... , ...) (.................... , ...)

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 22
Coordinate Geometry

1. The line is

The line is

$$x + y = 5$$

2. Calculate value

So

$$y = 5 - x$$

3. Substitute into the curve

Substitute into the curve:

$$2x^2 + 3y^2 = 210$$

$$2x^2 + 3(5 - x)^2 = 210$$

$$2x^2 + 3(x^2 - 10x + 25) = 210$$

$$5x^2 - 30x + 75 = 210$$

$$5x^2 - 30x - 135 = 0$$

4. Divide by 5

Divide by 5:

$$x^2 - 6x - 27 = 0$$

5. Factorise

Factorise:

$$(x - 9)(x + 3) = 0$$

6. Calculate value

So

$$x = 9 \quad \text{or} \quad x = -3$$

7. Calculate value

Using $y = 5 - x$:

$$x = 9 \Rightarrow y = -4$$

$$x = -3 \Rightarrow y = 8$$

Final answer

$(9, -4)$ and $(-3, 8)$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2024 | Q#: 21 | Marks: 4

TOPIC: **Coordinate Geometry**

Source: Jun 2024 P2HR

21 *ABCD* is a square.

The point A has coordinates $(-5, 2)$
The point B has coordinates $(3, 5)$

Find an equation of the line that passes through B and C
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(Total for Question 21 is 4 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jun 2024 P2HR | Question 21
Coordinate Geometry

1. Calculate value

$$A = (-5, 2), \qquad B = (3, 5)$$

2. The gradient of is

The gradient of AB is

$$\frac{5-2}{3-(-5)} = \frac{3}{8}$$

3. Find the gradient

Since $ABCD$ is a square, BC is perpendicular to AB.

4. Find the gradient

So the gradient of BC is

$$-\frac{8}{3}$$

5. The line through is

The line through $B = (3, 5)$ is

$$y - 5 = -\frac{8}{3}(x-3)$$

6. Multiply by 3

Multiply by 3:

$$3y - 15 = -8x + 24$$

$$8x + 3y - 39 = 0$$

Final answer
$8x + 3y - 39 = 0$

CLASSIFIED PROBLEM

Paper: P1H | Session: Jun 2025 | Q#: 25 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Jun 2025 P1H

25 $PQRS$ is a square.

PR is a diagonal of the square.

P is the point with coordinates (4, 7)
R is the point with coordinates (8, –5)

Find an equation of the straight line that passes through the points Q and S
Give your answer in the form $ay = bx + c$ where a, b and c are integers.

(Total for Question 25 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 25
Coordinate Geometry

1. Calculate value

$$P = (4, 7), \qquad R = (8, -5)$$

2. The gradient of diagonal is

The gradient of diagonal PR is

$$\frac{-5-7}{8-4} = -3$$

3. The other diagonal is perpendicular

The other diagonal QS is perpendicular to PR, so its gradient is

$$\frac{1}{3}$$

4. The diagonals of a square

The diagonals of a square bisect each other.

5. The midpoint of is

The midpoint of PR is

$$\left(\frac{4+8}{2}, \frac{7+(-5)}{2}\right) = (6, 1)$$

6. Find the gradient

So QS passes through $(6, 1)$ with gradient $\frac{1}{3}$.

$$y - 1 = \frac{1}{3}(x - 6)$$

$$y = \frac{1}{3}x - 1$$

7. Multiply by 3

Multiply by 3:

$$3y = x - 3$$

Final answer
$3y = x - 3$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2025 | Q#: 24 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Jun 2025 P2HR

24 PQ is a straight line drawn on a square grid, with a scale of 1 cm for 1 unit on each axis.

P has coordinates $(-5, a)$ and Q has coordinates $(7, 3a)$ where $a > 0$

The length of PQ is $4\sqrt{10}$ cm

Find an equation of the perpendicular bisector of PQ
Give your answer in the form $y = mx + c$ where m and c are integers.

(Total for Question 24 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | **Question 24**

Coordinate Geometry

1. Calculate value

$$P = (-5, a), \qquad Q = (7, 3a)$$

2. Simplify surd

The length of PQ is $4\sqrt{10}$, so

$$\sqrt{(7-(-5))^2 + (3a-a)^2} = 4\sqrt{10}$$

$$\sqrt{12^2 + (2a)^2} = 4\sqrt{10}$$

3. Square both sides

Square both sides:

$$144 + 4a^2 = 160$$

$$4a^2 = 16$$

$$a^2 = 4$$

4. Solve inequality

Since $a > 0$,

$$a = 2$$

5. Calculate value

So

$$P = (-5, 2), \qquad Q = (7, 6)$$

6. The midpoint of is

The midpoint of PQ is

$$\left(\frac{-5+7}{2}, \frac{2+6}{2}\right) = (1, 4)$$

7. The gradient of is

The gradient of PQ is

$$\frac{6-2}{7-(-5)} = \frac{4}{12} = \frac{1}{3}$$

8. Find the gradient

So the gradient of the perpendicular bisector is

$$-3$$

9. Calculate value

Using point $(1, 4)$:

$$y - 4 = -3(x - 1)$$

$$y = -3x + 7$$

Final answer

$\boxed{y = -3x + 7}$

CLASSIFIED PROBLEM

Paper: P1H | Session: MayNov 2020 | Q#: 22 | Marks: 4

TOPIC: **Coordinate Geometry**

Source: May-Nov 2020 P1H

22 *ABCD* is a rhombus.

The diagonals, AC and BD, intersect at the point M.
The coordinates of M are $(6, -11)$

The points A and C both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through B and D intersects the y-axis.

(.......................... , ...)

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1H | Question 22

Coordinate Geometry

1. Find the gradient

In a rhombus, the diagonals are perpendicular and bisect each other.

2. The diagonal lies on

The diagonal AC lies on

$$2y + 7x = 20$$

$$2y = -7x + 20$$

$$y = -\frac{7}{2}x + 10$$

3. Find the gradient

So the gradient of AC is

$$-\frac{7}{2}$$

4. Find the gradient

Therefore the gradient of BD is

$$\frac{2}{7}$$

5. Calculate value

The line BD passes through

$$M = (6, -11)$$

6. Evaluate fraction

So

$$y + 11 = \frac{2}{7}(x - 6)$$

7. To find where this line

To find where this line intersects the y-axis, put $x = 0$:

$$y + 11 = \frac{2}{7}(0 - 6)$$

$$y + 11 = -\frac{12}{7}$$

$$y = -11 - \frac{12}{7}$$

$$y = -\frac{89}{7}$$

Final answer

$\left(0, -\frac{89}{7}\right)$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2021 | Q#: 20 | Marks: 7

TOPIC: **Coordinate Geometry**

Source: Nov 2021 P1H

20 The straight line **L** passes through point A (– 6, 2) and point B (5, 3)
The straight line **M** is perpendicular to **L** and passes through the midpoint of A and B.
The line **M** intersects the line $x = -1$ at point C.

Calculate the area of triangle ABC.

(Total for Question 20 is 7 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2021 P1H | Question 20

Coordinate Geometry

1. Calculate value

$$A = (-6, 2), \qquad B = (5, 3)$$

2. The midpoint of is

The midpoint of AB is

$$\left(\frac{-6+5}{2}, \frac{2+3}{2}\right) = \left(-\frac{1}{2}, \frac{5}{2}\right)$$

3. The gradient of is

The gradient of AB is

$$\frac{3-2}{5-(-6)} = \frac{1}{11}$$

4. Find the gradient

So the perpendicular gradient is

$$-11$$

5. Evaluate fraction

Line M passes through $\left(-\frac{1}{2}, \frac{5}{2}\right)$, so

$$y - \frac{5}{2} = -11\left(x + \frac{1}{2}\right)$$

6. Evaluate fraction

At point C, $x = -1$:

$$y - \frac{5}{2} = -11\left(-\frac{1}{2}\right)$$

$$y - \frac{5}{2} = \frac{11}{2}$$

$$y = 8$$

7. Calculate value

So

$$C = (-1, 8)$$

8. Use vectors

Now use vectors from A:

$$\vec{AB} = (11, 1), \qquad \vec{AC} = (5, 6)$$

9. Area of triangle is

Area of triangle ABC is

$$\frac{1}{2}|11(6) - 1(5)|$$

$$= \frac{1}{2}(61)$$

$$= 30.5$$

Final answer

30.5

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2023 | Q#: 22 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Nov 2023 P1H

22 [In this question 1 cm = 1 unit on the x-axis and 1 cm = 1 unit on the y-axis]

P is a point on a circle with centre (0, 0)
The coordinates of P are (8, –10)

The line **L** is the tangent to the circle at the point P
L crosses the x-axis at the point Q and crosses the y-axis at the point R

Work out the length of RQ
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P1H | **Question 22**
Coordinate Geometry

1. The centre of the circle

The centre of the circle is $(0, 0)$, and

$$P = (8, -10)$$

2. The gradient of is

The gradient of OP is

$$\frac{-10}{8} = -\frac{5}{4}$$

3. Find the gradient

So the gradient of the tangent is

$$\frac{4}{5}$$

4. The tangent through is

The tangent through P is

$$y + 10 = \frac{4}{5}(x - 8)$$

$$y = \frac{4}{5}x - \frac{82}{5}$$

5. Evaluate fraction

At the x-axis, $y = 0$:

$$0 = \frac{4}{5}x - \frac{82}{5}$$

$$x = \frac{41}{2}$$

6. Evaluate fraction

So

$$Q = \left(\frac{41}{2}, 0\right)$$

7. Evaluate fraction

At the y-axis, $x = 0$:

$$R = \left(0, -\frac{82}{5}\right)$$

8. Simplify surd

Therefore

$$RQ = \sqrt{\left(\frac{41}{2}\right)^2 + \left(\frac{82}{5}\right)^2}$$

$$RQ = 26.252...$$

Final answer

26.3 cm

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2023 | Q#: 21 | Marks: 5

TOPIC: **Coordinate Geometry**

Source: Nov 2023 P2H

21 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | Question 21
Coordinate Geometry

1. The line is

The line is

$$x + 2y = 5$$

2. Calculate value

So

$$x = 5 - 2y$$

3. Calculate value

Substitute into

$$x^2 + 3y^2 = 13$$

$$(5 - 2y)^2 + 3y^2 = 13$$

$$25 - 20y + 4y^2 + 3y^2 = 13$$

$$7y^2 - 20y + 12 = 0$$

4. Factorise

Factorise:

$$(7y - 6)(y - 2) = 0$$

5. Evaluate fraction

So

$$y = \frac{6}{7} \quad \text{or} \quad y = 2$$

6. Evaluate fraction

Using $x = 5 - 2y$:

$$y = \frac{6}{7} \Rightarrow x = \frac{23}{7}$$

$$y = 2 \Rightarrow x = 1$$

Final answer

$\left(\frac{23}{7}, \frac{6}{7}\right)$ and $(1, 2)$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2025 | Q#: 20 | Marks: 3

TOPIC: **Coordinate Geometry**

Source: Nov 2025 P1H

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**
Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 20

Coordinate Geometry

1. Line has equation

Line L has equation

$$y = 4x + 7$$

2. Find the gradient

So its gradient is

$$4$$

3. Find the gradient

Line M is perpendicular to L, so its gradient is

$$-\frac{1}{4}$$

4. Evaluate fraction

Line M passes through $(8, 1)$.

$$y - 1 = -\frac{1}{4}(x - 8)$$

$$y - 1 = -\frac{1}{4}x + 2$$

$$y = -\frac{1}{4}x + 3$$

Final answer

$y = -\frac{1}{4}x + 3$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2025 | Q#: 28 | Marks: 6

TOPIC: **Coordinate Geometry**

Source: Nov 2025 P1H

28 The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are (0.2, 2.6)

Work out the coordinates of B
Show your working clearly.

(.............................. , ...)

(Total for Question 28 is 6 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Nov 2025 P1H | Question 28
Coordinate Geometry

1. The line is

The line is

$$y = 3x + e$$

2. Point lies on the line

Point $A = (0.2, 2.6)$ lies on the line.

$$2.6 = 3(0.2) + e$$

$$2.6 = 0.6 + e$$

$$e = 2$$

3. Calculate value

So

$$y = 3x + 2$$

4. The curve is

The curve is

$$x^2 + y^2 = d - 11x$$

5. Use point

Use point $A = (0.2, 2.6)$:

$$0.2^2 + 2.6^2 = d - 11(0.2)$$

$$0.04 + 6.76 = d - 2.2$$

$$d = 9$$

6. Substitute into the curve

Now substitute $y = 3x + 2$ into the curve:

$$x^2 + (3x + 2)^2 = 9 - 11x$$

$$x^2 + 9x^2 + 12x + 4 = 9 - 11x$$

$$10x^2 + 23x - 5 = 0$$

7. One root is

One root is $0.2 = \frac{1}{5}$.

8. The product of the roots

The product of the roots is

$$\frac{-5}{10} = -\frac{1}{2}$$

9. Evaluate fraction

So the other root is

$$\frac{-\frac{1}{2}}{\frac{1}{5}} = -\frac{5}{2}$$

10. Evaluate fraction

Then

$$y = 3\left(-\frac{5}{2}\right) + 2$$

$$y = -\frac{11}{2}$$

Final answer

$B = \left(-\frac{5}{2}, -\frac{11}{2}\right)$

Topic 10

Graphs of Functions

CLASSIFIED PROBLEM

Paper: P1H | Session: Jan 2020 | Q#: 20 | Marks: 4

TOPIC: **Graphs of Functions**

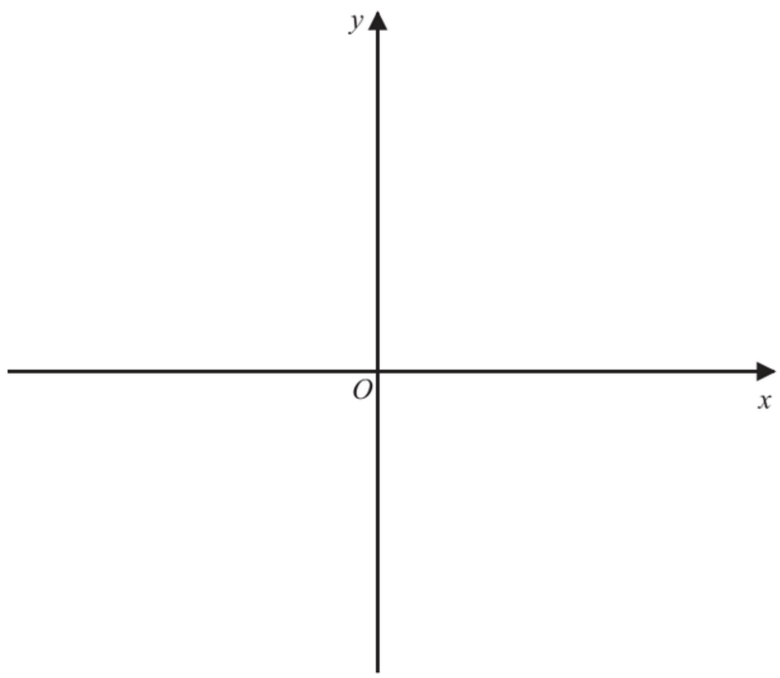
Source: Jan 2020 P1H

20 The curve **C** has equation $y = 4(x - 1)^2 - a$ where $a > 4$

Using the axes below, sketch the curve **C**.
On your sketch show clearly, in terms of a,

(i) the coordinates of any points of intersection of **C** with the coordinate axes,

(ii) the coordinates of the turning point.



(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1H | Question 20
Graphs of Functions

1. Calculate value

$$y = 4(x-1)^2 - a$$

2. The turning point is read

The turning point is read directly from the completed square form:

$$(1, -a)$$

3. Find intercepts

For the y-intercept, put $x = 0$:

$$y = 4(0-1)^2 - a$$

$$y = 4 - a$$

4. Find intercepts

So the y-intercept is

$$(0, 4-a)$$

5. Find intercepts

For the x-intercepts, put $y = 0$:

$$4(x-1)^2 - a = 0$$

$$4(x-1)^2 = a$$

$$(x-1)^2 = \frac{a}{4}$$

$$x - 1 = \pm\frac{\sqrt{a}}{2}$$

$$x = 1 \pm \frac{\sqrt{a}}{2}$$

6. Find intercepts

So the x-intercepts are

$$\left(1 - \frac{\sqrt{a}}{2}, 0\right) \quad \text{and} \quad \left(1 + \frac{\sqrt{a}}{2}, 0\right)$$

7. Solve inequality

Since $a > 4$, the turning point is below the x-axis and the curve crosses the x-axis twice.

Final answer

turning point $(1, -a)$, y-intercept $(0, 4 - a)$, x-intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

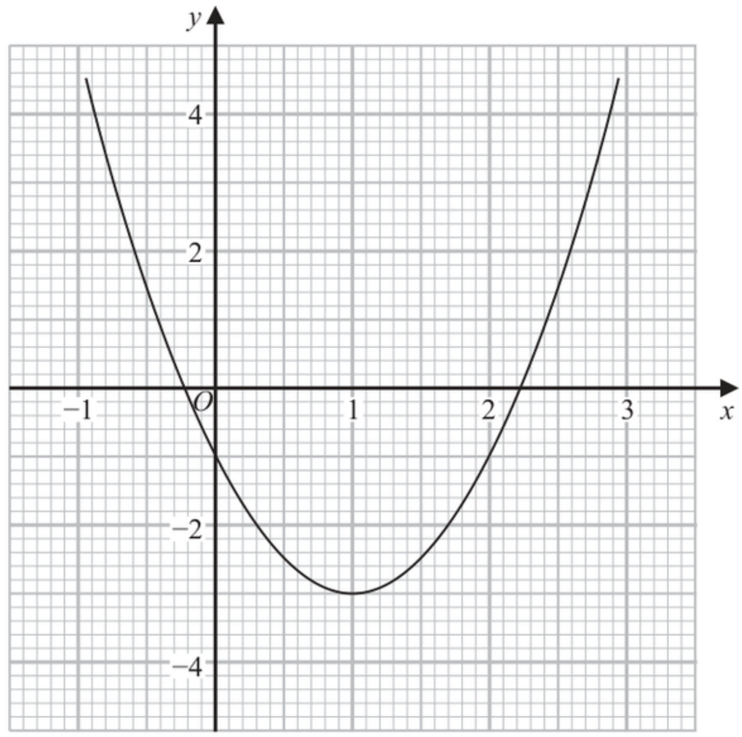
CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2022 | Q#: 21 | Marks: 5

TOPIC: **Graphs of Functions**

Source: Jan 2022 P2H

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



(a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

....

(2)

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

....

(3)

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 21

Graphs of Functions

1. Read the graph

For part (a), solve

$$2x^2 - 4x - 1 = 0$$

2. Read the graph

Using the graph gives approximately

$$x = -0.2, \quad x = 2.2$$

3. Read the graph

For part (b), we need to solve

$$x^2 - x - 1 = 0$$

4. Multiply by 2

Multiply by 2:

$$2x^2 - 2x - 2 = 0$$

5. The given graph is

The given graph is

$$y = 2x^2 - 4x - 1$$

6. Read the graph

Now

$$2x^2 - 2x - 2 = (2x^2 - 4x - 1) + 2x - 1$$

7. Read the graph

So draw the straight line

$$y = 1 - 2x$$

8. The intersections of this line

The intersections of this line with the curve give approximately

$$x = -0.6, \quad x = 1.6$$

Final answer

(a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

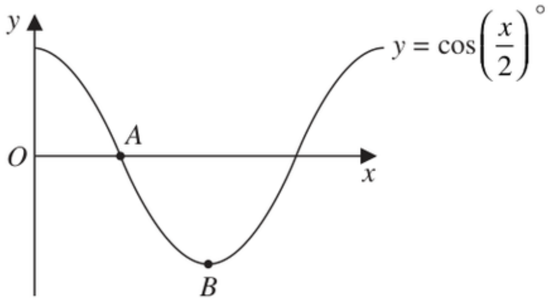
CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2022 | Q#: 23 | Marks: 2

TOPIC: **Graphs of Functions**

Source: Jun 2022 P2H

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^{\circ}$



(i) Find the coordinates of the point A

(........................ , ...)
(1)

(ii) Find the coordinates of the point B

(........................ , ...)
(1)

(Total for Question 23 is 2 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2H | Question 23

Graphs of Functions

1. The graph is

The graph is

$$y = \cos\left(\frac{x}{2}\right)^{\circ}$$

2. Point is the first positive

Point A is the first positive x-intercept, so $y = 0$.

$$\cos\left(\frac{x}{2}\right) = 0$$

$$\frac{x}{2} = 90^{\circ}$$

$$x = 180$$

3. Read the graph

So

$$A = (180, 0)$$

4. Point is the minimum point

Point B is the minimum point.

5. The minimum value of cosine

The minimum value of cosine is -1, which occurs when

$$\frac{x}{2} = 180^{\circ}$$

$$x = 360$$

6. Read the graph

So

$$B = (360, -1)$$

Final answer

$A = (180, 0),\ B = (360, -1)$

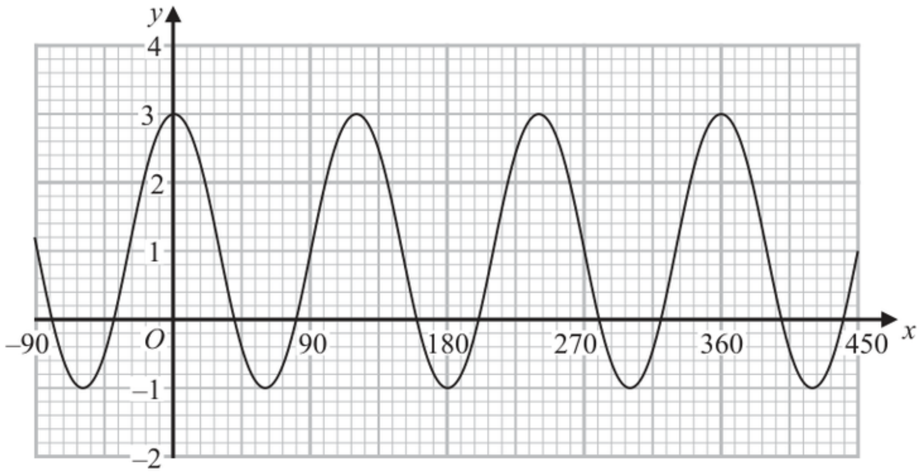
CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2023 | Q#: 26 | Marks: 3

TOPIC: **Graphs of Functions**

Source: Jun 2023 P2HR

26 Here is a sketch of the curve with equation $y = a\cos bx° + c$ where $-90 \leqslant x \leqslant 450$



Find the value of a, the value of b and the value of c

$a =$ ....

$b =$ ....

$c =$ ....

(Total for Question 26 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 26

Graphs of Functions

1. The curve is

The curve is

$$y = a\cos bx^\circ + c$$

2. Use trigonometry

From the graph, the maximum value is

$$3$$

3. Use trigonometry

and the minimum value is

$$-1$$

4. The amplitude is half the

The amplitude is half the difference:

$$a = \frac{3-(-1)}{2} = 2$$

5. The midline is the average

The midline is the average:

$$c = \frac{3+(-1)}{2} = 1$$

6. The distance between consecutive maxima

The distance between consecutive maxima is 120°, so the period is

$$120^\circ$$

7. Use trigonometry

For $y = a\cos bx^\circ + c$, the period is

$$\frac{360^\circ}{b}$$

8. Use trigonometry

So

$$\frac{360}{b} = 120$$

$$b = 3$$

Final answer

$\boxed{a = 2,\ b = 3,\ c = 1}$

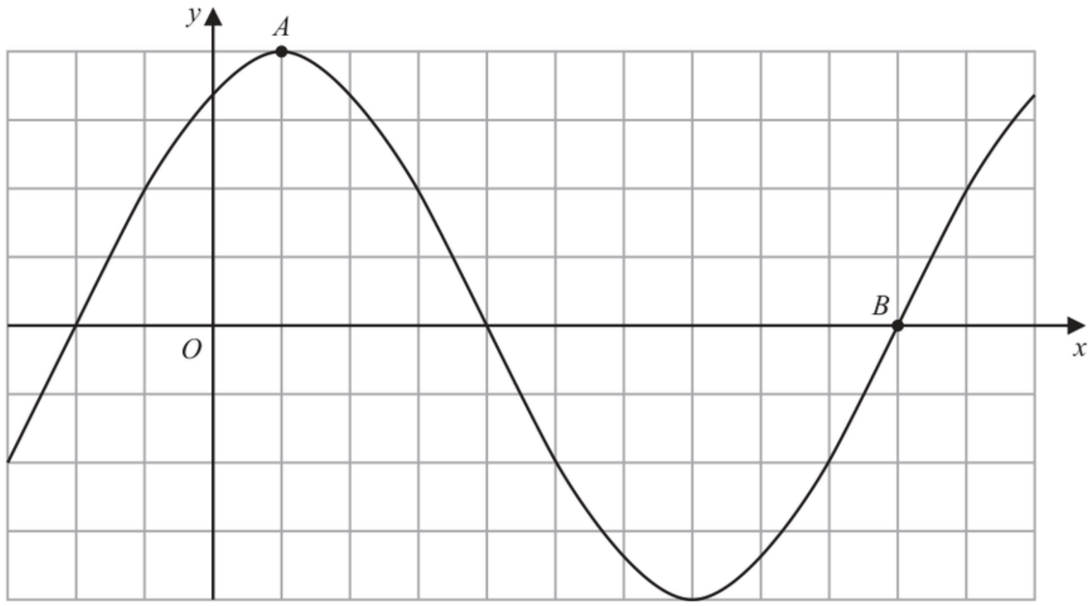
CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2024 | Q#: 25 | Marks: 2

TOPIC: **Graphs of Functions**

Source: Jun 2024 P2H

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)°$



(i) Find the coordinates of the point A

(.......................... , ...)
(1)

(ii) Find the coordinates of the point B

(.......................... , ...)
(1)

(Total for Question 25 is 2 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jun 2024 P2H | Question 25
Graphs of Functions

1. Use trigonometry

$$y = 2\sin(x + 60)^\circ$$

2. Point is the maximum point
Point A is the maximum point.

3. The maximum value is
The maximum value is

$$y = 2$$

4. This happens when
This happens when

$$x + 60 = 90$$

$$x = 30$$

5. Read the graph
So

$$A = (30, 2)$$

6. Point is the positive intercept
Point B is the positive x-intercept after the minimum.

$$2\sin(x + 60)^\circ = 0$$

$$\sin(x + 60)^\circ = 0$$

7. Read the graph
For this point,

$$x + 60 = 360$$

$$x = 300$$

8. Read the graph
So

$$B = (300, 0)$$

Final answer
$A = (30, 2), \ B = (300, 0)$

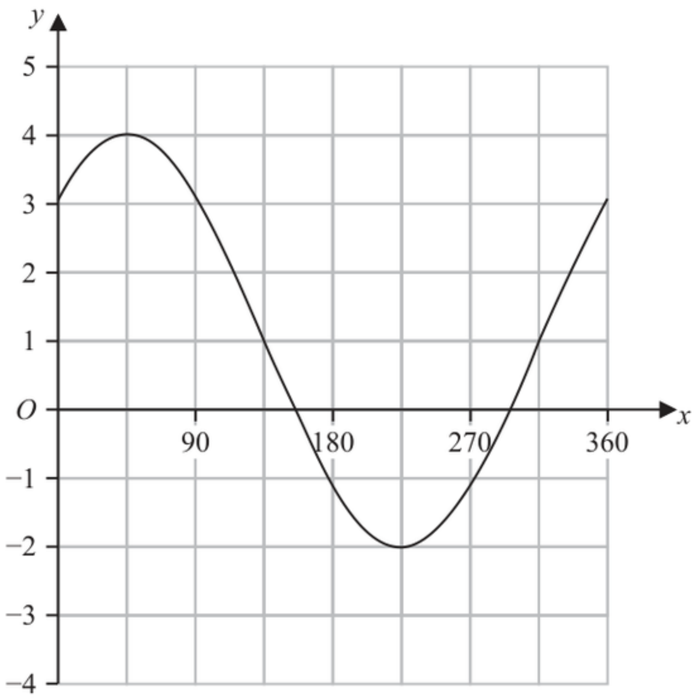
CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2025 | Q#: 25 | Marks: 3

TOPIC: **Graphs of Functions**

Source: Jun 2025 P2HR

25 The graph of $y = a\sin(x + b)^\circ + c$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Find a suitable value for a, for b and for c

a = ...

b = ...

c = ...

(Total for Question 25 is 3 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jun 2025 P2HR | **Question 25**
Graphs of Functions

1. Read the graph

From the graph,

$$\text{maximum} = 4$$

$$\text{minimum} = -2$$

2. The amplitude is

The amplitude is

$$a = \frac{4 - (-2)}{2} = 3$$

3. The centre line is

The centre line is

$$c = \frac{4 + (-2)}{2} = 1$$

4. Use trigonometry

The maximum is about $45°$. For a sine graph, a maximum happens when the angle inside the sine is $90°$, so

$$45 + b = 90$$

$$b = 45$$

Final answer
$a = 3,\ b = 45,\ c = 1$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2024 | Q#: 23 | Marks: 5

TOPIC: **Graphs of Functions**

Source: Nov 2024 P1H

23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)
The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k
Show clear algebraic working.

$k =$ ....

(Total for Question 23 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2024 P1H | **Question 23**
Graphs of Functions

1. The curve is

The curve is

$$y = x^2 - 8x - 9$$

2. The line is

The line is

$$y = k$$

3. Read the graph

At the intersection points,

$$x^2 - 8x - 9 = k$$

$$x^2 - 8x - (9 + k) = 0$$

4. Read the graph

The two roots are p and q.

5. Solve quadratic equation

For this quadratic,

$$p + q = 8$$

6. We are also given

We are also given

$$p - q = 14$$

7. Solve the two equations

Solve the two equations:

$$2p = 22$$

$$p = 11$$

$$q = -3$$

8. Read the graph

Use one root in $y = x^2 - 8x - 9$:

$$k = 11^2 - 8(11) - 9$$

$$k = 121 - 88 - 9$$

$$k = 24$$

Final answer
$k = 24$

Topic 11

Standard & Compound Units

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2022 | Q#: 21 | Marks: 6

TOPIC: **Standard & Compound Units**

Source: Jan 2022 P2HR

21 The diagram shows a solid made from a cylinder and a hemisphere.
The cylinder and the hemisphere are both made from the same metal.

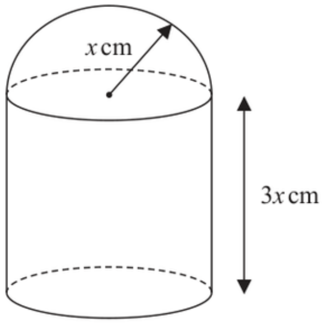


Diagram **NOT** accurately drawn

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The radius of the cylinder and the radius of the hemisphere are both x cm.
The height of the cylinder is $3x$ cm.

The total surface area of the solid is $81\pi \text{ cm}^2$
The mass of the solid is 840 grams.

The following table gives the density of each of four metals.

Metal	**Density (g/cm^3)**
Aluminium	2.7
Nickel	8.9
Gold	19.3
Silver	10.5

The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.
Show your working clearly.

(Total for Question 21 is 6 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jan 2022 P2HR | Question 21
Standard & Compound Units

1. The outside surface area is
The outside surface area is:

2. curved surface area of cylinder
curved surface area of cylinder:

$$2\pi x(3x) = 6\pi x^2$$

3. bottom of cylinder
bottom of cylinder:

$$\pi x^2$$

4. curved surface area of hemisphere
curved surface area of hemisphere:

$$2\pi x^2$$

5. Calculate value
So

$$6\pi x^2 + \pi x^2 + 2\pi x^2 = 81\pi$$

$$9\pi x^2 = 81\pi$$

$$x^2 = 9$$

$$x = 3$$

6. Volume of the cylinder
Volume of the cylinder:

$$\pi x^2(3x) = 3\pi x^3$$

7. Volume of the hemisphere
Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

8. Total volume
Total volume:

$$3\pi(3)^3 + \frac{2}{3}\pi(3)^3$$

$$= 81\pi + 18\pi = 99\pi$$

9. Density

Density:

$$\frac{840}{99\pi} = 2.70\ldots$$

10. Compare density

This matches aluminium, whose density is 2.7 g/cm^3.

Final answer

Aluminium

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2025 | Q#: 26 | Marks: 6

TOPIC: **Standard & Compound Units**

Source: Nov 2025 P2H

26 A solid cone is joined to a solid hemisphere to make the solid shown below.

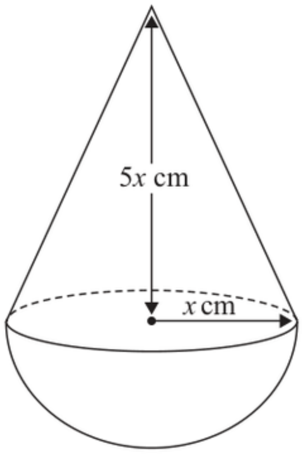


Diagram **NOT** accurately drawn

The cone is made from copper.
The density of copper is 9 g/cm^3

The hemisphere is made from a different metal.

The total mass of the solid is 4752π grams
The total volume of the solid is 504π cm^3

Work out the density of the hemisphere.
Show your working clearly.

.... g/cm^3

(Total for Question 26 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | Question 26

Standard & Compound Units

1. Volume of cone

Volume of cone:

$$\frac{1}{3}\pi x^2(5x) = \frac{5}{3}\pi x^3$$

2. Volume of hemisphere

Volume of hemisphere:

$$\frac{2}{3}\pi x^3$$

3. Total volume

Total volume:

$$\frac{5}{3}\pi x^3 + \frac{2}{3}\pi x^3 = 504\pi$$

$$\frac{7}{3}\pi x^3 = 504\pi$$

$$x^3 = 216$$

$$x = 6$$

4. Cone volume

Cone volume:

$$\frac{5}{3}\pi(6)^3 = 360\pi$$

5. Mass of copper cone

Mass of copper cone:

$$360\pi \times 9 = 3240\pi$$

6. Mass of hemisphere

Mass of hemisphere:

$$4752\pi - 3240\pi = 1512\pi$$

7. Hemisphere volume

Hemisphere volume:

$$\frac{2}{3}\pi(6)^3 = 144\pi$$

8. Density of the hemisphere

Density of the hemisphere:

$$\frac{1512\pi}{144\pi} = 10.5$$

Final answer

10.5 g/cm^3

Topic 12

Area & Perimeter

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 21 | Marks: 4

TOPIC: **Area & Perimeter**

Source: Nov 2025 4WM1H

21 The diagram shows an equilateral triangle and a circle.

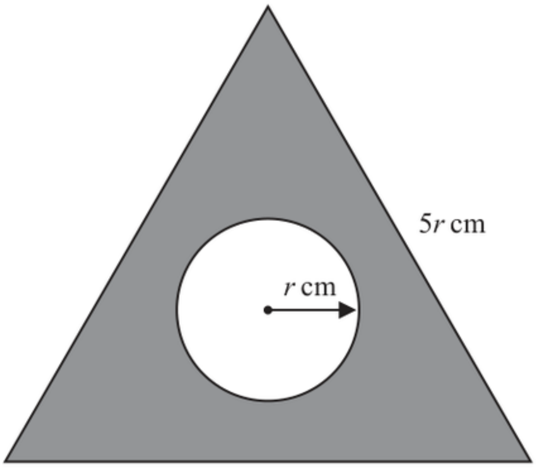


Diagram **NOT** accurately drawn

The equilateral triangle has sides of length $5r$ cm
The circle has radius r cm

The area of the shaded region is 610π cm^2
Work out the value of r

Give your answer correct to 3 significant figures.

r = ...

(Total for Question 21 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 21
Area & Perimeter

1. Triangle area

Triangle area:

$$\frac{\sqrt{3}}{4}(5r)^2 = \frac{25\sqrt{3}}{4}r^2$$

2. Circle area

Circle area:

$$\pi r^2$$

3. Simplify surd

So

$$\left(\frac{25\sqrt{3}}{4} - \pi\right) r^2 = 610\pi$$

$$r \approx 15.7926$$

Final answer
$r = 15.8$ to 3 significant figures.

Topic 13

Circles, Arcs & Sectors

CLASSIFIED PROBLEM

Paper: 4WM1HR | Session: May2025 | Q#: 22 | Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: May 2025 4WM1HR

22

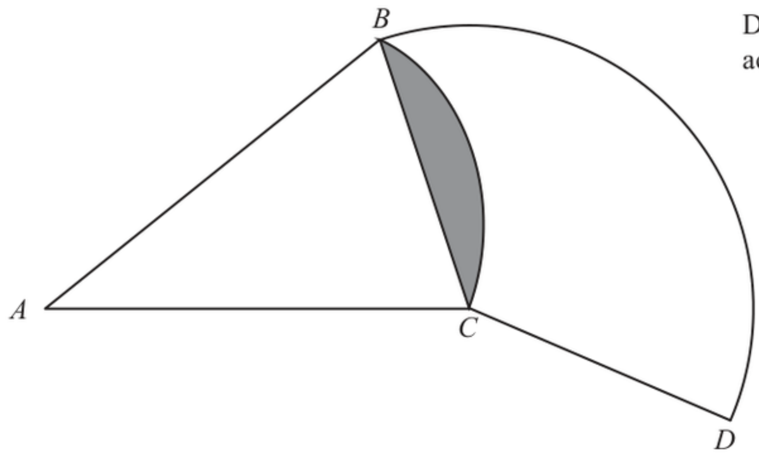


Diagram **NOT** accurately drawn

BAC is a sector of a circle, centre A

BCD is a sector of a circle, centre C

Angle $BAC = 40°$
Angle $BCD = 130°$
Area of shaded segment $= 28\,\text{cm}^2$

Find the length of the arc BD
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 22

Circles, Arcs & Sectors

1. Let the radius of sector

Let the radius of sector BAC be R.

2. The shaded segment is the

The shaded segment is the segment cut off by chord BC in the sector with centre A.

3. Its area is

Its area is

$$\text{sector area} - \text{triangle area}$$

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40^\circ$$

$$28 = R^2\left(\frac{\pi}{9} - \frac{1}{2}\sin 40^\circ\right)$$

$$R = 31.8096\ldots$$

4. Use trigonometry

Now find the chord BC:

$$BC = 2R\sin 20^\circ$$

$$BC = 2(31.8096\ldots)\sin 20^\circ = 21.7591\ldots$$

5. Calculate value

In sector BCD, the centre is C, so the radius is

$$CB = CD = 21.7591\ldots$$

6. Use trigonometry

The arc BD has angle 130°, so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591\ldots)$$

$$= 49.3697\ldots$$

Final answer

$\boxed{49.4 \text{ cm}}$

Topic 14

Area & Perimeter

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2021 | Q#: 20 | Marks: 4

TOPIC: **Area & Perimeter**

Source: Jan 2021 P2H

20

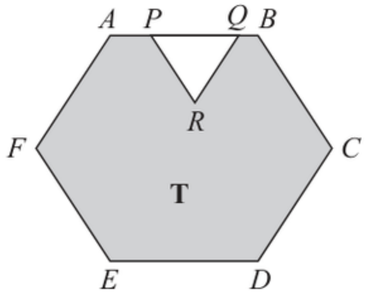


Diagram **NOT** accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3}$ cm²

work out the length of PQ.

.... cm

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2H | Question 20

Area & Perimeter

1. Let

Let

$$PQ = p$$

2. Given

Given

$$AB = 1.5PQ$$

3. Find the gradient

so the side length of the regular hexagon is

$$1.5p = \frac{3}{2}p$$

4. Area of a regular hexagon

Area of a regular hexagon with side length s is

$$\frac{3\sqrt{3}}{2}s^2$$

5. Find the gradient

So the area of the hexagon is

$$\frac{3\sqrt{3}}{2}\left(\frac{3}{2}p\right)^2$$

$$= \frac{27\sqrt{3}}{8}p^2$$

6. Find the gradient

The removed triangle PQR is equilateral, so its area is

$$\frac{\sqrt{3}}{4}p^2$$

7. The shaded area is

The shaded area is

$$\frac{27\sqrt{3}}{8}p^2 - \frac{\sqrt{3}}{4}p^2$$

$$= \frac{25\sqrt{3}}{8}p^2$$

8. Find the gradient

This is $72\sqrt{3}$, so

$$\frac{25\sqrt{3}}{8}p^2 = 72\sqrt{3}$$

$$\frac{25}{8}p^2 = 72$$

$$p^2 = \frac{576}{25}$$

$$p = 4.8$$

Final answer
$PQ = 4.8$ cm

CLASSIFIED PROBLEM

Paper: P1HR | Session: MayNov 2020 | Q#: 22 | Marks: 6

TOPIC: **Area & Perimeter**

Source: May-Nov 2020 P1HR

22 The diagram shows a regular octagon *ABCDEFGH*.

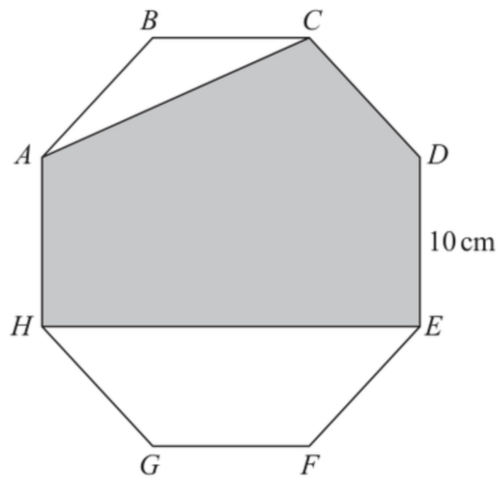


Diagram **NOT** accurately drawn

Each side of the octagon has length 10 cm.

Find the area of the shaded region *ACDEH*.
Give your answer correct to the nearest cm^2

.... cm^2

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | Question 22
Area & Perimeter

1. Each side of the regular

Each side of the regular octagon is 10 cm.

2. Calculate area

For a regular octagon with side length s,

$$\text{area} = 2(1+\sqrt{2})s^2$$

3. Calculate area

So the area of the whole octagon is

$$2(1+\sqrt{2})(10)^2 = 200 + 200\sqrt{2}$$

4. The unshaded triangle has

The unshaded triangle ABC has

$$AB = 10, \qquad BC = 10, \qquad \angle ABC = 135^\circ$$

5. Use trigonometry

So

$$\text{area of triangle } ABC = \frac{1}{2}(10)(10)\sin 135^\circ = 25\sqrt{2}$$

6. Find the gradient

At the bottom, the unshaded trapezium $HEFG$ has parallel sides

$$HE = 10 + 10\sqrt{2}, \qquad GF = 10$$

7. Simplify surd

and height

$$5\sqrt{2}$$

8. Calculate area

So

$$\text{area of trapezium } HEFG = \frac{1}{2}(10 + 10 + 10\sqrt{2})(5\sqrt{2})$$

$$= 50 + 50\sqrt{2}$$

9. The shaded area is

The shaded area is

$$(200 + 200\sqrt{2}) - 25\sqrt{2} - (50 + 50\sqrt{2})$$

$$= 150 + 125\sqrt{2}$$

$$= 326.7766\ldots$$

Final answer

$\boxed{327 \text{ cm}^2}$

Topic 15

Circles, Arcs & Sectors

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2021 | Q#: 22 | Marks: 6

TOPIC: Circles, Arcs & Sectors

Source: Jan 2021 P2H

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

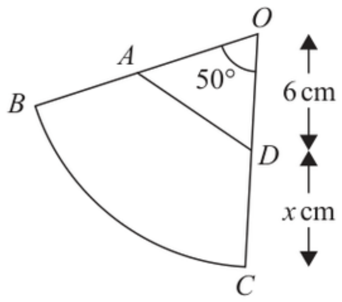


Diagram **NOT** accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50°$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x.
Give your answer correct to 3 significant figures.

$x =$ ....

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2H | Question 22

Circles, Arcs & Sectors

1. The radius of sector is

The radius of sector OBC is

$$6 + x$$

2. Use trigonometry

The angle is 50°, so the arc length of sector OBC is

$$\frac{50}{360} \times 2\pi(6 + x)$$

3. Evaluate fraction

Therefore the perimeter of sector OBC is

$$2(6 + x) + \frac{50}{360} \times 2\pi(6 + x)$$

$$= (6 + x)\left(2 + \frac{5\pi}{18}\right)$$

4. Use trigonometry

In triangle OAD,

$$OA = OD = 6, \qquad \angle AOD = 50^\circ$$

5. Use trigonometry

So

$$AD = 2(6)\sin 25^\circ = 12\sin 25^\circ$$

6. The perimeter of triangle is

The perimeter of triangle OAD is

$$12 + 12\sin 25^\circ$$

7. Given that

Given that

$$\text{perimeter of sector } OBC = 2 \times \text{perimeter of triangle } OAD$$

$$(6 + x)\left(2 + \frac{5\pi}{18}\right) = 2(12 + 12\sin 25^\circ)$$

$$x = \frac{2(12 + 12\sin 25^\circ)}{2 + \frac{5\pi}{18}} - 6$$

$$x = 5.8854\ldots$$

Final answer

$\boxed{5.89}$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2021 | Q#: 20 | Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Source: Jan 2021 P2HR

20 A, B and C are points on a circle with centre O.

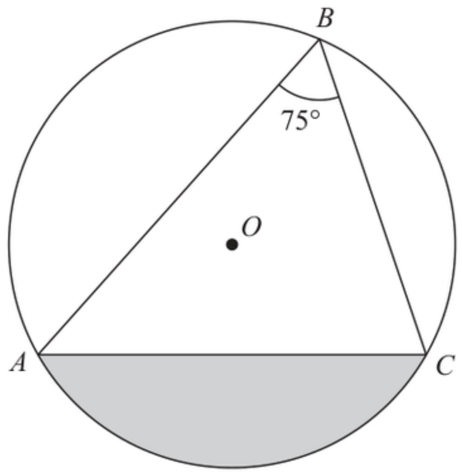


Diagram **NOT** accurately drawn

Angle $ABC = 75°$

The area of the shaded segment is $200\,\text{cm}^2$

Calculate the radius of the circle.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2HR | Question 20

Circles, Arcs & Sectors

1. Use trigonometry

$$\angle ABC = 75^\circ$$

2. The angle at the centre

The angle at the centre is twice the angle at the circumference, so

$$\angle AOC = 2(75^\circ) = 150^\circ$$

3. The shaded segment is

The shaded segment is:

$$\text{sector } AOC - \text{triangle } AOC$$

4. Let the radius be

Let the radius be r.

5. Sector area

Sector area:

$$\frac{150}{360}\pi r^2$$

6. Triangle area

Triangle area:

$$\frac{1}{2}r^2 \sin 150^\circ$$

7. Use trigonometry

So

$$\frac{150}{360}\pi r^2 - \frac{1}{2}r^2 \sin 150^\circ = 200$$

$$r^2\left(\frac{5\pi}{12} - \frac{1}{4}\right) = 200$$

$$r^2 = \frac{200}{\frac{5\pi}{12} - \frac{1}{4}}$$

$$r = 13.7426\ldots$$

Final answer

$\boxed{13.7\text{ cm}}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2022 | Q#: 20 | Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Source: Jun 2022 P2H

20 The diagram shows four identical circles drawn inside a square.

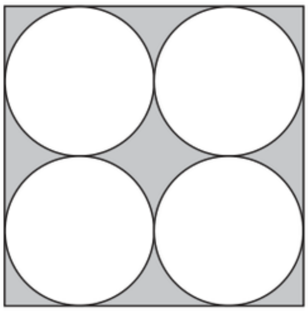


Diagram **NOT** accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of $40\,\text{cm}^2$

Work out the perimeter of the square.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P2H | Question 20

Circles, Arcs & Sectors

1. Let the radius of each

Let the radius of each circle be r.

2. There are two circles across

There are two circles across the width of the square, so the side length of the square is

$$4r$$

3. Area of the square

Area of the square:

$$(4r)^2 = 16r^2$$

4. Area of the four circles

Area of the four circles:

$$4\pi r^2$$

5. Calculate area

The shaded area is $40\ \text{cm}^2$, so

$$16r^2 - 4\pi r^2 = 40$$

$$r^2(16 - 4\pi) = 40$$

$$r = \sqrt{\frac{40}{16 - 4\pi}} = 3.4131\ldots$$

6. The perimeter of the square

The perimeter of the square is

$$4(4r) = 16r$$

$$16r = 54.6101\ldots$$

Final answer

$\boxed{54.6\ \text{cm}}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2024 | Q#: 21 | Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Source: Jun 2024 P2H

21 The diagram shows a square $ABCD$ and a circle.

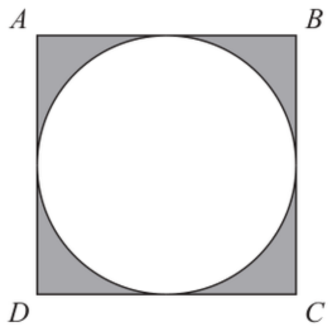


Diagram **NOT** accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is $80\,\text{cm}^2$

Work out the length of AC
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2024 P2H | Question 21

Circles, Arcs & Sectors

1. Let the side length

Let the side length of the square be s.

2. Square

The circle is inscribed in the square, so its radius is

$$\frac{s}{2}$$

3. The shaded area is the

The shaded area is the area of the square minus the area of the circle:

$$s^2 - \pi \left(\frac{s}{2}\right)^2 = 80$$

$$s^2 - \frac{\pi s^2}{4} = 80$$

$$s^2 \left(1 - \frac{\pi}{4}\right) = 80$$

$$s = \sqrt{\frac{80}{1 - \frac{\pi}{4}}} = 19.3076\ldots$$

4. The diagonal of the square

The diagonal of the square is

$$AC = s\sqrt{2}$$

$$AC = 19.3076\ldots\sqrt{2} = 27.3051\ldots$$

Final answer

$\boxed{27.3 \text{ cm}}$

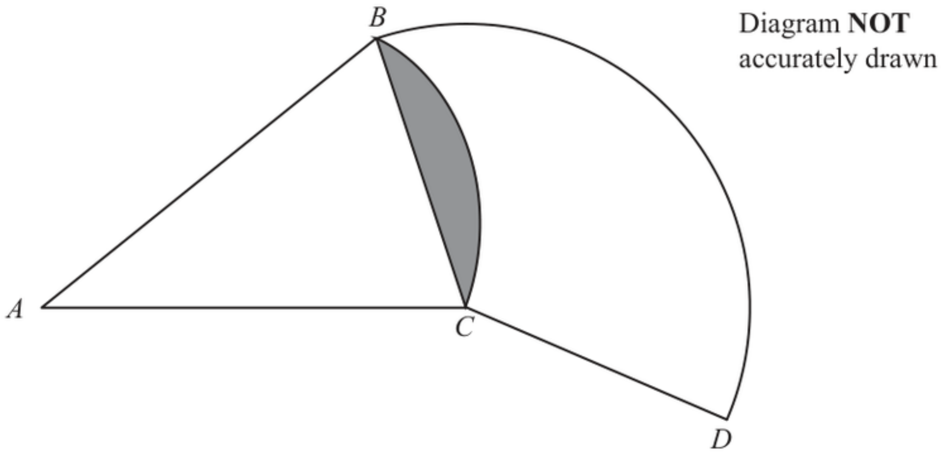
CLASSIFIED PROBLEM

Paper: P1HR | Session: Jun 2025 | Q#: 25 | Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: Jun 2025 P1HR

25



BAC is a sector of a circle, centre *A*

BCD is a sector of a circle, centre *C*

Angle $BAC = 40°$
Angle $BCD = 130°$
Area of shaded segment $= 28\,\text{cm}^2$

Find the length of the arc *BD*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 25

Circles, Arcs & Sectors

1. Let the radius of sector

Let the radius of sector BAC be R.

2. The shaded segment is the

The shaded segment is the segment cut off by chord BC in the sector with centre A.

3. Its area is

Its area is

$$\text{sector area} - \text{triangle area}$$

$$28 = \frac{40}{360}\pi R^2 - \frac{1}{2}R^2 \sin 40^\circ$$

$$28 = R^2\left(\frac{\pi}{9} - \frac{1}{2}\sin 40^\circ\right)$$

$$R = 31.8096\ldots$$

4. Use trigonometry

Now find the chord BC:

$$BC = 2R\sin 20^\circ$$

$$BC = 2(31.8096\ldots)\sin 20^\circ = 21.7591\ldots$$

5. Calculate value

In sector BCD, the centre is C, so the radius is

$$CB = CD = 21.7591\ldots$$

6. Use trigonometry

The arc BD has angle 130°, so

$$\text{arc } BD = \frac{130}{360} \times 2\pi(21.7591\ldots)$$

$$= 49.3697\ldots$$

Final answer

$\boxed{49.4 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P1H | Session: May 2021 | Q#: 25 | Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: May 2021 P1H

25 The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

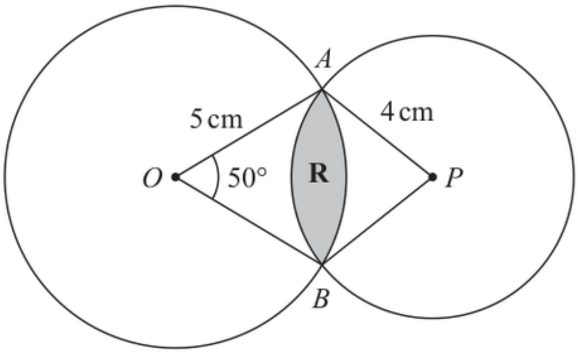


Diagram **NOT** accurately drawn

One of the circles has centre O and radius 5 cm.
The other circle has centre P and radius 4 cm.
Angle $AOB = 50°$

Calculate the area of region **R**.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 25
Circles, Arcs & Sectors

1. Use trigonometry

For the circle with centre O:

$$\angle AOB = 50^\circ, \qquad OA = OB = 5$$

2. The half chord length is

The half chord length is

$$\frac{AB}{2} = 5\sin 25^\circ$$

3. Use trigonometry

For the circle with centre P, let

$$\angle APB = \theta$$

4. Use trigonometry

The radius is 4, so

$$4\sin\left(\frac{\theta}{2}\right) = 5\sin 25^\circ$$

$$\sin\left(\frac{\theta}{2}\right) = \frac{5\sin 25^\circ}{4}$$

$$\theta = 63.7777\ldots^\circ$$

5. The common region is made

The common region is made from two minor segments.

6. Use trigonometry

For the circle with centre O:

$$\text{segment area} = \frac{50}{360}\pi(5)^2 - \frac{1}{2}(5)^2\sin 50^\circ = 1.3328\ldots$$

7. Use trigonometry

For the circle with centre P:

$$\text{segment area} = \frac{\theta}{360}\pi(4)^2 - \frac{1}{2}(4)^2\sin\theta$$

$$= 1.7284\ldots$$

8. Total area of region

Total area of region R:

$$1.3328\ldots + 1.7284\ldots = 3.0611\ldots$$

Final answer

$\boxed{3.06 \text{ cm}^2}$

CLASSIFIED PROBLEM

Paper: P1H | Session: May 2023 | Q#: 22 | Marks: 6

TOPIC: **Circles, Arcs & Sectors**

Source: May 2023 P1H

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

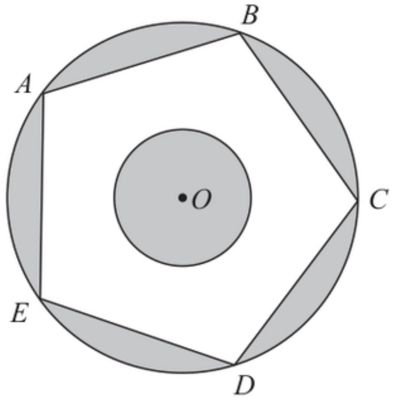


Diagram **NOT** accurately drawn

A, B, C, D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

shaded area = unshaded area

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 22 is 6 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

May 2023 P1H | Question 22
Circles, Arcs & Sectors

1. The regular pentagon has side

The regular pentagon has side length 8 cm.

2. Use trigonometry

For a regular pentagon,

$$R = \frac{s}{2\sin 36°}$$

3. Use trigonometry

where R is the radius of the larger circle.

$$R = \frac{8}{2\sin 36°} = 6.8052\ldots$$

4. Area of the larger circle

Area of the larger circle:

$$\pi R^2 = 145.4898\ldots$$

5. Area of the regular pentagon

Area of the regular pentagon:

$$\frac{5s^2}{4\tan 36°} = \frac{5(8)^2}{4\tan 36°} = 110.1106\ldots$$

6. Let the radius of the

Let the radius of the smaller circle be r.

7. The shaded area is

The shaded area is:

$$\text{larger circle area} - \text{pentagon area} + \pi r^2$$

8. The unshaded area is

The unshaded area is:

$$\text{pentagon area} - \pi r^2$$

9. Simplify surd

These are equal, so

$$145.4898 - 110.1106 + \pi r^2 = 110.1106 - \pi r^2$$

$$2\pi r^2 = 2(110.1106) - 145.4898$$

$$\pi r^2 = 37.3657\ldots$$

$$r = \sqrt{\frac{37.3657\ldots}{\pi}} = 3.4487\ldots$$

Final answer

$\boxed{3.45 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P1H | Session: May 2024 | Q#: 20 | Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Source: May 2024 P1H

20 The diagram shows a sector $OABC$ of a circle centre O

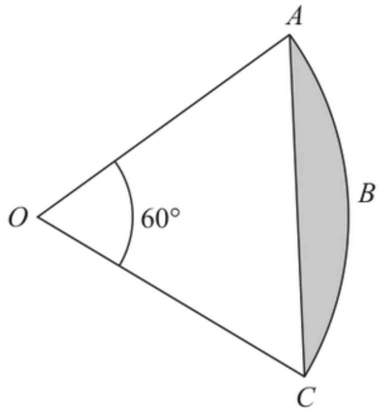


Diagram **NOT** accurately drawn

Angle $AOC = 60°$

The area of the shaded segment ABC is $38\,\text{cm}^2$

Work out the perimeter of the shaded segment ABC
Give your answer correct to one decimal place.

.... cm

(Total for Question 20 is 4 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

May 2024 P1H | Question 20
Circles, Arcs & Sectors

1. Let the radius of the
Let the radius of the sector be r.

2. The shaded segment is
The shaded segment is

$$\text{sector area} - \text{triangle area}$$

3. Use trigonometry
The angle is $60°$, so

$$38 = \frac{60}{360}\pi r^2 - \frac{1}{2}r^2 \sin 60°$$

$$38 = r^2\left(\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right)$$

$$r = 20.4815\ldots$$

4. The perimeter of the shaded
The perimeter of the shaded segment is the arc AC plus the chord AC.

5. Arc length
Arc length:

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

6. Chord length
Chord length:

$$2r \sin 30° = r$$

7. Evaluate fraction
So the perimeter is

$$\frac{\pi r}{3} + r$$

$$= \frac{\pi(20.4815\ldots)}{3} + 20.4815\ldots$$

$$= 41.9296\ldots$$

Final answer
$\boxed{41.9 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2023 | Q#: 20 | Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Source: Nov 2023 P2H

20 The diagram shows equilateral triangle ABC with sides of length 10 cm.
A circle is drawn inside the triangle.

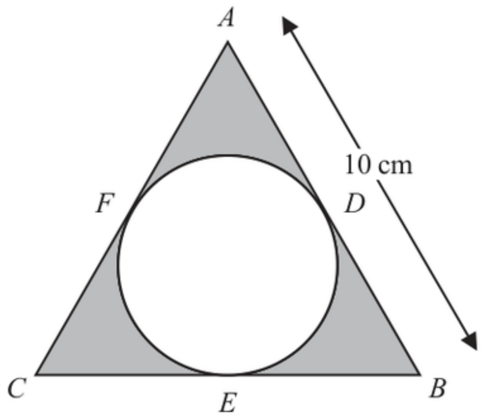


Diagram **NOT** accurately drawn

D, E and F are points on the circle.

ADB, BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P2H | Question 20

Circles, Arcs & Sectors

1. Use trigonometry

The triangle is equilateral with side length 10 cm.

2. Area of the triangle

Area of the triangle:

$$\frac{\sqrt{3}}{4}(10)^2 = 25\sqrt{3}$$

3. Use trigonometry

For an equilateral triangle, the inradius is

$$r = \frac{s\sqrt{3}}{6}$$

4. Simplify surd

So

$$r = \frac{10\sqrt{3}}{6} = \frac{5\sqrt{3}}{3}$$

5. Area of the circle

Area of the circle:

$$\pi r^2 = \pi\left(\frac{5\sqrt{3}}{3}\right)^2 = \frac{25\pi}{3}$$

6. The shaded area is

The shaded area is

$$25\sqrt{3} - \frac{25\pi}{3} = 17.1213\ldots$$

Final answer

$\boxed{17.1 \text{ cm}^2}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2024 | Q#: 25 | Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Source: Nov 2024 P2H

25 The diagram shows an equilateral triangle ABC and a circle with centre O

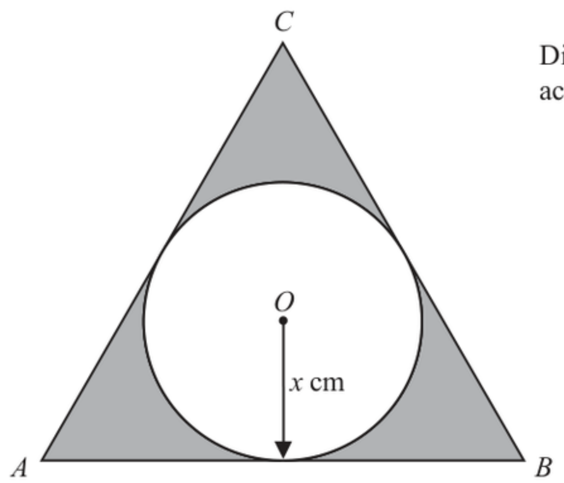


Diagram **NOT** accurately drawn

AB, BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2, of the regions shown shaded in the diagram is nx^2

Find the value of n
Give your answer correct to 3 significant figures.

$n =$ ....

(Total for Question 25 is 5 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Nov 2024 P2H | Question 25
Circles, Arcs & Sectors

1. Use trigonometry
The circle has radius x, so the inradius of the equilateral triangle is x.

2. Use trigonometry
For an equilateral triangle with side length a,

$$\text{inradius} = \frac{a\sqrt{3}}{6}$$

3. Simplify surd
So

$$x = \frac{a\sqrt{3}}{6}$$

$$a = \frac{6x}{\sqrt{3}} = 2\sqrt{3}x$$

4. Area of the triangle
Area of the triangle:

$$\frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}(2\sqrt{3}x)^2$$

$$= \frac{\sqrt{3}}{4}(12x^2) = 3\sqrt{3}x^2$$

5. Area of the circle
Area of the circle:

$$\pi x^2$$

6. The shaded area is
The shaded area is

$$3\sqrt{3}x^2 - \pi x^2$$

$$= (3\sqrt{3} - \pi)x^2$$

7. Simplify surd
Therefore

$$n = 3\sqrt{3} - \pi$$

Final answer

$\boxed{3\sqrt{3} - \pi}$

Topic 16

Sine, Cosine Rule & Area of Triangles

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: May2025 | Q#: 21 | Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: May 2025 4WM1H

21

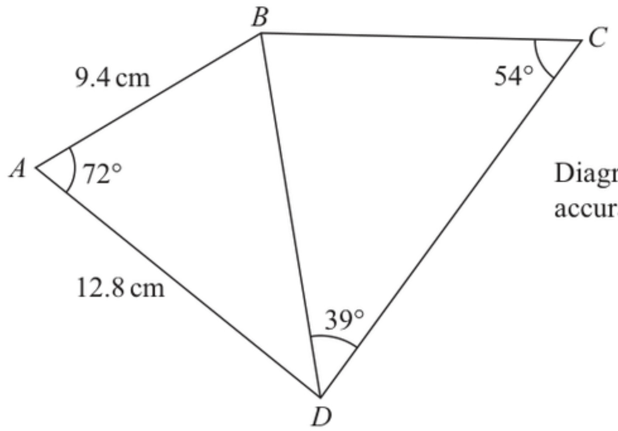


Diagram **NOT** accurately drawn

Work out the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 21

Sine, Cosine Rule & Area of Triangles

1. Use trigonometry

First find BD in triangle ABD using the cosine rule.

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ$$

$$BD = 13.3356\ldots$$

2. Use trigonometry

In triangle BCD, use the sine rule.

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ}$$

$$BC = 13.3356\ldots \times \frac{\sin 39^\circ}{\sin 54^\circ}$$

$$BC = 10.3735\ldots$$

3. Use trigonometry

To 3 significant figures,

$$BC = 10.4$$

Final answer

10.4 cm

Topic 17

3D Pythagoras & Trigonometry

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: May2025 | Q#: 23 | Marks: 6

TOPIC: **3D Pythagoras & Trigonometry**

Source: May 2025 4WM1H

23 The diagram shows cuboid *ABCDEFGH*

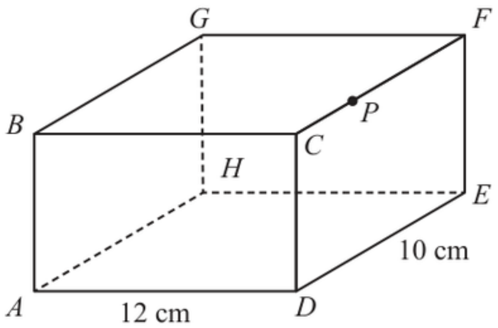


Diagram **NOT** accurately drawn

The point P lies on CF

$AD = 12$ cm $\quad DE = 10$ cm $\quad CP = 3$ cm

The angle of elevation of P from A is $24°$

Calculate the size of angle APG
Give your answer correct to the nearest degree.
Show your working clearly.

.... °

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | Question 23

3D Pythagoras & Trigonometry

1. Point is cm from along

Point P is 3 cm from C along CF. The horizontal distance from A to the point directly below P is

$$\sqrt{12^2 + 3^2} = \sqrt{153}$$

2. Use trigonometry

Using the angle of elevation,

$$h = \sqrt{153}\tan 24^\circ = 5.507\ldots$$

3. Use trigonometry

So

$$AP = \frac{\sqrt{153}}{\cos 24^\circ} = 13.5399\ldots$$

4. Use trigonometry

On the top face,

$$PG = \sqrt{12^2 + 7^2} = \sqrt{193} = 13.8924\ldots$$

5. Use trigonometry

Also,

$$AG = \sqrt{12^2 + 10^2 + h^2} = 16.561\ldots$$

6. Use cosine rule

Use the cosine rule in triangle APG.

$$\cos \angle APG = \frac{AP^2 + PG^2 - AG^2}{2(AP)(PG)}$$

$$\angle APG = 49.163\ldots^\circ$$

7. Use trigonometry

To the nearest degree,

$$49^\circ$$

Final answer

49°

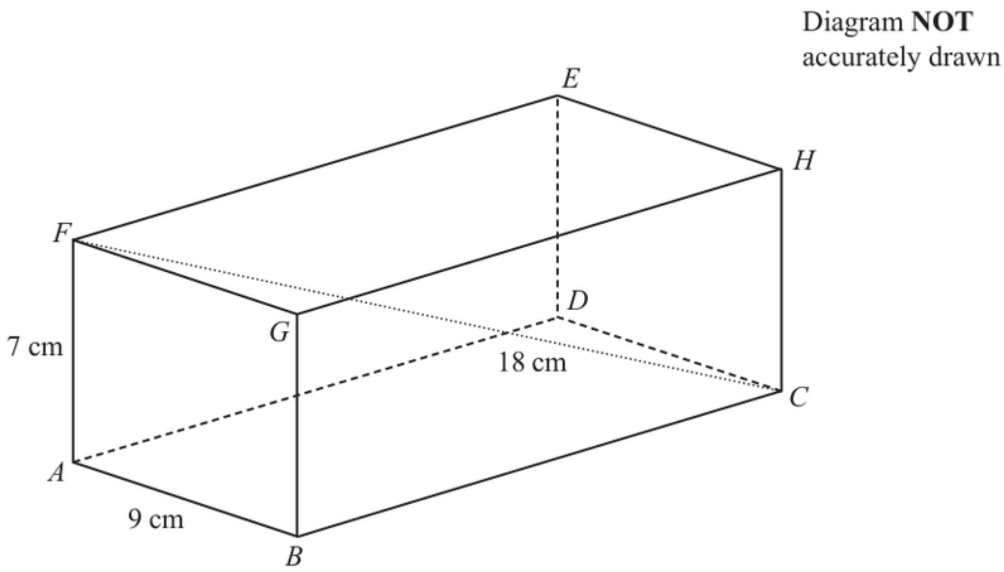
CLASSIFIED PROBLEM

Paper: 4WM1HR | Session: May2025 | Q#: 20 | Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Source: May 2025 4WM1HR

20 The diagram shows cuboid *ABCDEFGH*



$AB = 9\,\text{cm}$ $\quad AF = 7\,\text{cm}$ $\quad FC = 18\,\text{cm}$

Calculate the length of BC
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 20
3D Pythagoras & Trigonometry

1. Use trigonometry

In the rectangle $ABCF$, use Pythagoras with the diagonal FC:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

2. Use trigonometry

Now use Pythagoras in triangle ABC:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

Final answer
13.9 cm to 3 significant figures

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 26 | Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Source: Nov 2025 4WM1H

26 The diagram shows prism *ABCDEFGH*

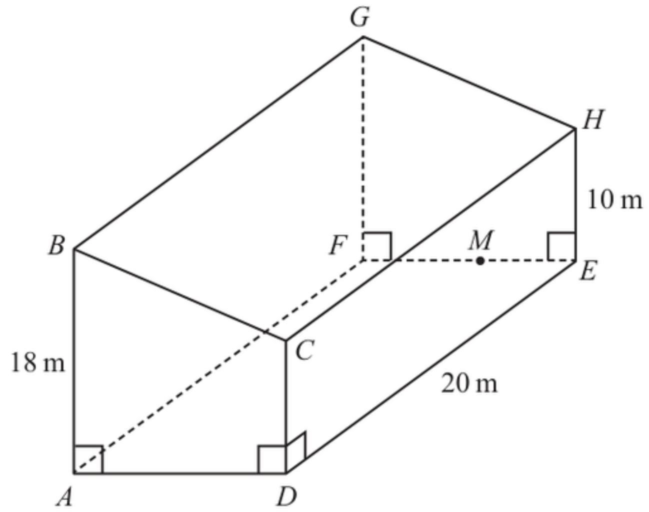


Diagram **NOT** accurately drawn

$AB = FG = 18\,\text{m}$ $\quad DC = EH = 10\,\text{m}$ $\quad AD = FE$

$DE = CH = AF = BG = 20\,\text{m}$

angle BAD = angle ADC = angle CDE = 90°

M is the point on FE such that $FM = \frac{3}{5}FE$

Given that angle $GMF = 60°$

calculate the angle of elevation of G from D
Give your answer correct to one decimal place.
Show your working clearly.

.... °

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 26
3D Pythagoras & Trigonometry

1. Use trigonometry

From triangle GMF,

$$\tan 60^\circ = \frac{FG}{FM} = \frac{18}{FM}$$

$$FM = \frac{18}{\tan 60^\circ} = 6\sqrt{3}$$

2. Simplify surd

Since $FM = \frac{3}{5}FE$,

$$FE = 10\sqrt{3}$$

3. Simplify surd

The horizontal distance from D to the point below G is

$$\sqrt{20^2 + (10\sqrt{3})^2} = 10\sqrt{7}$$

4. Use trigonometry

If θ is the angle of elevation,

$$\tan\theta = \frac{18}{10\sqrt{7}}$$

$$\theta \approx 34.2289^\circ$$

Final answer

34.2°.

Topic 18

Sine, Cosine Rule & Area of Triangles

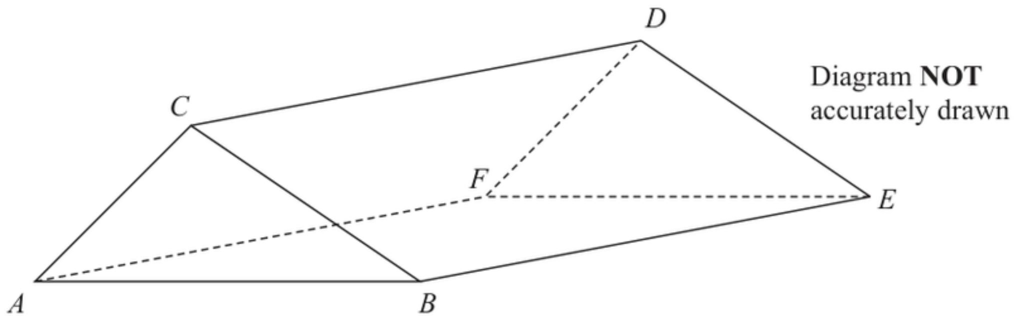
CLASSIFIED PROBLEM

Paper: P1HR | Session: Jan 2020 | Q#: 21 | Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: Jan 2020 P1HR

21 The diagram shows the prism *ABCDEF* with cross section triangle *ABC*.



Angle $BEC = 40°$ and angle ACB is obtuse.
$AC = 6$ cm and $CE = 13$ cm

The area of triangle ABC is 22 cm^2

Calculate the length of AB.
Give your answer correct to one decimal place.

.... cm

(Total for Question 21 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2020 P1HR | Question 21

Sine, Cosine Rule & Area of Triangles

1. Use trigonometry

In triangle BEC, $CE = 13$ and $\angle BEC = 40°$.

2. Find the gradient

Since this is a prism, BE is perpendicular to the cross section, so triangle BEC is right-angled at B.

$$BC = 13 \sin 40°$$

$$BC = 8.356238\ldots$$

3. The area of triangle is

The area of triangle ABC is 22 cm^2.

4. Use trigonometry

Using

$$\text{Area} = \frac{1}{2}ab\sin C$$

$$22 = \frac{1}{2}(AC)(BC)\sin(\angle ACB)$$

$$22 = \frac{1}{2}(6)(8.356238\ldots)\sin(\angle ACB)$$

$$\sin(\angle ACB) = 0.877587\ldots$$

5. Use trigonometry

The angle ACB is obtuse, so

$$\angle ACB = 180° - \sin^{-1}(0.877587\ldots)$$

$$\angle ACB = 118.647\ldots°$$

6. Use the cosine rule

Use the cosine rule:

$$AB^2 = AC^2 + BC^2 - 2(AC)(BC)\cos(\angle ACB)$$

$$AB^2 = 6^2 + (8.356238\ldots)^2 - 2(6)(8.356238\ldots)\cos(118.647\ldots°)$$

$$AB = 12.4056\ldots$$

7. Correct to one decimal place

Correct to one decimal place:

$$12.4$$

Final answer

$\boxed{12.4 \text{ cm}}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2022 | Q#: 23 | Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: Jan 2022 P2H

23 The diagram shows triangle PQR

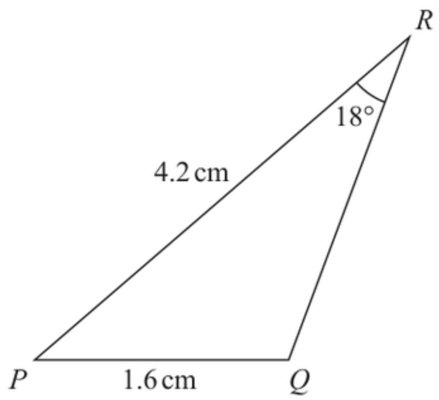


Diagram **NOT** accurately drawn

$PQ = 1.6\,\text{cm}$ $PR = 4.2\,\text{cm}$ Angle $PRQ = 18°$

Given that angle PQR is obtuse,

work out the area of triangle PQR
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 23 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 23
Sine, Cosine Rule & Area of Triangles

1. Use trigonometry

In triangle PQR,

$$PQ = 1.6, \qquad PR = 4.2, \qquad \angle PRQ = 18^\circ$$

2. Use the sine rule

Use the sine rule:

$$\frac{\sin Q}{PR} = \frac{\sin R}{PQ}$$

$$\frac{\sin Q}{4.2} = \frac{\sin 18^\circ}{1.6}$$

$$\sin Q = \frac{4.2 \sin 18^\circ}{1.6}$$

$$\sin Q = 0.811169\ldots$$

3. Use trigonometry

Since angle PQR is obtuse,

$$Q = 180^\circ - \sin^{-1}(0.811169\ldots)$$

$$Q = 125.7896\ldots^\circ$$

4. Use trigonometry

Now find angle P:

$$P = 180^\circ - 18^\circ - 125.7896\ldots^\circ$$

$$P = 36.2103\ldots^\circ$$

5. Area of triangle

Area of triangle PQR:

$$\frac{1}{2}(PQ)(PR)\sin P$$

$$= \frac{1}{2}(1.6)(4.2)\sin(36.2103\ldots^\circ)$$

$$= 1.984925\ldots$$

6. Correct to 3 significant figures

Correct to 3 significant figures:

$$1.98$$

Final answer

$\boxed{1.98 \text{ cm}^2}$

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2023 | Q#: 22 | Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: Jun 2023 P2H

22 The diagram shows a triangle ABC and a flagpole BF

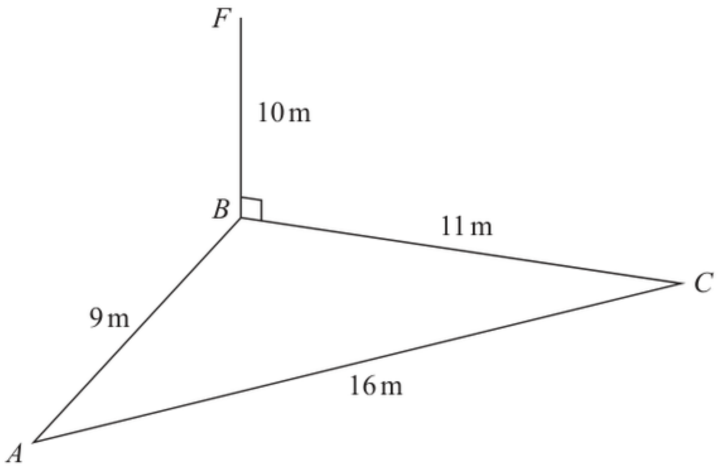


Diagram **NOT** accurately drawn

A, B and C are points on horizontal ground.

BF is vertical.

$AB = 9\,\text{m}$ $\quad$ $BC = 11\,\text{m}$ $\quad$ $AC = 16\,\text{m}$ $\quad$ $BF = 10\,\text{m}$

D is the point on AC such that angle $BDC = 90°$

Work out the size of the angle of elevation of the point F from the point D
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 22

Sine, Cosine Rule & Area of Triangles

1. Use trigonometry

In triangle ABC,

$$AB = 9, \qquad BC = 11, \qquad AC = 16$$

2. Use Heron s formula

Use Heron's formula.

$$s = \frac{9 + 11 + 16}{2} = 18$$

$$\text{area} = \sqrt{18(18-9)(18-11)(18-16)}$$

$$= \sqrt{18 \cdot 9 \cdot 7 \cdot 2} = 47.6235\ldots$$

3. is the foot of the

D is the foot of the perpendicular from B to AC, so BD is the height of triangle ABC.

$$\text{area} = \frac{1}{2} \times AC \times BD$$

$$47.6235\ldots = \frac{1}{2} \times 16 \times BD$$

$$BD = 5.9529\ldots$$

4. Use trigonometry

In right-angled triangle BDF,

$$\tan\theta = \frac{BF}{BD}$$

$$\tan\theta = \frac{10}{5.9529\ldots}$$

$$\theta = \tan^{-1}\left(\frac{10}{5.9529\ldots}\right) = 59.2349\ldots^\circ$$

Final answer

$\boxed{59.2^\circ}$

CLASSIFIED PROBLEM

Paper: P1H | Session: Jun 2025 | Q#: 20 | Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: Jun 2025 P1H

20

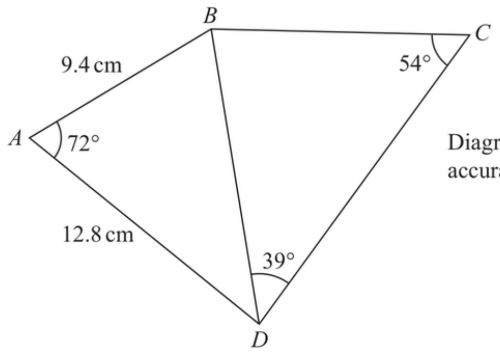


Diagram **NOT** accurately drawn

Work out the length of *BC*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 20
Sine, Cosine Rule & Area of Triangles

1. Use trigonometry

First use triangle ABD.

$$AB = 9.4, \quad AD = 12.8, \quad \angle BAD = 72^\circ$$

2. Use cosine rule

Use the cosine rule to find BD:

$$BD^2 = 9.4^2 + 12.8^2 - 2(9.4)(12.8)\cos 72^\circ$$

$$BD = 13.3356\ldots$$

3. Use trigonometry

In triangle BCD,

$$\angle BDC = 39^\circ, \quad \angle BCD = 54^\circ$$

4. Use the sine rule

Use the sine rule:

$$\frac{BC}{\sin 39^\circ} = \frac{BD}{\sin 54^\circ}$$

$$BC = \frac{13.3356\ldots \sin 39^\circ}{\sin 54^\circ}$$

$$BC = 10.3735\ldots$$

5. Use trigonometry

Correct to 3 significant figures,

$$BC = 10.4$$

Final answer
10.4 cm

CLASSIFIED PROBLEM

Paper: P1HR | Session: May 2023 | Q#: 25 | Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Source: May 2023 P1HR

25 Here is a triangle ABC

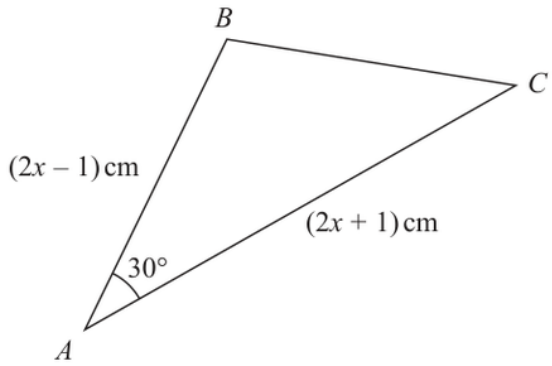


Diagram **NOT** accurately drawn

The area of the triangle is $(x^2 + x - 3.75)\,\text{cm}^2$

Find the size of the largest angle in triangle ABC
Give your answer correct to the nearest degree.

.... °

(Total for Question 25 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 25

Sine, Cosine Rule & Area of Triangles

1. The two sides around the

The two sides around the $30°$ angle are

$$2x - 1 \quad \text{and} \quad 2x + 1$$

2. Use trigonometry

Using the area formula,

$$\text{area} = \frac{1}{2}ab\sin C$$

$$x^2 + x - 3.75 = \frac{1}{2}(2x-1)(2x+1)\sin 30°$$

3. Use trigonometry

Since $\sin 30° = \frac{1}{2}$,

$$x^2 + x - 3.75 = \frac{1}{4}(2x-1)(2x+1)$$

$$x^2 + x - 3.75 = \frac{1}{4}(4x^2-1)$$

$$x^2 + x - 3.75 = x^2 - 0.25$$

$$x = 3.5$$

4. Use trigonometry

So

$$AB = 2x - 1 = 6, \qquad AC = 2x + 1 = 8$$

5. Use cosine rule

Use the cosine rule to find BC:

$$BC^2 = 6^2 + 8^2 - 2(6)(8)\cos 30°$$

$$BC = 4.1063\ldots$$

6. Use trigonometry

The largest side is $AC = 8$, so the largest angle is $\angle B$.

7. Use trigonometry

Using the cosine rule again:

$$\cos B = \frac{6^2 + BC^2 - 8^2}{2(6)(BC)}$$

$$B = 103.0643\ldots^\circ$$

Final answer

$\boxed{103^\circ}$

Topic 19

3D Pythagoras & Trigonometry

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2021 | Q#: 22 | Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jan 2021 P2HR

22 ABC is an isosceles triangle in a horizontal plane.
The point T is vertically above B.

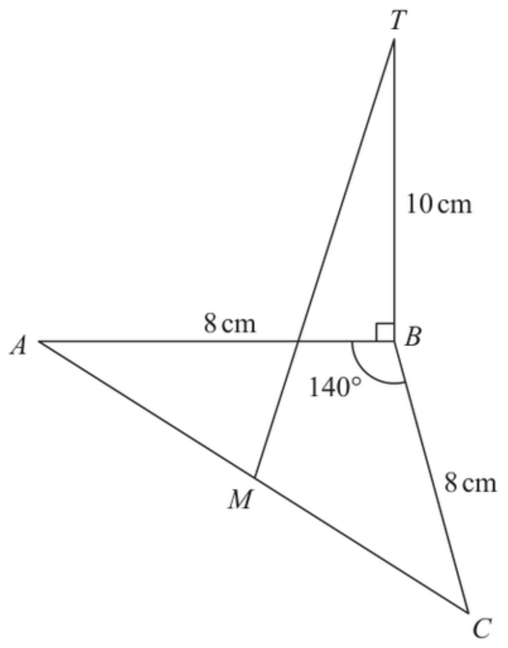


Diagram **NOT** accurately drawn

Angle $ABC = 140°$
$AB = BC = 8$ cm
$TB = 10$ cm
M is the midpoint of AC.

Calculate the size of the angle between MT and the horizontal plane ABC.
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2HR | Question 22

3D Pythagoras & Trigonometry

1. Use trigonometry

Since $AB = BC = 8$, triangle ABC is isosceles.

2. The line from to the

The line from B to the midpoint M bisects angle ABC.

$$\angle ABM = \frac{140^\circ}{2} = 70^\circ$$

3. Use trigonometry

In right triangle ABM,

$$\cos 70^\circ = \frac{BM}{8}$$

$$BM = 8\cos 70^\circ$$

4. Use trigonometry

Since T is vertically above B, the projection of MT onto the horizontal plane is MB.

5. Let the required angle be

Let the required angle be θ.

$$\tan\theta = \frac{TB}{BM}$$

$$\tan\theta = \frac{10}{8\cos 70^\circ}$$

$$\theta = 74.7^\circ$$

Final answer

74.7°

CLASSIFIED PROBLEM Paper: P1H | Session: Jan 2022 | Q#: 21 | Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jan 2022 P1H

21 The diagram shows the prism *ABCDEFGHJK* with horizontal base *AEFG*

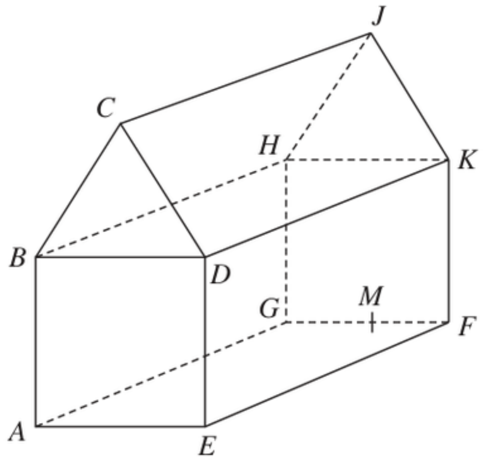


Diagram **NOT** accurately drawn

ABCDE is a cross section of the prism where
ABDE is a square
BCD is an equilateral triangle

$EF = 2 \times AE$
M is the midpoint of GF so that JM is vertical.

Angle $MAJ = y°$

Given that $\tan y° = T$

find the value of T, giving your answer in the form $\dfrac{\sqrt{p}+\sqrt{q}}{17}$ where p and q are integers.

$T =$ ....

(Total for Question 21 is 5 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jan 2022 P1H | Question 21
3D Pythagoras & Trigonometry

1. Let
Let

$$AE = x$$

2. Use trigonometry
Since $ABDE$ is a square,

$$AB = BD = DE = AE = x$$

3. Use trigonometry
Since BCD is an equilateral triangle, the height from C to BD is

$$\frac{\sqrt{3}}{2}x$$

4. Use trigonometry
Therefore the vertical height from the base to J is

$$JM = x + \frac{\sqrt{3}}{2}x$$

$$JM = \frac{x(2+\sqrt{3})}{2}$$

5. Use trigonometry
Also,

$$EF = 2AE = 2x$$

6. Find the midpoint
Since M is the midpoint of GF, the horizontal distance from A to M is

$$AM = \sqrt{\left(\frac{x}{2}\right)^2 + (2x)^2}$$

$$AM = \sqrt{\frac{x^2}{4} + 4x^2}$$

$$AM = \frac{x\sqrt{17}}{2}$$

7. Use trigonometry
In right triangle AMJ,

$$\tan y^\circ = \frac{JM}{AM}$$

$$T = \frac{\frac{x(2+\sqrt{3})}{2}}{\frac{x\sqrt{17}}{2}}$$

$$T = \frac{2+\sqrt{3}}{\sqrt{17}}$$

8. Rationalise the denominator

Rationalise the denominator:

$$T = \frac{(2+\sqrt{3})\sqrt{17}}{17}$$

$$T = \frac{2\sqrt{17}+\sqrt{51}}{17}$$

$$T = \frac{\sqrt{68}+\sqrt{51}}{17}$$

Final answer

$T = \dfrac{\sqrt{68}+\sqrt{51}}{17}$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2023 | Q#: 22 | Marks: 5

TOPIC: 3D Pythagoras & Trigonometry

Source: Jun 2023 P2HR

22 Here is a cuboid *ABCDEFGH*

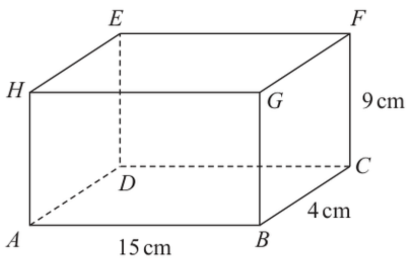


Diagram **NOT** accurately drawn

$AB = 15$ cm $\quad BC = 4$ cm $\quad CF = 9$ cm

(a) Work out the length of *BE*
Give your answer correct to 3 significant figures.

.... cm
(2)

Here is a cuboid *PQRSTUVW*

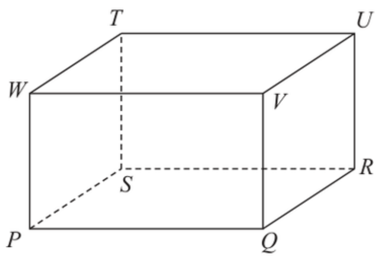


Diagram **NOT** accurately drawn

$PR = 42$ cm

The size of the angle between *PU* and the plane *PQRS* is 30°

M is the midpoint of *PR*

(b) Work out the size of angle *UMR*
Give your answer correct to 3 significant figures.

.... °
(3)

(Total for Question 22 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2HR | Question 22

3D Pythagoras & Trigonometry

1. Find the gradient

For part (a), BE is the space diagonal using the three perpendicular lengths:

$$AB = 15, \qquad BC = 4, \qquad CF = 9$$

$$BE^2 = 15^2 + 4^2 + 9^2$$

$$BE^2 = 225 + 16 + 81 = 322$$

$$BE = \sqrt{322} = 17.9 \text{ cm}$$

2. Use trigonometry

For part (b), the projection of PU onto the plane $PQRS$ is PR.

$$PR = 42$$

3. The angle between and the

The angle between PU and the plane is $30°$, so

$$\tan 30° = \frac{RU}{PR}$$

$$RU = 42 \tan 30°$$

4. Find the midpoint

Since M is the midpoint of PR,

$$MR = 21$$

5. Use trigonometry

In right triangle MRU,

$$\tan \angle UMR = \frac{RU}{MR}$$

$$\tan \angle UMR = \frac{42 \tan 30°}{21}$$

$$\tan \angle UMR = 2 \tan 30°$$

$$\angle UMR = 49.1°$$

Final answer

$BE = 17.9$ cm, $\angle UMR = 49.1°$

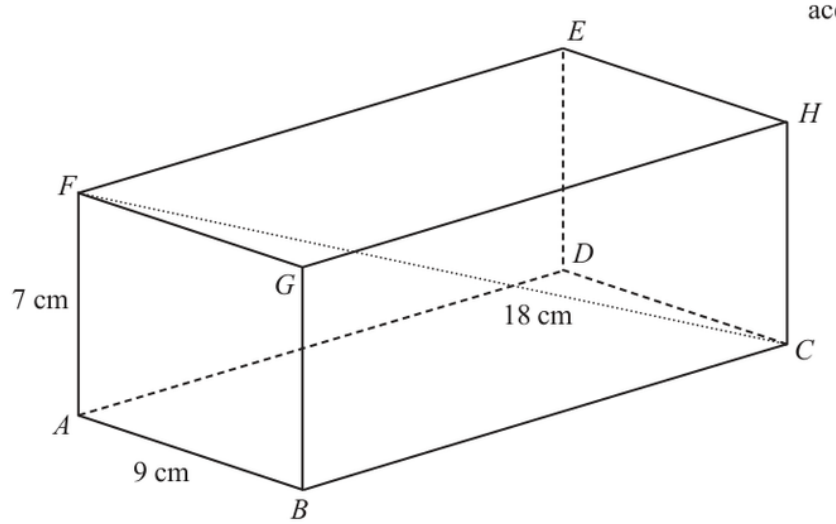
CLASSIFIED PROBLEM Paper: P1HR | Session: Jun 2025 | Q#: 20 | Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jun 2025 P1HR

20 The diagram shows cuboid *ABCDEFGH*

Diagram **NOT** accurately drawn



$AB = 9$ cm $\quad AF = 7$ cm $\quad FC = 18$ cm

Calculate the length of BC
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1HR | Question 20

3D Pythagoras & Trigonometry

1. Use trigonometry

In the rectangle $ABCF$, use Pythagoras with the diagonal FC:

$$AC^2 = FC^2 - AF^2$$

$$AC^2 = 18^2 - 7^2 = 324 - 49 = 275$$

2. Use trigonometry

Now use Pythagoras in triangle ABC:

$$BC^2 = AC^2 - AB^2$$

$$BC^2 = 275 - 9^2 = 275 - 81 = 194$$

$$BC = \sqrt{194} = 13.928\ldots$$

Final answer

13.9 cm to 3 significant figures

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2025 | Q#: 24 | Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jun 2025 P2H

24 The diagram shows a square-based pyramid *ABCDE*

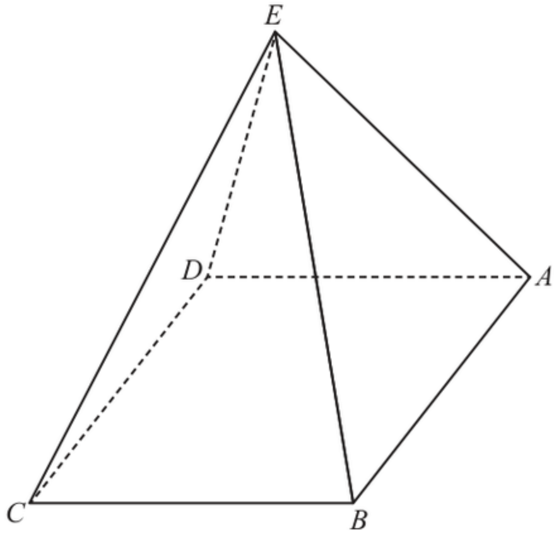


Diagram **NOT** accurately drawn

$EA = EB = EC = ED$

M is the centre of the horizontal square base $ABCD$
Q is the midpoint of AB
Angle $EQM = 80°$

$EA : AB = n : 1$

Find the value of n
Give your answer correct to 3 significant figures.

$n =$ ....

(Total for Question 24 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 24

3D Pythagoras & Trigonometry

1. Let

Let

$$AB = s$$

2. Find the midpoint

Since M is the centre of the square base and Q is the midpoint of AB,

$$QM = \frac{s}{2}$$

3. Use trigonometry

Since $EA = EB = EC = ED$, the point E is vertically above M.

4. Use trigonometry

In right triangle EQM,

$$\tan 80^\circ = \frac{EM}{QM}$$

$$EM = \frac{s}{2}\tan 80^\circ$$

5. The distance from the centre

The distance from the centre of a square to a vertex is half the diagonal:

$$AM = \frac{s\sqrt{2}}{2}$$

6. Use trigonometry

In right triangle EMA,

$$EA^2 = EM^2 + AM^2$$

$$EA^2 = \left(\frac{s}{2}\tan 80^\circ\right)^2 + \left(\frac{s\sqrt{2}}{2}\right)^2$$

$$EA = s\sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

7. Use trigonometry

So

$$\frac{EA}{AB} = \sqrt{\frac{\tan^2 80^\circ}{4} + \frac{1}{2}}$$

$$n = 2.92$$

Final answer

$n = 2.92$

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2025 | Q#: 23 | Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Source: Jun 2025 P2HR

23 The diagram shows a triangular prism, *ABCDEF*, with a horizontal rectangular base *ABCD*

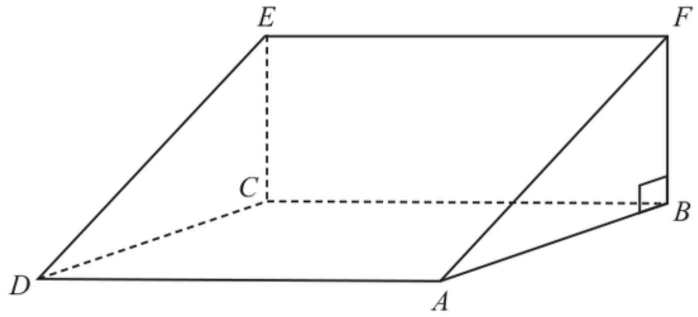


Diagram **NOT** accurately drawn

M is the midpoint of the line *AC*

$AC = 40\,\text{cm}$ angle $CAE = 35°$ angle $ABF = 90°$

Work out the size of angle *CME*
Give your answer correct to 3 significant figures.

.... °

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 23

3D Pythagoras & Trigonometry

1. Use trigonometry

$$AC = 40$$

2. The point is vertically above

The point E is vertically above C, so triangle ACE is right-angled at C.

$$\angle CAE = 35^\circ$$

$$\tan 35^\circ = \frac{CE}{AC}$$

$$CE = 40 \tan 35^\circ$$

3. Find the midpoint

Since M is the midpoint of AC,

$$CM = 20$$

4. Use trigonometry

In right triangle CME,

$$\tan \angle CME = \frac{CE}{CM}$$

$$\tan \angle CME = \frac{40 \tan 35^\circ}{20}$$

$$\tan \angle CME = 2 \tan 35^\circ$$

$$\angle CME = 54.5^\circ$$

Final answer

54.5°

CLASSIFIED PROBLEM

Paper: P1H | Session: May 2021 | Q#: 22 | Marks: 4

TOPIC: **3D Pythagoras & Trigonometry**

Source: May 2021 P1H

22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

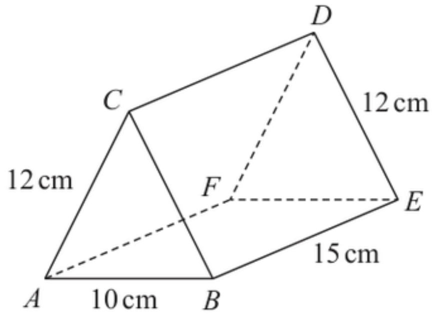


Diagram **NOT** accurately drawn

$AC = BC = FD = ED = 12\,\text{cm}$ $\quad AB = 10\,\text{cm}$ $\quad BE = 15\,\text{cm}$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

.... °

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2021 P1H | Question 22
3D Pythagoras & Trigonometry

1. Calculate value

The triangular end is isosceles, with

$$AC = BC = 12, \qquad AB = 10$$

2. The perpendicular height of the

The perpendicular height of the triangle is found by halving AB:

$$AM = 5$$

$$CM^2 = 12^2 - 5^2 = 144 - 25 = 119$$

$$CM = \sqrt{119}$$

3. The projection of on the

The projection of AD on the horizontal base goes from A to the midpoint of FE. Its length is

$$\sqrt{15^2 + 5^2} = \sqrt{250}$$

4. Use trigonometry

So, for the angle θ between AD and the base,

$$\tan\theta = \frac{\sqrt{119}}{\sqrt{250}}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{119}}{\sqrt{250}}\right) = 34.602\ldots$$

Final answer
34.6°

CLASSIFIED PROBLEM

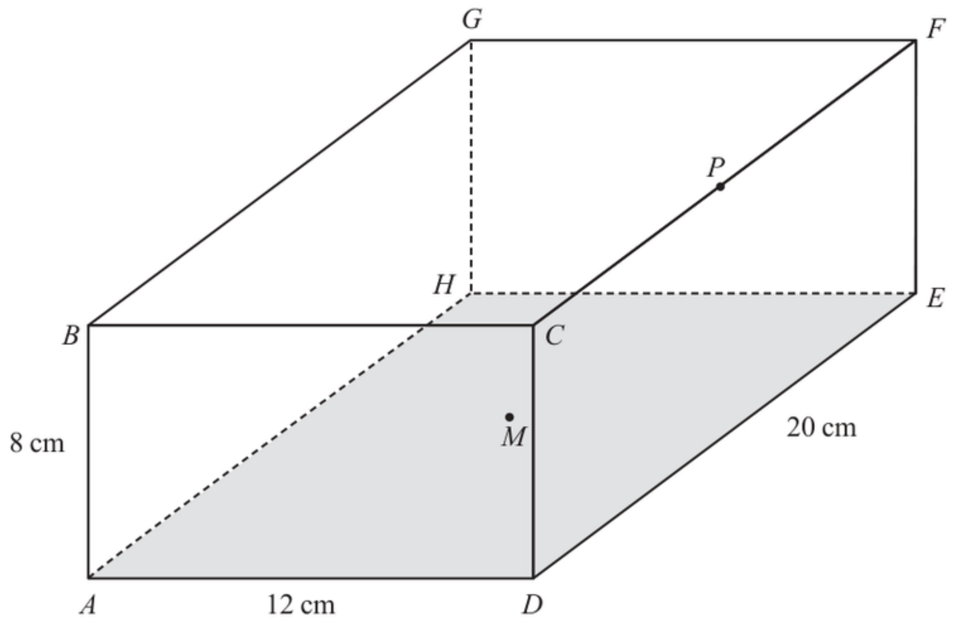
Paper: P1HR | Session: May 2024 | Q#: 22 | Marks: 6

TOPIC: 3D Pythagoras & Trigonometry

Source: May 2024 P1HR

22 The diagram shows a cuboid *ABCDEFGH* with horizontal base *ADEH*

Diagram **NOT** accurately drawn



$AB = 8$ cm $\quad AD = 12$ cm $\quad DE = 20$ cm

M is the midpoint of the base *ADEH* and *P* is the midpoint of the edge *CF*

Work out the size of angle *BMP*
Give your answer correct to one decimal place.

.... °

(Total for Question 22 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1HR | Question 22
3D Pythagoras & Trigonometry

1. Use trigonometry
Use coordinates with

$$A = (0, 0, 0)$$

2. Use trigonometry
Since

$$AD = 12, \qquad DE = 20, \qquad AB = 8$$

3. take
take

$$B = (0, 0, 8)$$

4. The centre of the rectangular
The centre of the rectangular base $ADEH$ is

$$M = (6, 10, 0)$$

5. Find the midpoint
Point P is the midpoint of CF.

$$C = (12, 0, 8), \qquad F = (12, 20, 8)$$

$$P = (12, 10, 8)$$

6. Use trigonometry
Now

$$\vec{MB} = B - M = (-6, -10, 8)$$

$$\vec{MP} = P - M = (6, 0, 8)$$

7. Use the scalar product formula
Use the scalar product formula:

$$\vec{MB} \cdot \vec{MP} = (-6)(6) + (-10)(0) + (8)(8)$$

$$\vec{MB} \cdot \vec{MP} = 28$$

$$|\vec{MB}| = \sqrt{(-6)^2 + (-10)^2 + 8^2} = \sqrt{200}$$

$$|\vec{MP}| = \sqrt{6^2 + 0^2 + 8^2} = 10$$

$$\cos \angle BMP = \frac{28}{10\sqrt{200}}$$

$$\angle BMP = 78.6^\circ$$

Final answer
78.6°

CLASSIFIED PROBLEM Paper: P1H | Session: Nov 2025 | Q#: 26 | Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Source: Nov 2025 P1H

26 Here is a prism *ABCDEFGH* with a horizontal, rectangular base *ABGF*

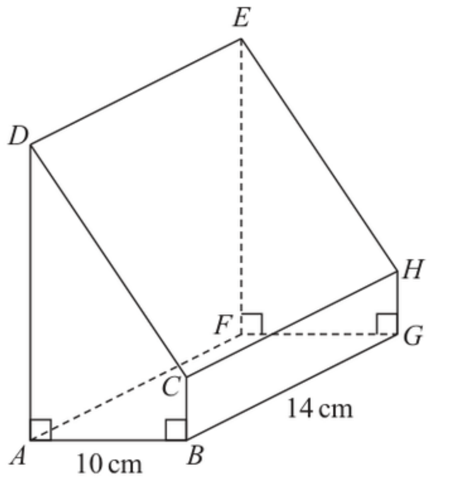


Diagram **NOT** accurately drawn

$AB = 10\text{ cm}$ $BG = 14\text{ cm}$

angle DAB = angle ABC = angle EFG = angle $FGH = 90°$

$BC = \frac{1}{5}AD$

The angle of elevation of E from B is $50°$

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

.... cm^3

(Total for Question 26 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 26

3D Pythagoras & Trigonometry

1. The horizontal base is a

The horizontal base $ABGF$ is a rectangle.

$$AB = 10, \qquad BG = 14$$

2. Use trigonometry

So the horizontal distance from B to F is

$$BF = \sqrt{10^2 + 14^2} = \sqrt{296}$$

3. Use trigonometry

The angle of elevation of E from B is $50°$.

4. Use trigonometry

Since E is vertically above F,

$$\tan 50° = \frac{FE}{BF}$$

$$FE = \sqrt{296}\tan 50°$$

5. The prism has the same

The prism has the same cross section throughout, so

$$AD = FE = \sqrt{296}\tan 50°$$

6. Use trigonometry

Also

$$BC = \frac{1}{5}AD$$

7. Area of the trapezium cross

Area of the trapezium cross section $ABCD$:

$$\frac{1}{2}(AD + BC) \times AB$$

$$= \frac{1}{2}\left(AD + \frac{1}{5}AD\right) \times 10$$

$$= 6AD$$

8. Volume of the prism

Volume of the prism:

$$6AD \times 14 = 84AD$$

$$= 84\sqrt{296}\tan 50^\circ$$

$$= 1722.311\ldots$$

9. Correct to 3 significant figures

Correct to 3 significant figures:

$$1720$$

Final answer

$\boxed{1720\ \text{cm}^3}$

Topic 20

Histograms

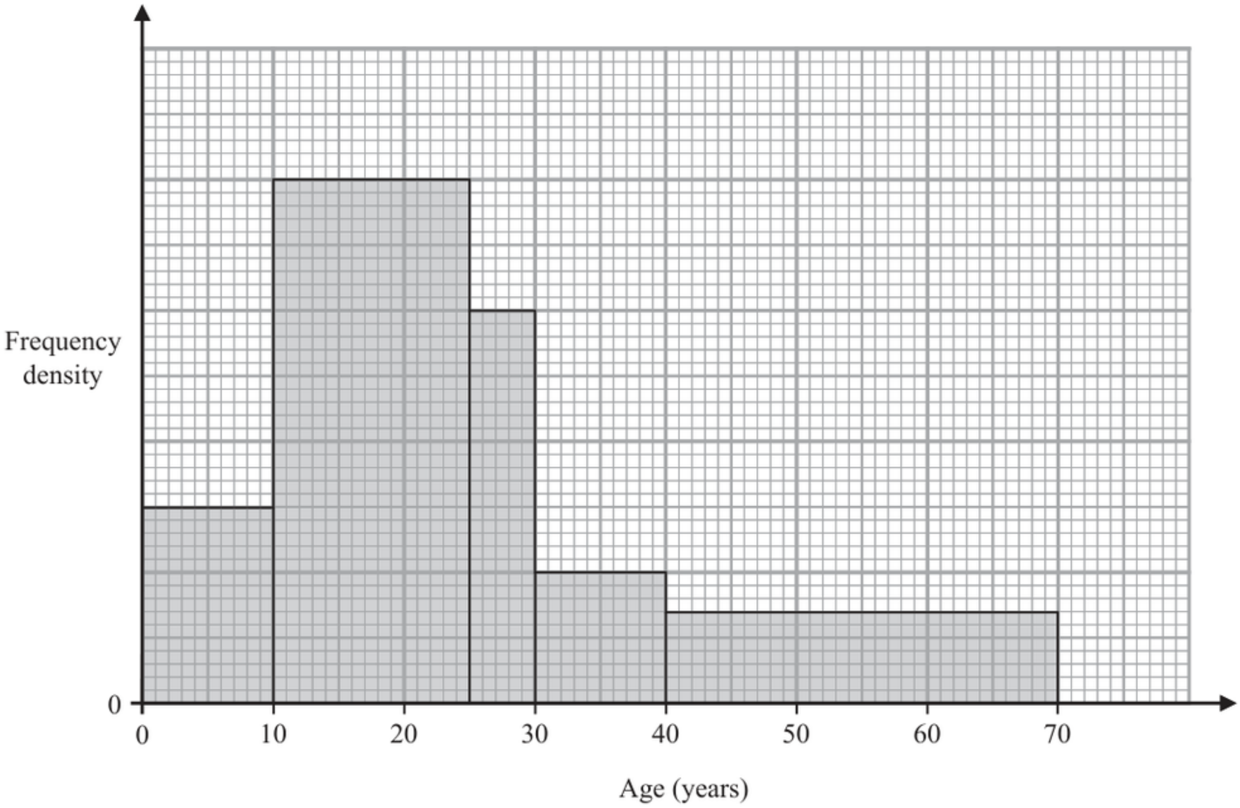
CLASSIFIED PROBLEM

Paper: 4WM1HR | Session: May2025 | Q#: 21 | Marks: 3

TOPIC: **Histograms**

Source: May 2025 4WM1HR

21 The histogram shows information about the ages of the members of a chess club.



There are 60 members aged between 10 and 25
There are no members older than 70

Work out an estimate for the number of members who are over 35 years of age.

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1HR | Question 21

Histograms

1. The interval to has width

The interval 10 to 25 has width 15 and frequency 60, so its frequency density is

$$\frac{60}{15} = 4$$

2. Use histogram

Using that scale from the histogram:

$$35 \text{ to } 40 : 5 \times 1 = 5$$

$$40 \text{ to } 70 : 30 \times 0.75 = 22.5$$

$$5 + 22.5 = 27.5$$

Final answer

about 28 members.

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: Nov2025 | Q#: 20 | Marks: 3

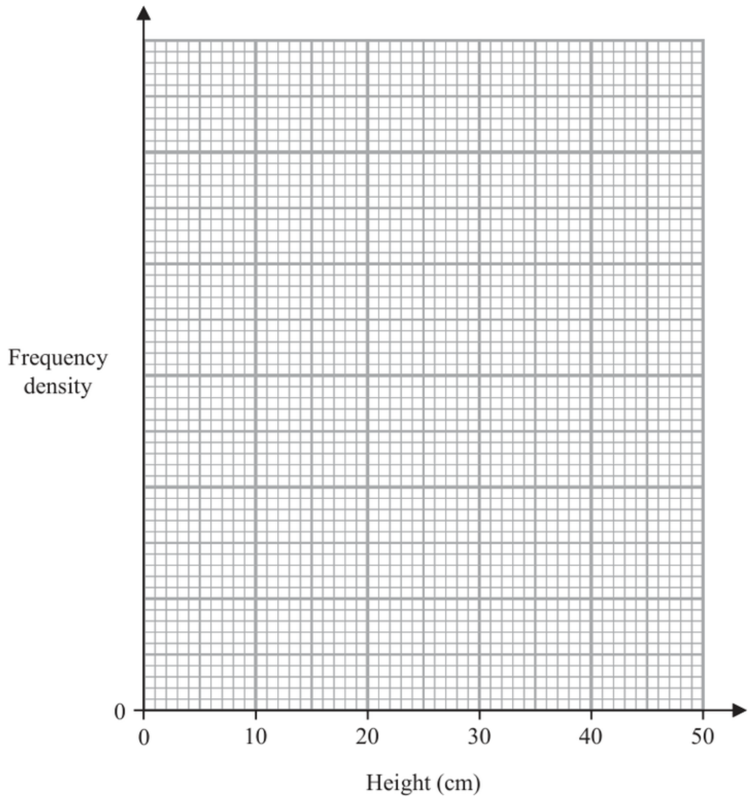
TOPIC: **Histograms**

Source: Nov 2025 4WM1H

20 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	**Frequency**
$0 < h \leqslant 5$	12
$5 < h \leqslant 15$	34
$15 < h \leqslant 30$	45
$30 < h \leqslant 50$	28

On the grid, draw a histogram for this information.
You must include a scale on the vertical axis.



(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | **Question 20**

Histograms

1. Frequency density

Frequency density $= \dfrac{\text{frequency}}{\text{class width}}$.

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

$$5 < h \leq 15 : \frac{34}{10} = 3.4$$

$$15 < h \leq 30 : \frac{45}{15} = 3$$

$$30 < h \leq 50 : \frac{28}{20} = 1.4$$

Final answer

draw bars with densities $2.4, 3.4, 3, 1.4$.

Topic 21

Probability Toolkit

CLASSIFIED PROBLEM

Paper: 4WM1H | Session: May2025 | Q#: 22 | Marks: 3

TOPIC: **Probability Toolkit**

Source: May 2025 4WM1H

22 A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

(Total for Question 22 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2025 4WM1H | **Question 22**
Probability Toolkit

1. Total ways to choose counters

Total ways to choose 3 counters:

$$\binom{20}{3} = 1140$$

2. Exactly two red

Exactly two red:

$$\binom{9}{2}\binom{11}{1} = 396$$

3. Exactly two yellow

Exactly two yellow:

$$\binom{7}{2}\binom{13}{1} = 273$$

4. Exactly two green

Exactly two green:

$$\binom{4}{2}\binom{16}{1} = 96$$

5. Favourable outcomes

Favourable outcomes:

$$396 + 273 + 96 = 765$$

$$P = \frac{765}{1140} = \frac{51}{76}$$

Final answer
$\frac{51}{76}$

Topic 22

Histograms

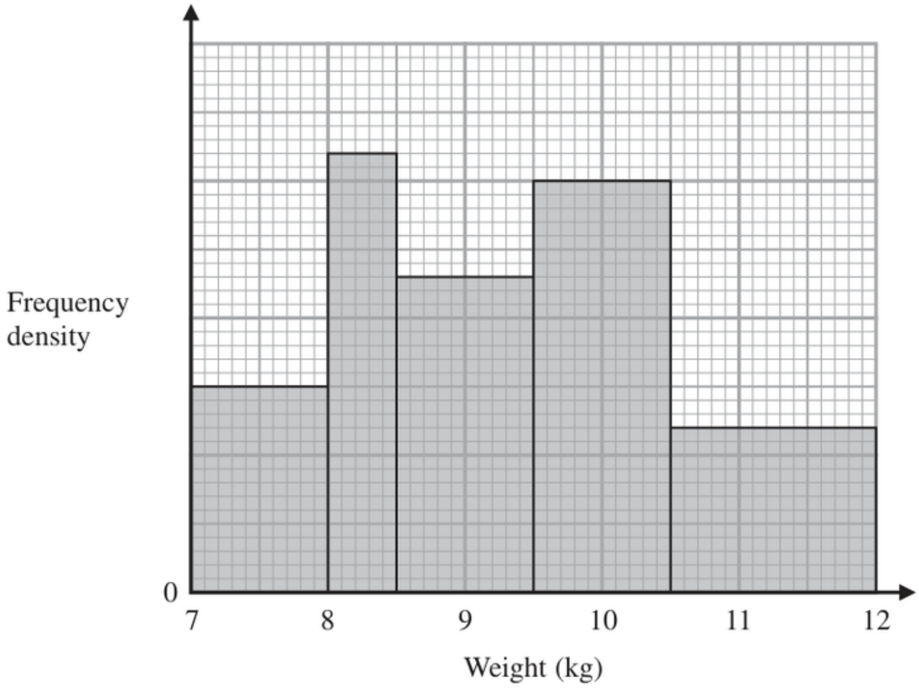
CLASSIFIED PROBLEM

Paper: P1HR | Session: Jan 2022 | Q#: 21 | Marks: 3

TOPIC: **Histograms**

Source: Jan 2022 P1HR

21



The histogram gives information about the weights, in kg, of all the watermelons in a field.

There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P1HR | Question 21
Histograms

1. Use histogram

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency table

For $8 \leq w < 8.5$, the frequency is 16 and the class width is 0.5, so

$$\text{frequency density} = \frac{16}{0.5} = 32$$

3. Use histogram

Using this scale from the histogram, the frequency densities are:

$$15,\ 32,\ 23,\ 30,\ 12$$

4. The total frequency is

The total frequency is

$$(1)(15) + (0.5)(32) + (1)(23) + (1)(30) + (1.5)(12)$$

$$= 15 + 16 + 23 + 30 + 18$$

$$= 102$$

Final answer
102 watermelons

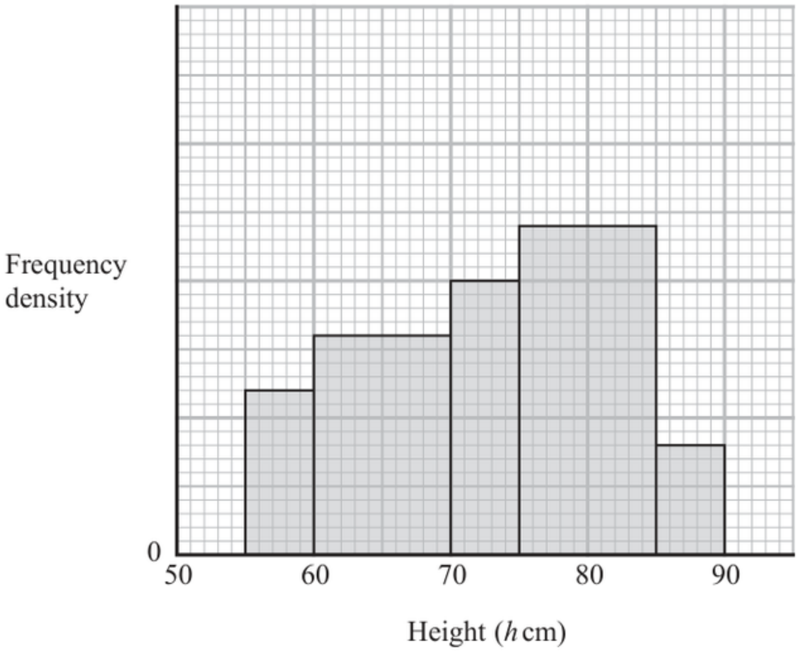
CLASSIFIED PROBLEM

Paper: P2H | Session: Jan 2022 | Q#: 20 | Marks: 4

TOPIC: **Histograms**

Source: Jan 2022 P2H

20 The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leqslant 85$
One of the tomato plants is selected at random.

Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2022 P2H | Question 20

Histograms

1. Use histogram

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency table

For $75 < h \leq 85$, the frequency is 12 and the class width is 10, so

$$\text{frequency density} = \frac{12}{10} = 1.2$$

3. Use histogram

Using this scale from the histogram, the frequency densities are:

$$0.6, \ 0.8, \ 1.0, \ 1.2, \ 0.4$$

4. The total number of tomato

The total number of tomato plants is

$$(5)(0.6) + (10)(0.8) + (5)(1.0) + (10)(1.2) + (5)(0.4)$$

$$= 3 + 8 + 5 + 12 + 2 = 30$$

5. Solve inequality

For $h > 82.5$:

$$(2.5)(1.2) + (5)(0.4) = 3 + 2 = 5$$

6. Calculate probability

So the probability is

$$\frac{5}{30} = \frac{1}{6}$$

Final answer

$\frac{1}{6}$

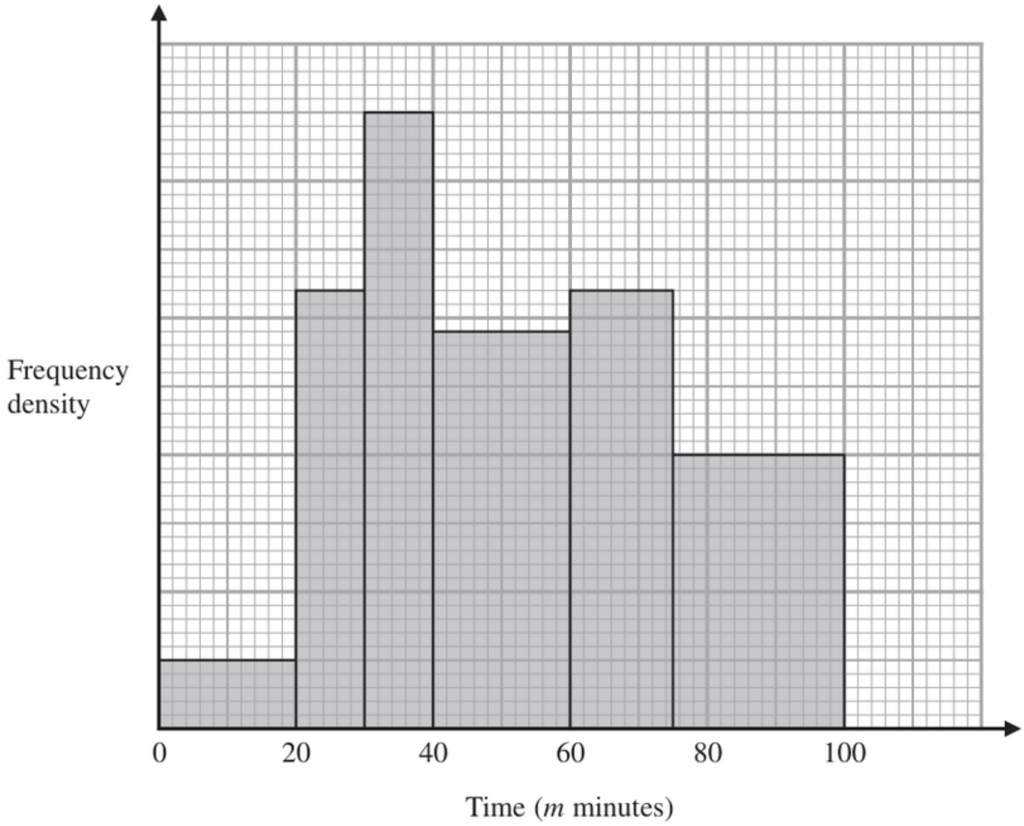
CLASSIFIED PROBLEM

Paper: P1H | Session: Jun 2022 | Q#: 21 | Marks: 3

TOPIC: **Histograms**

Source: Jun 2022 P1H

21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$
There are 10 children for whom $m \leqslant 20$

Work out an estimate for the number of children for whom $50 < m \leqslant 80$

(Total for Question 21 is 3 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

Jun 2022 P1H | Question 21
Histograms

1. Use histogram

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency table

For $m \leq 20$, the frequency is 10 and the class width is 20, so

$$\text{frequency density} = \frac{10}{20} = 0.5$$

3. Use histogram

Using this scale from the histogram:

$$40 < m \leq 60 \quad \text{has frequency density } 2.9$$

$$60 < m \leq 75 \quad \text{has frequency density } 3.2$$

$$75 < m \leq 100 \quad \text{has frequency density } 2.0$$

4. Use part of each relevant

For $50 < m \leq 80$, use part of each relevant bar:

$$(10)(2.9) + (15)(3.2) + (5)(2.0)$$

$$= 29 + 48 + 10$$

$$= 87$$

Final answer
87 children

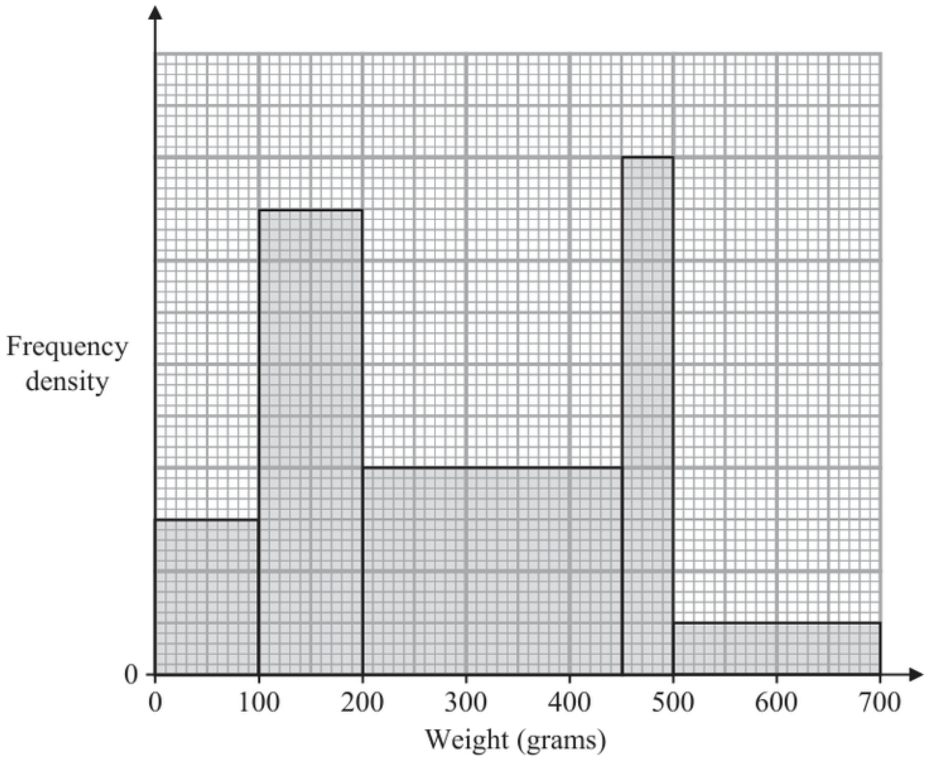
CLASSIFIED PROBLEM

Paper: P2HR | Session: Jun 2025 | Q#: 20 | Marks: 4

TOPIC: **Histograms**

Source: Jun 2025 P2HR

20 The histogram gives some information about the weights, in grams, of some books.



75 books weigh less than 100 grams.

A book is chosen at random.

Find an estimate for the probability that this book weighs between 400 grams and 600 grams.

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2HR | Question 20

Histograms

1. Use histogram

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency table

For books weighing less than 100 g, the frequency is 75 and the class width is 100, so

$$\text{frequency density} = \frac{75}{100} = 0.75$$

3. Use histogram

Using this scale from the histogram, the frequency densities are:

$$0.75,\ 2.25,\ 1.0,\ 2.5,\ 0.25$$

4. The total number of books

The total number of books is

$$(100)(0.75) + (100)(2.25) + (250)(1.0) + (50)(2.5) + (200)(0.25)$$

$$= 75 + 225 + 250 + 125 + 50 = 725$$

5. Calculate value

For weights between 400 g and 600 g:

$$(50)(1.0) + (50)(2.5) + (100)(0.25)$$

$$= 50 + 125 + 25 = 200$$

6. Calculate probability

So the probability is

$$\frac{200}{725} = \frac{8}{29}$$

Final answer

$\frac{8}{29}$ or 0.276 to 3 significant figures

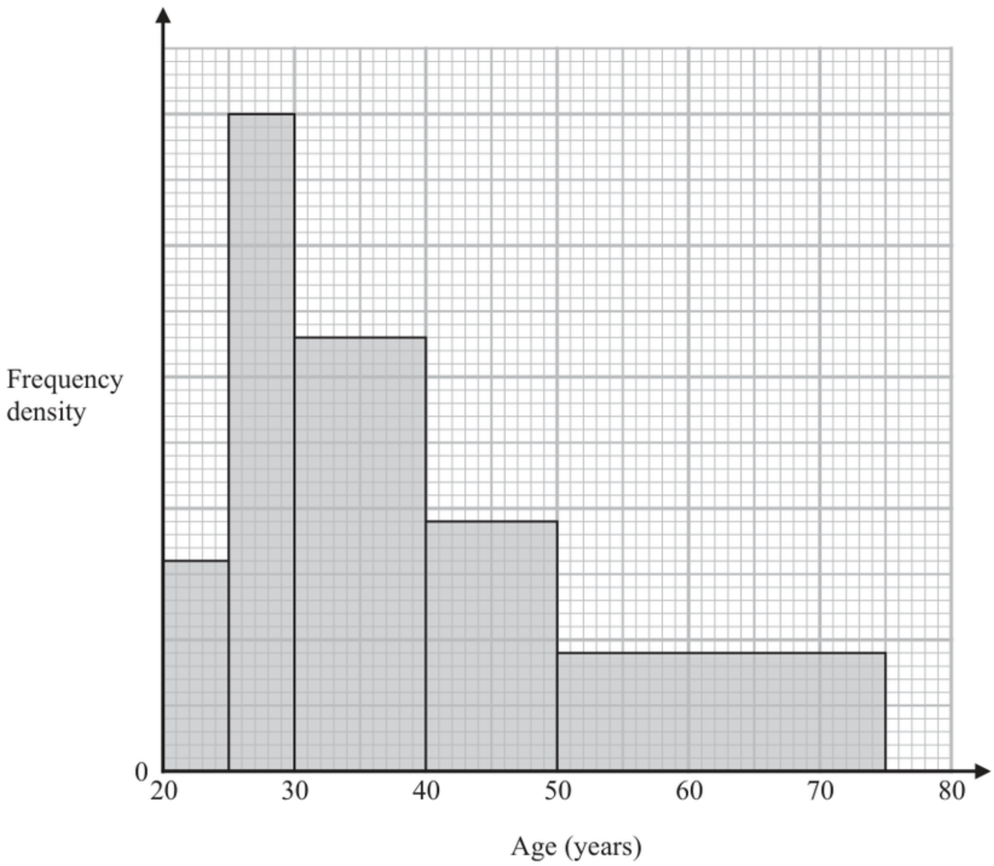
CLASSIFIED PROBLEM

Paper: P1HR | Session: May 2023 | Q#: 22 | Marks: 4

TOPIC: **Histograms**

Source: May 2023 P1HR

22 Some people attend a concert.
The histogram shows information about the ages of these people.



Work out an estimate for the percentage of these people who are aged more than 55 years.
Give your answer correct to one decimal place.

....%

(Total for Question 22 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2023 P1HR | Question 22

Histograms

1. The scale of the frequency

The scale of the frequency density axis is not needed because the percentage is found from areas.

2. Calculate area

Using the bar heights from the histogram, the relative areas are:

$$(5)(277) + (5)(866) + (10)(571) + (10)(329) + (25)(155)$$

$$= 1385 + 4330 + 5710 + 3290 + 3875$$

$$= 18590$$

3. Calculate value

For ages more than 55, use the part of the last bar from 55 to 75:

$$(20)(155) = 3100$$

4. Calculate percentage

So the estimated percentage is

$$\frac{3100}{18590} \times 100 = 16.675\ldots$$

Final answer

16.7%

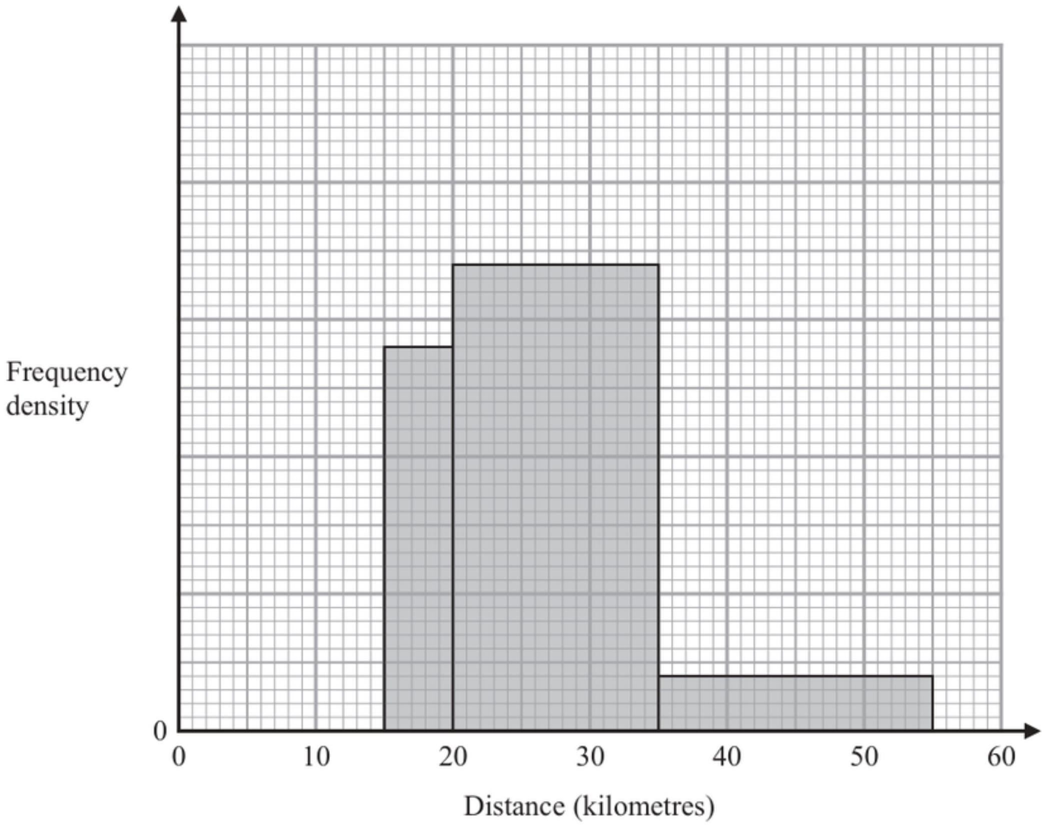
CLASSIFIED PROBLEM

Paper: P1H | Session: May 2024 | Q#: 22 | Marks: 3

TOPIC: **Histograms**

Source: May 2024 P1H

22 The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.
None of the adults ran more than 55 kilometres.
14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

(Total for Question 22 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1H | Question 22

Histograms

1. Use histogram

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency table

For $15 \leq d < 20$, the frequency is 14 and the class width is 5, so

$$\text{frequency density} = \frac{14}{5} = 2.8$$

3. Use histogram

Using this scale from the histogram:

$$20 \leq d < 35 \quad \text{has frequency density } 3.4$$

$$35 \leq d \leq 55 \quad \text{has frequency density } 0.4$$

4. Calculate value

So the known frequencies are

$$14 + (15)(3.4) + (20)(0.4)$$

$$= 14 + 51 + 8 = 73$$

5. Use frequency table

There are 100 adults in total, so the missing frequency from 5 km to 15 km is

$$100 - 73 = 27$$

6. The missing class width is

The missing class width is 10, so its frequency density is

$$\frac{27}{10} = 2.7$$

Final answer

draw the missing bar from 5 to 15 with frequency density 2.7

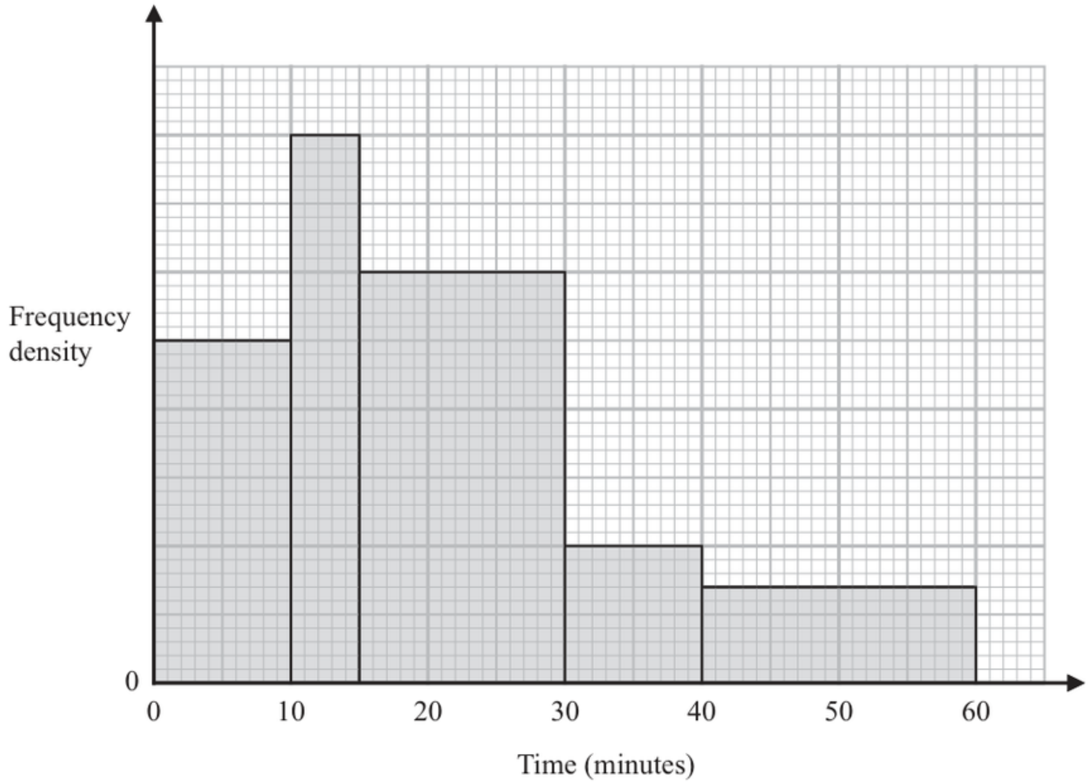
CLASSIFIED PROBLEM

Paper: P2H | Session: Nov 2025 | Q#: 21 | Marks: 3

TOPIC: **Histograms**

Source: Nov 2025 P2H

21 The histogram shows information about the times, in minutes, that some trains arrived late at a station one day.



20 of these trains arrived between 10 minutes late and 15 minutes late.

No trains arrived more than 60 minutes late.

Work out an estimate for the number of these trains that arrived at least 25 minutes late.

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P2H | **Question 21**

Histograms

1. Use histogram

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

2. Use frequency table

For $10 < t \leq 15$, the frequency is 20 and the class width is 5, so

$$\text{frequency density} = \frac{20}{5} = 4$$

3. Use histogram

Using this scale from the histogram:

$$15 < t \leq 30 \quad \text{has frequency density } 3$$

$$30 < t \leq 40 \quad \text{has frequency density } 1$$

$$40 < t \leq 60 \quad \text{has frequency density } 0.7$$

4. Calculate value

For trains at least 25 minutes late:

$$(5)(3) + (10)(1) + (20)(0.7)$$

$$= 15 + 10 + 14$$

$$= 39$$

Final answer

39 trains

Topic 23

Combined & Conditional Probability

CLASSIFIED PROBLEM Paper: 4WM1H | Session: Nov2025 | Q#: 23 | Marks: 3

TOPIC: **Combined & Conditional Probability**

Source: Nov 2025 4WM1H

23 There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 23 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 4WM1H | Question 23
Combined & Conditional Probability

1. There are 2 pink buttons

There are 2 pink buttons.

2. At least one pink button

At least one pink button is still in the box unless both pink buttons are taken.

3. Use the complement

So use the complement.

4. The probability that both pink

The probability that both pink buttons are taken is

$$\frac{\binom{2}{2}\binom{13}{1}}{\binom{15}{3}}$$

$$= \frac{13}{455}$$

$$= \frac{1}{35}$$

5. Evaluate fraction

Therefore,

$$P(\text{at least one pink is still in the box}) = 1 - \frac{1}{35}$$

$$= \frac{34}{35}$$

Final answer
$\frac{34}{35}$

Topic 24

Probability Toolkit

CLASSIFIED PROBLEM

Paper: P2HR | Session: Jan 2021 | Q#: 21 | Marks: 6

TOPIC: **Probability Toolkit**

Source: Jan 2021 P2HR

21 A bag contains n beads.
6 of the beads are red and the rest are blue.

Ravi is going to take at random 2 beads from the bag.

The probability that the 2 beads will be of the same colour is $\frac{9}{17}$

Using algebra, and showing each stage of your working, calculate the value of n.

n = ...

(Total for Question 21 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jan 2021 P2HR | **Question 21**
Probability Toolkit

1. There are beads altogether

There are n beads altogether.

2. Red beads

Red beads:

$$6$$

3. Blue beads

Blue beads:

$$n - 6$$

4. The probability of taking two

The probability of taking two beads of the same colour is

$$P(RR) + P(BB) = \frac{9}{17}$$

5. Use trigonometry

So

$$\frac{6}{n} \cdot \frac{5}{n-1} + \frac{n-6}{n} \cdot \frac{n-7}{n-1} = \frac{9}{17}$$

$$\frac{30 + (n-6)(n-7)}{n(n-1)} = \frac{9}{17}$$

6. Expand

Expand:

$$(n-6)(n-7) = n^2 - 13n + 42$$

7. Use trigonometry

So

$$\frac{n^2 - 13n + 72}{n(n-1)} = \frac{9}{17}$$

8. Cross multiply

Cross multiply:

$$17(n^2 - 13n + 72) = 9n(n-1)$$

$$17n^2 - 221n + 1224 = 9n^2 - 9n$$

$$8n^2 - 212n + 1224 = 0$$

9. Divide by 4

Divide by 4:

$$2n^2 - 53n + 306 = 0$$

10. Factor

Factor:

$$(2n - 17)(n - 18) = 0$$

11. Use trigonometry

So

$$n = \frac{17}{2} \quad \text{or} \quad n = 18$$

12. The number of beads must

The number of beads must be a whole number, so

$$n = 18$$

Final answer

$\boxed{18}$

CLASSIFIED PROBLEM

Paper: P1HR | Session: Jun 2022 | Q#: 24 | Marks: 5

TOPIC: **Probability Toolkit**

Source: Jun 2022 P1HR

24 Elliot has x counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.
He then picks at random a second counter and turns the counter over.

The probability that there are still 5 counters showing a red face is $\frac{19}{32}$

Work out the value of x
Show clear algebraic working.

$x =$ ....

(Total for Question 24 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2022 P1HR | Question 24

Probability Toolkit

1. There are counters. Initially 5

There are x counters. Initially 5 show red, so $x - 5$ show green.

2. List probability cases

After two turns, there are still 5 red faces if:

3. the same counter is picked

- the same counter is picked twice, or
- one red counter and one green counter are picked.

4. The probability of picking the

The probability of picking the same counter twice is

$$\frac{1}{x}$$

5. The probability of picking one

The probability of picking one red and one green in either order is

$$\frac{5}{x} \cdot \frac{x-5}{x} + \frac{x-5}{x} \cdot \frac{5}{x}$$
$$= \frac{10(x-5)}{x^2}$$

6. Evaluate fraction

So

$$\frac{1}{x} + \frac{10(x-5)}{x^2} = \frac{19}{32}$$
$$\frac{11x-50}{x^2} = \frac{19}{32}$$

$$32(11x - 50) = 19x^2$$

$$19x^2 - 352x + 1600 = 0$$

$$(x - 8)(19x - 200) = 0$$

7. Calculate value

Since x must be a whole number,

$$x = 8$$

Final answer

$x = 8$

CLASSIFIED PROBLEM

Paper: P1H | Session: Jun 2025 | Q#: 21 | Marks: 3

TOPIC: **Probability Toolkit**

Source: Jun 2025 P1H

21 A box contains 20 counters.

9 of the counters are red
7 of the counters are yellow
4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

(Total for Question 21 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P1H | Question 21

Probability Toolkit

1. There are

There are

$$\binom{20}{3} = 1140$$

2. ways to choose 3 counters

ways to choose 3 counters.

3. Exactly two counters are the

Exactly two counters are the same colour:

$$\binom{9}{2}(11) + \binom{7}{2}(13) + \binom{4}{2}(16)$$

$$= 36(11) + 21(13) + 6(16)$$

$$= 396 + 273 + 96 = 765$$

4. Calculate probability

So the probability is

$$\frac{765}{1140} = \frac{51}{76}$$

Final answer

$\frac{51}{76}$

CLASSIFIED PROBLEM Paper: P2H | Session: May 2021 | Q#: 23 | Marks: 5

TOPIC: **Probability Toolkit**

Source: May 2021 P2H

23 Pippa has a box containing N pens.

There are only black pens and red pens in the box.
The number of black pens in the box is 3 more than the number of red pens.

Pippa is going to take at random 2 pens from the box.

The probability that she will take a black pen **followed** by a red pen is $\frac{9}{35}$

Find the possible values of N.
Show clear algebraic working.

(Total for Question 23 is 5 marks)

WORKED SOLUTION *Dr. Eslam Ahmed*

May 2021 P2H | Question 23
Probability Toolkit

1. Let the number of red

Let the number of red pens be r.

2. List probability cases

Then the number of black pens is

$$r + 3$$

3. Calculate value

and

$$N = 2r + 3$$

4. Evaluate fraction

So

$$r = \frac{N-3}{2}$$

5. Evaluate fraction

and

$$r + 3 = \frac{N+3}{2}$$

6. The probability of black followed

The probability of black followed by red is

$$\frac{(N+3)/2}{N} \cdot \frac{(N-3)/2}{N-1} = \frac{9}{35}$$

$$\frac{N^2-9}{4N(N-1)} = \frac{9}{35}$$

$$35(N^2-9) = 36N(N-1)$$

$$35N^2 - 315 = 36N^2 - 36N$$

$$N^2 - 36N + 315 = 0$$

$$(N-15)(N-21) = 0$$

Final answer
$N = 15$ or $N = 21$

CLASSIFIED PROBLEM

Paper: P1HR | Session: May 2024 | Q#: 21 | Marks: 5

TOPIC: **Probability Toolkit**

Source: May 2024 P1HR

21 There are 25 counters in a bag such that

6 counters are blue
x counters are orange, where $x > 9$
the rest of the counters are pink

Maalam takes at random two of the counters from the bag.

The probability that Maalam takes one orange counter and one pink counter is $\dfrac{22}{75}$

Calculate the probability that Maalam takes 2 pink counters from the bag.
Show clear algebraic working.

(Total for Question 21 is 5 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May 2024 P1HR | Question 21
Probability Toolkit

1. There are 25 counters

There are 25 counters.

2. Blue counters

Blue counters:

$$6$$

3. Orange counters

Orange counters:

$$x$$

4. Pink counters

Pink counters:

$$25 - 6 - x = 19 - x$$

5. The probability of taking one

The probability of taking one orange and one pink in either order is

$$\frac{x}{25} \cdot \frac{19 - x}{24} + \frac{19 - x}{25} \cdot \frac{x}{24}$$

$$= \frac{2x(19 - x)}{25 \cdot 24}$$

6. Evaluate fraction

This is equal to $\frac{22}{75}$, so

$$\frac{2x(19 - x)}{600} = \frac{22}{75}$$

$$\frac{x(19 - x)}{300} = \frac{22}{75}$$

$$x(19 - x) = 88$$

$$19x - x^2 = 88$$

$$x^2 - 19x + 88 = 0$$

$$(x - 8)(x - 11) = 0$$

7. Solve inequality

Since $x > 9$,

$$x = 11$$

8. Calculate value

So the number of pink counters is

$$19 - 11 = 8$$

9. The probability of taking 2

The probability of taking 2 pink counters is

$$\frac{8}{25} \cdot \frac{7}{24} = \frac{56}{600}$$

$$= \frac{7}{75}$$

Final answer

$\frac{7}{75}$

CLASSIFIED PROBLEM

Paper: P1HR | Session: MayNov 2020 | Q#: 23 | Marks: 4

TOPIC: **Probability Toolkit**

Source: May-Nov 2020 P1HR

23 In a bag, there are only

3 blue beads
4 white beads
and x orange beads.

Jean is going to take at random two beads from the bag.

The probability that Jean will take two beads of the same colour is $\frac{3}{8}$

Find the total number of beads in the bag.
Show clear algebraic working.

(Total for Question 23 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

May-Nov 2020 P1HR | Question 23

Probability Toolkit

1. The total number of beads

The total number of beads is

$$x+7$$

2. The probability of choosing two

The probability of choosing two beads of the same colour is

$$\frac{\binom{3}{2}+\binom{4}{2}+\binom{x}{2}}{\binom{x+7}{2}}$$

3. Evaluate fraction

This is equal to $\frac{3}{8}$, so

$$\frac{3+6+\frac{x(x-1)}{2}}{\frac{(x+7)(x+6)}{2}}=\frac{3}{8}$$

$$\frac{x^2-x+18}{(x+7)(x+6)}=\frac{3}{8}$$

$$8(x^2-x+18)=3(x+7)(x+6)$$

$$8x^2-8x+144=3x^2+39x+126$$

$$5x^2-47x+18=0$$

$$(x-9)(5x-2)=0$$

4. Calculate value

Since x must be a whole number,

$$x=9$$

5. Calculate value

So the total number of beads is

$$9+7=16$$

Final answer

16 beads

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2023 | Q#: 20 | Marks: 4

TOPIC: **Probability Toolkit**

Source: Nov 2023 P1H

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag.
He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

(Total for Question 20 is 4 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2023 P1H | Question 20
Probability Toolkit

1. Let the probability of taking

Let the probability of taking a 10 cent coin be p.

2. The mean is 10 cents

The mean is 10 cents only when both coins are 10 cent coins.

$$p^2 = \frac{49}{121}$$

$$p = \frac{7}{11}$$

3. Calculate probability

So the probability of taking a 20 cent coin is

$$1 - \frac{7}{11} = \frac{4}{11}$$

4. The mean is 15 cents

The mean is 15 cents when one coin is 10 cents and the other is 20 cents.

5. There are two possible orders

There are two possible orders:

$$P(10, 20) + P(20, 10)$$

$$= \frac{7}{11} \cdot \frac{4}{11} + \frac{4}{11} \cdot \frac{7}{11}$$

$$= \frac{56}{121}$$

Final answer
$\frac{56}{121}$

Topic 25

Probability Diagrams - Venn & Tree Diagrams

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2025 | Q#: 20 | Marks: 6

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Source: Jun 2025 P2H

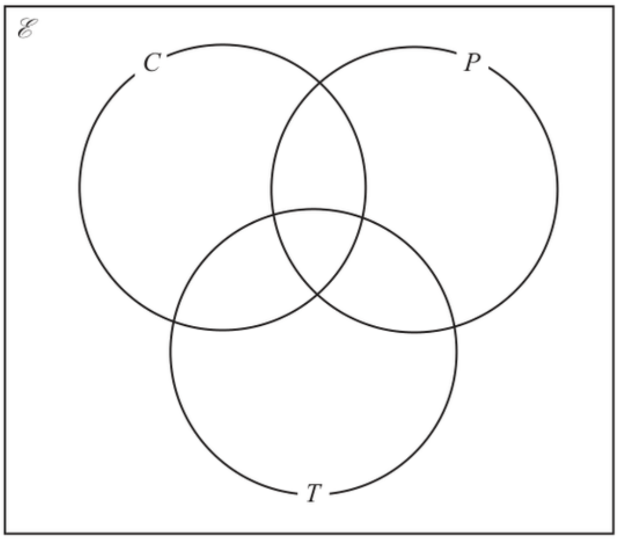
20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots
12 grow carrots and potatoes and tomatoes
18 grow carrots and potatoes
27 grow carrots and tomatoes
32 grow potatoes and tomatoes
29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

....

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

....

(2)

(Total for Question 20 is 6 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2025 P2H | Question 20

Probability Diagrams - Venn & Tree Diagrams

1. Calculate value

There are 120 gardeners in total and 29 are outside all three sets, so

$$n(C \cup P \cup T) = 120 - 29 = 91$$

2. The triple intersection is

The triple intersection is

$$n(C \cap P \cap T) = 12$$

3. Calculate value

So the pair-only regions are:

$$C \cap P \text{ only} = 18 - 12 = 6$$

$$C \cap T \text{ only} = 27 - 12 = 15$$

$$P \cap T \text{ only} = 32 - 12 = 20$$

4. Calculate value

Since 43 grow carrots,

$$C \text{ only} = 43 - 6 - 15 - 12 = 10$$

5. Let the number who grow

Let the number who grow only potatoes be y. The number who grow only tomatoes is also y.

6. Add the regions union

Now add the regions in the union:

$$10 + y + y + 6 + 15 + 20 + 12 = 91$$

$$63 + 2y = 91$$

$$2y = 28$$

$$y = 14$$

7. Calculate value

So the Venn regions are:

$$C \text{ only} = 10, \quad P \text{ only} = 14, \quad T \text{ only} = 14$$

$$C \cap P \text{ only} = 6, \quad C \cap T \text{ only} = 15, \quad P \cap T \text{ only} = 20$$

$$C \cap P \cap T = 12, \quad \text{outside} = 29$$

8. Calculate value

For part (b),

$$n(T' \cap C) = C \text{ only} + C \cap P \text{ only}$$

$$= 10 + 6 = 16$$

9. Evaluate fraction

For part (c), one gardener who grows carrots is chosen.

$$P(\text{also grows potatoes}) = \frac{n(C \cap P)}{n(C)}$$

$$= \frac{18}{43}$$

Final answer

(b) 16, (c) $\frac{18}{43}$

Topic 26

Combined & Conditional Probability

CLASSIFIED PROBLEM

Paper: P2H | Session: Jun 2023 | Q#: 20 | Marks: 3

TOPIC: **Combined & Conditional Probability**

Source: Jun 2023 P2H

20 There are 12 counters in a bag.

3 of the counters are red
9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

(Total for Question 20 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Jun 2023 P2H | Question 20
Combined & Conditional Probability

1. There are 3 red counters

There are 3 red counters.

2. At least one red counter

At least one red counter is still in the bag unless all 3 red counters are taken.

3. Use the complement

So use the complement:

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

$$= \frac{6}{1320}$$

$$= \frac{1}{220}$$

4. Evaluate fraction

Therefore,

$$P(\text{at least one red is still in the bag}) = 1 - \frac{1}{220}$$

$$= \frac{219}{220}$$

Final answer

$\frac{219}{220}$

CLASSIFIED PROBLEM

Paper: P1H | Session: Nov 2025 | Q#: 22 | Marks: 3

TOPIC: **Combined & Conditional Probability**

Source: Nov 2025 P1H

22 There are 15 buttons in a box.

3 of the buttons are red
2 of the buttons are pink
10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 22 is 3 marks)

WORKED SOLUTION

Dr. Eslam Ahmed

Nov 2025 P1H | Question 22

Combined & Conditional Probability

1. There are 2 pink buttons

There are 2 pink buttons.

2. At least one pink button

At least one pink button is still in the box unless both pink buttons are taken.

3. Use the complement

So use the complement.

4. The probability that both pink

The probability that both pink buttons are taken is

$$\frac{\binom{2}{2}\binom{13}{1}}{\binom{15}{3}}$$

$$= \frac{13}{455}$$

$$= \frac{1}{35}$$

5. Evaluate fraction

Therefore,

$$P(\text{at least one pink is still in the box}) = 1 - \frac{1}{35}$$

$$= \frac{34}{35}$$

Final answer

$\frac{34}{35}$